The Rhythm of Numbers

Continued Fractions, Primes, Music and Self-Similarity

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Abstract

This monograph grows out of one guess: that π is close to $3 + k/(7k+1)$ for a suitable whole number k. The guess is right at $k = 16$, where it gives $355/113$, the fraction Zu Chongzhi recorded in the fifth century AD, and it is right for a reason: $k/(7k+1)$ is a continued fraction cut off after two steps. We follow that reason in three directions.

First, *rhythm*. Playing a number α as a beat (a hit wherever $\lfloor n\alpha \rfloor$ steps up) produces a Sturmian word, and we give a self-contained proof that its blocks nest exactly as the continued fraction says: the blocks at level n are q_n beats long and hold p_n hits. For $\pi - 3 = [0; 7, 15, 1, 292, \dots]$ this gives bars of 7 beats, phrases of 106 and 113, and a single short phrase that begins at beat 32,996 and closes a block of 33,102 beats. Reading the levels as bar, phrase, section and movement turns a continued fraction into musical form; Teddy Rook used it to compose the six-movement *Continued-Fraction Suite* for piano, which we describe as a method.

Second, *primes*. The same machine (the Euclidean algorithm) generates Euclidean rhythms, tunings from $\log_2(3/2)$, mode-locking staircases and the factoring method of Lehmer, Powers, Morrison and Brillhart. Coprimality ties these to $6/\pi^2$, and we derive the exact correlation of two pulse trains, $(\gcd(a, b) - 1)/\sqrt{(a-1)(b-1)}$. We then pull two open threads with real computations. (a) *Prime convergents.* Combining Lévy's theorem with the equidistribution of convergent denominators modulo m (Szüsz; Moeckel; Jager and Liardet), we derive the heuristic that a typical real number has about $2 \ln 2 \cdot \ln N \approx 1.3863 \ln N$ prime denominators among its first N convergents. A sample of 400 random reals fits the predicted growth; π has 9 prime denominators among its first 5,000 convergents against 10.9 expected, which is typical. (b) *Spacings.* We place a crystal, the rhythm of π (a one-dimensional quasicrystal with two gaps), 5,000 zeros of the Riemann zeta function, random GUE matrices, the primes and a Poisson process on one scale of rigidity, using nearest-neighbor spacings, pair correlation and number variance. The picture organizes the Montgomery–Dyson connection; it proves nothing about the Riemann hypothesis.

Third, *self-similarity*. A continued fraction contains a copy of itself after every term, which is the idea behind Ford circles, the Stern–Brocot tree, Minkowski's question-mark function, the devil's staircase, continued-fraction Cantor sets (whose dimension we recompute as 0.531280506277), the Farey bulbs of the Mandelbrot set and the Fibonacci word fractal. A chi-square test of π's first 10,000 terms against the Gauss–Kuzmin law gives $p = 0.33$ (12 degrees of freedom).

Returning to six open questions (Chapter 15), we expand π to twelve million certified terms, search its convergents for primes to $n = 20{,}000$, and prove that for a random real number the *expected* number of prime denominators among the first N convergents is $2 \ln 2 \cdot \ln N + O(1)$, with exactly the heuristic's constant; the almost-everywhere version remains open. Another chapter audits what is classical, what is known to specialists, what is a new presentation, and what is new. The mathematics is classical or known; what we did not find elsewhere is the reading of a number's continued fraction as a complete musical form, the Suite as an artwork, the prime-convergent experiment at this scale with an explicit constant, and the single spacing

spectrum. Every number in the text was computed by the scripts listed in the appendices.

Keywords: continued fractions; Sturmian words; Christoffel words; Euclidean rhythms; musical form; coprimality; prime numbers; Lévy's constant; pair correlation; random matrices; quasicrystals; fractals; Gauss–Kuzmin law.

How to read this work

Each chapter is written to stand alone and to become one segment of a narrated video; a boxed note *For the video* at the end of each chapter lists what to show. Chapters 1–3 build the main tool. Chapters 4–7 follow it into music, coprimality and the primes, mostly retelling classical results with proofs or precise references. Chapters 8 and 9 are the two research threads, with new computations. Chapter 10 is about self-similarity and fractals, Chapter 11 about the music, Chapter 12 about the wider Connections diagram. Chapter 13 says what is new and what is not; Chapter 14 discusses limits and open questions, and Chapter 15 reports further work on those questions.

Results that are classical carry a citation in their heading or the sentence that introduces them. Short standard proofs are included so that the text can be read by a strong undergraduate with calculus and a first course in number theory; longer ones are cited. A "Heuristic" is an argument that predicts a number without proving it, and we always say so. Interactive versions of most figures are on the companion pages [84, 85].

Every number in this monograph was computed or checked by the scripts in Appendix C. Numbers that come from data, from the terms of π to the last decimal of a p-value, are inserted into the text automatically from the scripts' output; small numbers that appear in the prose (such as $355/113$ or $292 \times 113 = 32{,}996$) are asserted by the same scripts. All figures were drawn for this work from computed data.

Contents

Abstract **i**

How to read this work **iii**

1 Introduction: a guess about pi **1**
1.1 The guess . . . 1
1.2 Why the guess works . . . 1
1.3 Three directions . . . 2
1.4 What is new and what is not . . . 3
1.5 Conventions . . . 3

2 Continued fractions: the machine **4**
2.1 The algorithm . . . 4
2.2 Convergents and the recurrence . . . 5
2.3 How close the convergents are . . . 5
2.4 Patterns in the terms . . . 6
2.5 The statistics of a typical number . . . 7

3 The rhythm of a number **8**
3.1 Playing a number as a beat . . . 8
3.2 Sturmian and Christoffel words . . . 8
3.3 The nested-block theorem . . . 9
3.4 Pi's levels, and the beat where the surprise lands . . . 10
3.5 From levels to musical form . . . 11
3.6 Three gaps . . . 12

4 Music from the same ladder **14**
4.1 Tuning: the fifth and its convergents . . . 14
4.2 Euclidean rhythms . . . 15
4.3 Polyrhythm and coprimality . . . 16
4.4 Mode locking and the circle map . . . 16

5 Coprimality and $6/\pi^2$ **18**
5.1 The orchard . . . 18
5.2 Coprime means uncorrelated . . . 18
5.3 Hydrogen's $1/n^2$: a shared formula, not shared physics . . . 20

6 Primes in sound and space **22**
6.1 Primes as notes, and the bias of neighbors . . . 22

6.2 A sunflower seeded with primes . . . 22
6.3 Diffraction: a quasicrystal and the primes . . . 24
6.4 Primes as a sum of waves . . . 24

7 Tools that hunt primes **27**
7.1 Factoring with the continued fraction of $\sqrt{N}$ . . . 27
7.2 Mersenne necklaces and Fermat's little theorem . . . 28
7.3 The prime hunt . . . 28

8 Thread (a): prime convergents **31**
8.1 The question . . . 31
8.2 A heuristic with an explicit constant . . . 31
8.3 The test on random reals . . . 33
8.4 Is pi typical? . . . 34
8.5 Do the site's tools help? . . . 35
8.6 What is known, and what is not . . . 36

9 Thread (b): one spectrum, from crystal to randomness **37**
9.1 The question . . . 37
9.2 The point sets . . . 37
9.3 Nearest-neighbor spacings . . . 38
9.4 Pair correlation . . . 39
9.5 Number variance: a scale of rigidity . . . 39
9.6 The spectrum, and Dyson's bridge . . . 40
9.7 What this does and does not show . . . 41

10 Self-similarity and fractals **42**
10.1 Teddy Rook's picture . . . 42
10.2 Ford circles . . . 43
10.3 The Stern–Brocot tree . . . 44
10.4 Minkowski's question-mark function . . . 44
10.5 The devil's staircase . . . 45
10.6 Cantor dust: numbers with small terms . . . 46
10.7 The Mandelbrot set's Farey bulbs . . . 47
10.8 The Fibonacci word fractal . . . 47
10.9 Six numbers, six pictures . . . 47
10.10 Is pi typical? . . . 49

11 Composition: the Continued-Fraction Suite as a method **51**
11.1 The work . . . 51
11.2 The method . . . 51
11.3 What the method gives a composer . . . 53

12 Connections: music, patterns, math and chemistry **55**
12.1 Music and math: ratios, clocks and rhythms . . . 55
12.2 Music and chemistry: octaves of elements . . . 55
12.3 Music and patterns: sand that draws sound . . . 55
12.4 Math and patterns, patterns and chemistry . . . 56
12.5 Math and chemistry: why 2, 8, 18, 32 . . . 56

12.6 All four: waves and symmetry . . . 56
12.7 Where the Rhythm of Numbers sits . . . 56

13 What is new here, and what is re-found **58**
13.1 The audit . . . 58
13.2 The two questions the owner asked . . . 62

14 Discussion **63**
14.1 What the work shows . . . 63
14.2 Limitations . . . 63
14.3 Open questions . . . 64
14.4 Future work . . . 64

15 Working on the open questions **66**
15.1 Twelve million terms of pi . . . 66
15.2 Question 1: the prime convergents of pi . . . 67
15.3 Question 2: Fibonacci and Pell primes . . . 69
15.4 Question 3: a theorem in the model . . . 70
15.5 Question 4: is pi typical? . . . 73
15.6 Question 5: the spectrum of the correlation matrix . . . 76
15.7 Question 6: ranking the equal temperaments . . . 79
15.8 Status after this work . . . 81

A Algorithms **83**

B Computational methods **86**

C Reproducibility **88**

Bibliography **90**

List of Figures

1.1 The starting guess $\pi \approx 3 + k/(7k+1)$ . . . 2

3.1 The nested blocks of π's rhythm . . . 11
3.2 The three-gap theorem for $\pi - 3$ . . . 13

4.1 The fifth in equal temperaments . . . 15
4.2 Euclidean rhythms as necklaces . . . 16
4.3 Arnold tongues of the circle map . . . 17

5.1 The orchard of visible lattice points . . . 19
5.2 The pulse-train correlation matrix . . . 20
5.3 Two roads to $\pi^2/6$ . . . 21

6.1 The last digits of consecutive primes . . . 23
6.2 Pi's sunflower, seeded with primes . . . 23
6.3 Diffraction of π's rhythm and of the primes . . . 25
6.4 The prime staircase rebuilt from zeta zeros . . . 26

7.1 The prime hunt . . . 30

8.1 Convergent denominators avoid small primes . . . 32
8.2 Prime denominators among convergents . . . 34
8.3 Where π falls among random reals . . . 35

9.1 Nearest-neighbor spacings . . . 38
9.2 Pair correlation of zeta zeros . . . 39
9.3 Number variance: one scale of rigidity . . . 40
9.4 The zeros diffract at logarithms of prime powers . . . 41

10.1 The nested tail of π . . . 43
10.2 Ford circles near $\pi - 3$ . . . 44
10.3 The Stern–Brocot tree . . . 45
10.4 Minkowski's question-mark function . . . 45
10.5 The devil's staircase of the circle map . . . 46
10.6 Cantor dust of continued fractions with terms 1 and 2 . . . 47
10.7 The Farey bulbs of the Mandelbrot set . . . 48
10.8 The Fibonacci word fractal and its relatives . . . 48
10.9 The Suite's numbers as square-cutting ladders . . . 49
10.10 Pi's terms against the Gauss–Kuzmin law . . . 50
10.11 Khinchin and Lévy for π . . . 50

11.1 The forms of the six movements 52

12.1 Chladni figures 56

15.1 Pi's prime convergents to $n = 20{,}000$ 68
15.2 A theorem checked by simulation 74
15.3 Twelve million terms against Gauss–Kuzmin 75
15.4 Khinchin, Lévy and the records of pi 76
15.5 The spectrum of the arithmetic correlation matrix 79
15.6 Prime sizes do not stand out 81

List of Tables

2.1 The first convergents of π . . . 6

3.1 The levels of the rhythm of $\pi - 3$ . . . 11

4.1 Euclidean rhythms . . . 16

7.1 Continued-fraction factoring of $2^{32} + 1$ . . . 28
7.2 The prime hunt in rhythm prefixes . . . 29

8.1 Divisibility of convergent denominators by small primes . . . 32
8.2 Prime convergents of five constants . . . 34

11.1 The six movements . . . 52

13.1 The prior-art audit . . . 58

15.1 Pi's prime convergents against the heuristic . . . 67
15.2 Fibonacci and Pell primes against the heuristic . . . 70
15.3 The exact expectation against simulation . . . 73
15.4 The largest term against Galambos's law . . . 75
15.5 The spectrum of the correlation matrix . . . 78
15.6 The largest eigenvalue for large m . . . 78
15.7 The best equal temperaments, 5 to 200 notes . . . 80
15.8 Prime against composite sizes . . . 80
15.9 Status of the six questions . . . 82

C.1 The scripts . . . 89

CHAPTER 1

Introduction: a guess about pi

1.1 The guess

This work began with a guess written on a scrap of paper. Pi is a little more than 3. The leftover, $\pi - 3 = 0.14159\ldots$, is a little less than $1/7$. So try

$$\pi \approx 3 + \frac{k}{7k+1} \tag{1.1}$$

for some whole number k. When k is large, $k/(7k+1)$ is just under $1/7$; when k is small it is further below. Somewhere in between it might hit $\pi - 3$ well. Teddy Rook tried values of k one at a time, and Fig. 1.1 shows what happens. The error falls steadily, reaches a sharp minimum at $k = 16$, and then rises again. At $k = 16$ the guess gives

$$3 + \frac{16}{113} = \frac{355}{113} = 3.14159292\ldots,$$

which differs from π by 2.67e-7. That is six correct decimal places from a fraction with a three-digit denominator.

The fraction 355/113 is famous. The Chinese mathematician Zu Chongzhi recorded it in the fifth century AD; it is called *Milü*, "the close ratio" [65, p. 281]; see also Berggren et al. [8]. Two other values of k give classical fractions too: $k = 10$ gives 223/71, the lower bound that Archimedes proved, and the limit $k \to \infty$ gives 22/7, his upper bound. So the guess (1.1) quietly contains three classical fractions for π.

1.2 Why the guess works

The guess works because of a small piece of algebra:

$$\frac{k}{7k+1} = \frac{1}{7 + \dfrac{1}{k}}. \tag{1.2}$$

The right-hand side is a *continued fraction* stopped after two steps. Solving $1/(7+1/k) = \pi - 3$ for k gives

$$k^* = \frac{\pi - 3}{1 - 7(\pi - 3)} = 15.99659,$$

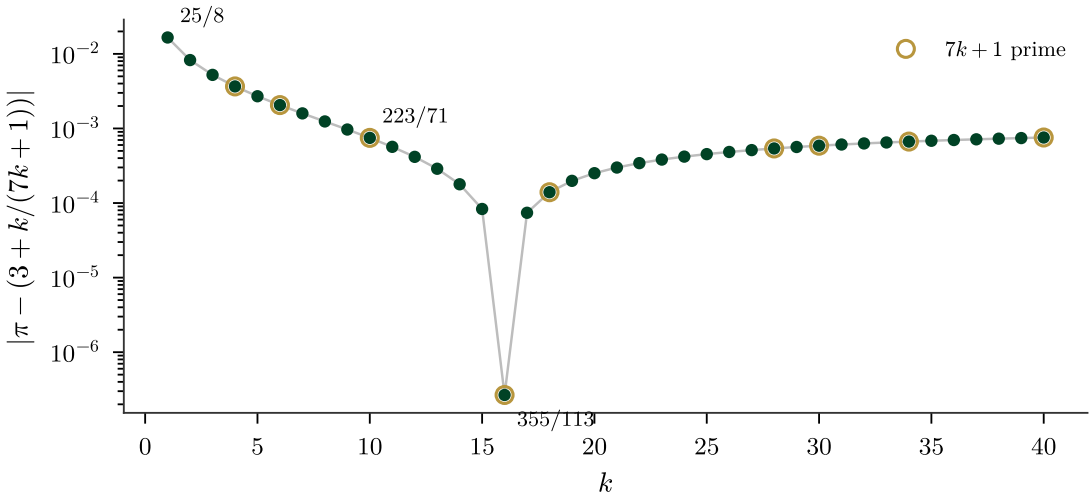


Figure 1.1. The starting guess. The error $|\pi - (3 + k/(7k + 1))|$ for $k = 1, \ldots, 40$ on a logarithmic scale. The sharp dip at $k = 16$ is 355/113; $k = 10$ gives Archimedes' lower bound 223/71 and $k = 1$ gives 25/8. Gold rings mark the k for which $7k + 1$ is prime.

not a whole number but very close to 16. Rounding k^* to the nearest integer is exactly the next step of the continued fraction of π, and the reason the rounding error is so small is that the step after that is huge. Writing out the first few steps,

$$\pi = 3 + \cfrac{1}{7 + \cfrac{1}{15 + \cfrac{1}{1 + \cfrac{1}{292 + \cdots}}}}, \tag{1.3}$$

and $k^* = 15 + 1/(1 + 1/(292 + \cdots)) \approx 16$. The terms of π's continued fraction begin

$$\pi = [3;\ 7, 15, 1, 292, 1, 1, 1, 2, 1, 3, 1, 14, 2, 1, 1, 2, 2, 2, 2, 1, 84, \ldots].$$

Chapter 2 develops this machinery from the beginning.

1.3 Three directions

The rest of this work follows the continued fraction of π, and of a few other numbers, in three directions.

Rhythm and music. Play a number α between 0 and 1 as a beat: on beat n, sound a hit if $\lfloor (n+1)\alpha \rfloor > \lfloor n\alpha \rfloor$, and stay silent otherwise. For $\alpha = \pi - 3$ the hits come every 7 beats, with an occasional gap of 8. Listen longer and the gaps of 8 come in a pattern of their own, and that pattern has a pattern, and so on. Chapter 3 proves that the levels of this rhythm are exactly the levels of the continued fraction: the terms $7, 15, 1, 292, \ldots$ are repeat counts. Read as bar, phrase, section and movement, they give a musical form; Chapter 4 connects the same machinery to tuning, Euclidean rhythms and mode locking, and Chapter 11 describes the *Continued-Fraction Suite*, six piano pieces Teddy Rook composed this way.

Primes. The algorithm that builds a continued fraction is Euclid's algorithm, and Euclid's algorithm is the oldest test for whether two numbers share a prime factor. Chapters 5–7 follow that link through coprime pairs and $6/\pi^2$, the patterns of primes in residues, spirals and diffraction, the zeros of the zeta function, factoring with continued fractions, and a hunt for primes in rhythms. Two questions from the companion site get chapters of their own. Does π's continued fraction have infinitely many prime denominators (Chapter 8)? And what is the right way to picture the famous match between zeros of the zeta function and energy levels of heavy nuclei (Chapter 9)?

Self-similarity. Teddy Rook pictured π as "a number, then a smaller number after it, then a smaller one after that, and so on forever." Chapter 10 makes this precise: the tail of a continued fraction is again a continued fraction, built by the same rule. That one fact underlies a family of classical fractals, and we draw each of them for π.

1.4 What is new and what is not

Almost all of the mathematics here is classical, and we cite it as such. The rhythm of a number is a Sturmian word [63]; the even spreading of beats is the Euclidean rhythm of Bjorklund and Toussaint [96]; Ford circles, the Stern–Brocot tree and the question-mark function date from 1858 to 1938. Chapter 13 carries out a careful audit, idea by idea. In brief, the parts we did not find elsewhere are: the systematic reading of a number's whole continued fraction as nested musical form (bar, phrase, section, movement), together with its use as a compositional method; the Suite itself, which is new as an artwork whatever the prior art on its method; the explicit constant $2\ln 2$ in the heuristic count of prime convergents and its test on hundreds of random reals; and the placement of several point sets on a single spacing spectrum. Even for these we say "we did not find prior work," not "this is new," wherever the literature is too wide to search completely.

1.5 Conventions

Throughout, $\lfloor x \rfloor$ is the floor of x and $\{x\} = x - \lfloor x \rfloor$ its fractional part. A continued fraction is written $[a_0; a_1, a_2, \dots]$. For a number α between 0 and 1 we often write $\alpha = [0; a_1, a_2, \dots]$ and work with $\pi - 3$, $\varphi - 1$, $\sqrt{2} - 1$, $e - 2$ and $\log_2(3/2)$, the five numbers of the Suite. We count beats from 1 when we state the rhythm rule, but we index positions in a beat string from 0, as the companion site does; Section 3.4 says exactly what each count means. "Prime" for numbers with more than a few hundred digits means *probable prime* by the Baillie–PSW test (Appendix B).

For the video. Open on the scrap of paper: $3 + k/(7k+1)$. Slide k from 1 to 40 while Fig. 1.1 draws itself, and stop at the dip. Show the identity (1.2) as two nested fractions. Instrument: "The starting guess" on the Rhythm of Numbers page. About 6 minutes.

CHAPTER 2

Continued fractions: the machine

This chapter collects the facts about continued fractions that the rest of the work uses. All of them are classical; good sources are Khinchin's short book [48] and Chapter X of Hardy and Wright [42]. We include the short proofs because the identities in them are the ones that turn into rhythms later.

2.1 The algorithm

Take a real number x. Write it as its whole part plus a leftover, and turn the leftover upside down:

$$x = a_0 + \frac{1}{x_1}, \qquad a_0 = \lfloor x \rfloor, \quad x_1 = \frac{1}{x - a_0} > 1. \tag{2.1}$$

Now do the same to x_1, then to x_2, and so on: $x_n = a_n + 1/x_{n+1}$ with $a_n = \lfloor x_n \rfloor$. The whole numbers $a_0, a_1, a_2, \ldots$ are the *terms* (or partial quotients) of x, and the real numbers $x_n > 1$ are its *complete quotients*. The process stops exactly when some leftover is zero, which happens exactly when x is rational. Writing the steps inside one another gives

$$x = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{\ddots + \cfrac{1}{x_n}}}} = [a_0; a_1, \ldots, a_{n-1}, x_n].$$

For π the first complete quotients are $x_1 = 7.06251$, $x_2 = 15.9966$, $x_3 = 1.00342$ and $x_4 = 292.6346$. The size of x_4 is the famous 292 at work.

When $x = a/b$ is rational, the algorithm is Euclid's algorithm in disguise: dividing a by b gives quotient a_0 and remainder r_1, then b by r_1 gives a_1, and so on. Running it on 355 and 113 gives the quotients $3, 7, 16$, so $355/113 = [3; 7, 16]$, and the last nonzero remainder is $\gcd(355, 113) = 1$.

For $0 < x < 1$ the step from x_n to x_{n+1} is one iteration of the *Gauss map*

$$T(x) = \frac{1}{x} - \left\lfloor \frac{1}{x} \right\rfloor, \qquad 0 < x < 1, \tag{2.2}$$

which reads off the term $\lfloor 1/x \rfloor$ and keeps the leftover. Most of the statistics in this work are statements about long runs of T.

2.2 Convergents and the recurrence

Stopping after the term a_n gives a fraction $p_n/q_n = [a_0; a_1, \dots, a_n]$, the nth *convergent*. Its numerator and denominator obey a two-term recurrence that is the engine of everything that follows.

Proposition 2.1 (the recurrence; see 48, Thm. 1)**.** *Set $p_{-1} = 1$, $q_{-1} = 0$, $p_0 = a_0$, $q_0 = 1$, and for $n \geq 1$*

$$p_n = a_n p_{n-1} + p_{n-2}, \qquad q_n = a_n q_{n-1} + q_{n-2}. \tag{2.3}$$

Then $[a_0; a_1, \dots, a_n] = p_n/q_n$ for every n, and more generally $[a_0; a_1, \dots, a_{n-1}, t] = (tp_{n-1} + p_{n-2})/(tq_{n-1} + q_{n-2})$ for every real $t > 0$.

Proof. In matrix form, (2.3) says

$$\begin{pmatrix} p_n & p_{n-1} \\ q_n & q_{n-1} \end{pmatrix} = \begin{pmatrix} a_0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} a_1 & 1 \\ 1 & 0 \end{pmatrix} \cdots \begin{pmatrix} a_n & 1 \\ 1 & 0 \end{pmatrix}. \tag{2.4}$$

A matrix $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ acts on numbers by $t \mapsto (at + b)/(ct + d)$, and products of matrices act by composition. The matrix $\left(\begin{smallmatrix} a & 1 \\ 1 & 0 \end{smallmatrix}\right)$ sends t to $a + 1/t$, which is one layer of a continued fraction. Composing the layers gives the general statement, and $t = a_n$ in the last layer gives the first. □

Taking determinants in (2.4), each factor has determinant -1, so

$$p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}. \tag{2.5}$$

Two consequences will matter again and again. First, any common divisor of p_n and q_n divides ± 1, so every convergent is in lowest terms. Second, the same is true of neighboring denominators: $\gcd(q_n, q_{n-1}) = 1$. In the language of Chapter 3, the length of a rhythmic block and its number of hits never share a factor, and neither do the lengths of neighboring blocks.

2.3 How close the convergents are

Proposition 2.2 (the error of a convergent; 48, Thms. 9 and 13)**.** *If x is irrational, then*

$$x - \frac{p_n}{q_n} = \frac{(-1)^n}{q_n \left(x_{n+1} q_n + q_{n-1}\right)}, \tag{2.6}$$

and therefore

$$\frac{1}{q_n(q_n + q_{n+1})} < \left| x - \frac{p_n}{q_n} \right| < \frac{1}{q_n q_{n+1}} \leq \frac{1}{q_n^2}. \tag{2.7}$$

Proof. By Proposition 2.1 with $t = x_{n+1}$, $x = (x_{n+1}p_n + p_{n-1})/(x_{n+1}q_n + q_{n-1})$. Subtract p_n/q_n, put everything over a common denominator, and use (2.5). For the bounds, $a_{n+1} < x_{n+1} < a_{n+1} + 1$ gives $q_{n+1} < x_{n+1}q_n + q_{n-1} < q_{n+1} + q_n$. □

The formula (2.6) explains the 292. The error of $p_3/q_3 = 355/113$ is about $1/(113 \cdot x_4 \cdot 113)$, and $x_4 = 292.6346$ is large, so the error is tiny. A big term is a sign that the convergent just before it is unusually good. Table 2.1 lists the first convergents of π with their errors and the normalized error $q_n^2 |\pi - p_n/q_n|$, which by (2.6) is roughly $1/a_{n+1}$.

Convergents are not only close; they are the best possible for their size.

Table 2.1. The first convergents of π. The last column is $q_n^2\,|\pi - p_n/q_n|$, which is close to $1/a_{n+1}$; it is smallest just before the large term 292.

n	a_n	p_n	q_n	$\lvert\pi - p_n/q_n\rvert$	$q_n^2\,\lvert\pi - p_n/q_n\rvert$
0	3	3	1	0.142	0.1416
1	7	22	7	0.001,26	0.06196
2	15	333	106	8.32×10^{-5}	0.9351
3	1	355	113	2.67×10^{-7}	0.003406
4	292	103,993	33,102	5.78×10^{-10}	0.6332
5	1	104,348	33,215	3.32×10^{-10}	0.3659
6	1	208,341	66,317	1.22×10^{-10}	0.5381
7	1	312,689	99,532	2.91×10^{-11}	0.2887
8	2	833,719	265,381	8.72×10^{-12}	0.6138

Theorem 2.3 (best approximation; Lagrange, see 48, Thms. 16–17)**.** *If $1 \le q < q_{n+1}$, then $|qx - p| \ge |q_n x - p_n|$ for every integer p, with equality only for $p/q = p_n/q_n$. In particular no fraction with denominator smaller than q_{n+1} is closer to x than p_n/q_n.*

For π this means that no fraction with denominator below 33,102 beats 355/113; our script checks it directly by scanning every denominator. A partial converse is due to Legendre.

Theorem 2.4 (Legendre; 42, Thm. 184)**.** *If p/q is in lowest terms and $|x - p/q| < 1/(2q^2)$, then p/q is a convergent of x.*

We confirmed it for π by brute force: among all fractions with denominator up to 3,000, the only ones within $1/(2q^2)$ of π are convergents. Legendre's theorem reappears in Chapter 10 as a statement about circles.

2.4 Patterns in the terms

Some numbers have continued fractions with simple patterns:

$$\varphi = \tfrac{1+\sqrt{5}}{2} = [1;1,1,1,\dots], \qquad \sqrt{2} = [1;2,2,2,\dots],$$
$$e = [2;1,2,1,1,4,1,1,6,1,\dots], \quad \log_2 \tfrac{3}{2} = [0;1,1,2,2,3,1,5,2,23,2,2,1,1,55,\dots].$$

For φ the recurrence (2.3) with all terms equal to 1 is the Fibonacci recurrence, and the denominators are $1,1,2,3,5,8,13,21,34,55,89,144,\dots$. For $\sqrt{2}$ they are the Pell numbers $1,2,5,12,29,70,169,408,985,\dots$. The pattern of e, with $2,4,6,\dots$ in every third place, is due to Euler; our scripts verify it for the first 3,900 terms. The number $\log_2(3/2)$, which measures a musical fifth (Chapter 4), shows no pattern, and neither does π.

Theorem 2.5 (Euler and Lagrange; 42, Thms. 176–177)**.** *The continued fraction of x is eventually periodic if and only if x is a quadratic irrational, that is, an irrational root of a quadratic equation with integer coefficients.*

So φ and $\sqrt{2}$ have repeating terms, while π and e, which are not quadratic irrationals, do not. This dichotomy returns in every chapter: repeating terms give exactly self-similar rhythms, music and pictures; non-repeating terms give the same *rule* at every level but a new *shape*.

2.5 The statistics of a typical number

Pick x at random between 0 and 1. What do its terms look like? Gauss conjectured, and Kuzmin proved in 1928 [53], with a sharper version by Lévy [59], that for almost every x the terms follow one fixed law.

Theorem 2.6 (Gauss–Kuzmin law)**.** *For almost every $x \in (0,1)$, the proportion of indices $n \le N$ with $a_n = k$ tends, as $N \to \infty$, to*

$$P(k) = -\log_2\!\left(1 - \frac{1}{(k+1)^2}\right). \tag{2.8}$$

So about 41.5% of the terms are 1, 17.0% are 2 and 9.3% are 3. Behind the law is the *Gauss measure* $d\mu = dx/((1+x)\ln 2)$, which the Gauss map (2.2) leaves invariant, and the fact that T is ergodic for it; the frequency of $a_n = k$ is the Gauss measure of the interval $(1/(k+1), 1/k]$, which is (2.8). Two more laws follow from the same ergodic theory.

Theorem 2.7 (Khinchin 1935; Lévy 1936)**.** *For almost every x,*

$$(a_1 a_2 \cdots a_n)^{1/n} \to K_0 = 2.685452001\ldots \qquad \textit{and} \qquad \frac{\ln q_n}{n} \to \frac{\pi^2}{12\ln 2} = 1.18657\ldots \tag{2.9}$$

The first limit is Khinchin's constant [49]; the second is Lévy's constant [60], and it says that the denominators grow like $e^{1.18657n} \approx 3.27582^n$. Lévy's theorem is the key input to the prime-convergent heuristic of Chapter 8. No one knows whether π obeys any of these laws; nobody has even proved that its terms are unbounded. What we can do is count. From π computed to 11,600 digits, our scripts extract its first 10,000 terms (two precisions agree on every one), and find 4,206 ones (42.1%), a geometric mean of 2.6632 and $\ln q_{10000}/10000 = 1.17941$. Chapter 10 turns these counts into a formal test.

For the video. Animate the algorithm on π: peel off 3, flip 0.14159 to 7.0625, peel off 7, flip again. Show Table 2.1 filling in, with the error column shrinking and dipping at 355/113. End with the slogan "a big term means the fraction before it was unusually good." Instrument: "The ladder." About 8 minutes.

CHAPTER 3

The rhythm of a number

3.1 Playing a number as a beat

Let $0 < \alpha < 1$. Imagine a drummer who keeps a running total, adding α on every beat, and strikes the drum whenever the total passes a whole number. On beat n the total goes from $n\alpha$ to $(n+1)\alpha$, so the rule is:

$$\text{beat } n \text{ is a hit} \iff \lfloor (n+1)\alpha \rfloor > \lfloor n\alpha \rfloor , \qquad n = 1, 2, 3, \ldots \tag{3.1}$$

We write a hit as `x` and a rest as `.`, or as 1 and 0. The resulting infinite word $c_\alpha = c_\alpha(1)\, c_\alpha(2)\, c_\alpha(3) \cdots$ is called the *characteristic word* of α. Since the total grows by less than 1 per beat, there is at most one hit per beat, and after L beats there have been exactly $\lfloor (L+1)\alpha \rfloor$ hits. The hits are spread as evenly as the whole numbers allow: the density of hits is α, and any two stretches of the same length differ by at most one hit. For $\alpha = \pi - 3$ the first 40 beats are

```
......x......x......x......x......x.....
```

a hit every 7 beats, because $1/\alpha = 7.06251$ lies between 7 and 8. The leftover $0.0625 \approx 1/16$ means that roughly every 16th gap is an 8 instead of a 7. That is already the start of a hierarchy, and this chapter shows that it continues forever, level by level, exactly as the continued fraction says.

3.2 Sturmian and Christoffel words

Words like c_α have a long history in combinatorics, where they are called *Sturmian words*; the standard reference is Chapter 2 of Lothaire's book, written by Berstel and Séébold [63], and the books of Allouche and Shallit [2, Ch. 9] and Berstel, Lauve, Reutenauer and Saliola [10].

Definition 3.1. A binary word is *balanced* if any two factors (blocks of consecutive letters) of the same length contain numbers of 1s that differ by at most 1. An infinite word is *Sturmian* if it is balanced and not eventually periodic. A *mechanical word* of slope α and intercept ρ has nth letter $\lfloor (n+1)\alpha + \rho \rfloor - \lfloor n\alpha + \rho \rfloor$; the characteristic word is the case $\rho = 0$. For a rational slope p/q in lowest terms, one period of the mechanical word with $\rho = 0$, read as $\lfloor (i+1)p/q \rfloor - \lfloor ip/q \rfloor$ for $i = 0, \ldots, q-1$, is the *lower Christoffel word* of slope p/q.

For irrational α, the mechanical words of slope α are exactly the Sturmian words [63, Ch. 2]. A Christoffel word is the finite, rational version: it spreads p hits over q beats as evenly as possible. In music these are the maximally even rhythms of Clough and Douthett [22], and, up to rotation, the Euclidean rhythms of Chapter 4.

3.3 The nested-block theorem

The construction that the companion site uses to play a number builds the rhythm from its continued fraction rather than from the floor rule (3.1). It is the classical construction of *standard words*.

Definition 3.2 (standard words)**.** Let $\alpha = [0; a_1, a_2, \dots]$. Set $s_{-1} = \texttt{x}$, $s_0 = \texttt{.}$, and

$$s_1 = s_0^{a_1 - 1}\, s_{-1}, \qquad s_n = s_{n-1}^{a_n}\, s_{n-2} \quad (n \geq 2), \tag{3.2}$$

where w^a means a copies of w in a row.

In words: the level-1 block is $a_1 - 1$ rests and a hit; each later block is *(term) copies of the block just built, followed by the block before it.* For $\pi - 3$, s_1 is the 7-beat block `.....x`; s_2 is fifteen of those and one extra rest, 106 beats; $s_3 = s_2 s_1$ is 113 beats; $s_4 = s_3^{292} s_2$ is 33,102 beats. In pseudocode (Algorithm 3 in Appendix A):

```
this, prev = "."*(a1-1) + "x", "."; for a in a2, a3, ...: this, prev = this*a
                                    + prev, this
```

The theorem below says that this construction and the floor rule (3.1) produce the same rhythm, and that the block lengths are the convergent denominators.

Lemma 3.3 (floors near a convergent)**.** *Let $n \geq 1$ and $1 \leq m < q_n + q_{n+1}$. Then $\lfloor m\alpha \rfloor = \lfloor mp_n/q_n \rfloor - \varepsilon_n(m)$, where $\varepsilon_n(m) = 1$ if q_n divides m and n is odd, and $\varepsilon_n(m) = 0$ otherwise.*

Proof. By (2.5), p_n/q_n and p_{n+1}/q_{n+1} satisfy $|p_n q_{n+1} - p_{n+1} q_n| = 1$, and α lies strictly between them (Proposition 2.2). A fraction a/b strictly between two such fractions has $b \geq q_n + q_{n+1}$: indeed $|a/b - p_n/q_n| \geq 1/(bq_n)$ and $|p_{n+1}/q_{n+1} - a/b| \geq 1/(bq_{n+1})$, and the two distances add up to $1/(q_n q_{n+1})$, which forces $b \geq q_n + q_{n+1}$. Now take $m < q_n + q_{n+1}$. If an integer k lay strictly between $m\alpha$ and mp_n/q_n, then k/m would lie strictly between α and p_n/q_n, which is impossible. So if mp_n/q_n is not an integer, both numbers have the same floor. If it is an integer (that is, $q_n \mid m$), the floors agree when $\alpha > p_n/q_n$ and differ by one when $\alpha < p_n/q_n$; by (2.6) the convergent lies above α exactly when n is odd. □

Lemma 3.4 (near-periodicity)**.** *For $n \geq 1$ and $1 \leq j \leq q_{n+1} - 2$, $c_\alpha(j + q_n) = c_\alpha(j)$.*

Proof. Let $f(m) = \lfloor mp_n/q_n \rfloor - \varepsilon_n(m)$. Then $f(m + q_n) = f(m) + p_n$, because ε_n has period q_n. By Lemma 3.3, $c_\alpha(j) = f(j+1) - f(j)$ whenever $j+1 < q_n + q_{n+1}$. For $j \leq q_{n+1} - 2$ both $j+1$ and $j + q_n + 1$ are below $q_n + q_{n+1}$, so $c_\alpha(j + q_n) = f(j + q_n + 1) - f(j + q_n) = f(j+1) - f(j) = c_\alpha(j)$. □

Theorem 3.5 (nested blocks; Fraenkel, Mushkin and Tassa; Brown)**.** *Let $\alpha = [0; a_1, a_2, \dots]$ be irrational, with convergents p_n/q_n and standard words s_n. Then for every $n \geq 0$:*

(i) s_n is q_n beats long and contains p_n hits;

(ii) *the length and the number of hits of s_n are coprime, and so are the lengths of s_n and s_{n-1};*

(iii) *for $n \geq 1$, s_n is the beginning of the characteristic word c_α. Hence the first q_{n+1} beats of the rhythm consist of a_{n+1} copies of the level-n block followed by one level-$(n-1)$ block, and $c_\alpha = \lim_{n\to\infty} s_n$.*

Part (iii) is due to Fraenkel, Mushkin and Tassa [34] and Brown [15]; a proof through Sturmian morphisms is in Lothaire [63, Ch. 2]. We give a direct proof from the two lemmas, since the same counting reappears in every musical application.

Proof. (i) The lengths satisfy $|s_{-1}| = |s_0| = 1$, $|s_1| = a_1 = q_1$, and $|s_n| = a_n|s_{n-1}| + |s_{n-2}|$, which is the recurrence (2.3) for q_n with $q_0 = 1$. The hit counts satisfy $0, 1, a_n \cdot (\text{previous}) + (\text{one before})$, starting from s_0 with $0 = p_0$ hits and s_1 with $1 = p_1$ hit, which is the recurrence for p_n.

(ii) follows from (i) and the determinant identity (2.5).

(iii) For $n = 1$: α lies between $1/(a_1+1)$ and $1/a_1$, so $(j+1)\alpha < 1$ for $j < a_1$ and $(a_1+1)\alpha > 1$; the first a_1 beats are $a_1 - 1$ rests and a hit, which is s_1. For $n = 2$ when $a_1 = 1$ (so $q_1 = 1$ and the lemma below does not apply): then $1-\alpha$ lies between $1/(a_2+2)$ and $1/(a_2+1)$, hence $\lfloor j\alpha \rfloor = j - 1$ for $1 \leq j \leq a_2 + 1$ and $\lfloor (a_2+2)\alpha \rfloor = a_2$; the first $a_2 + 1$ beats are a_2 hits and a rest, which is $s_2 = s_1^{a_2} s_0$.

For the inductive step, suppose $n \geq 1$, $q_n \geq 2$, and s_n and s_{n-1} are beginnings of c_α (for $n = 1$ we use $s_0 = \,.\, = c_\alpha(1)$, which holds when $a_1 \geq 2$, the only case with $q_1 \geq 2$). Since $q_n \geq 2$, every position $j \leq q_{n+1} - q_n$ satisfies $j \leq q_{n+1} - 2$, so Lemma 3.4 lets us shift by q_n repeatedly: the first $q_{n+1} = a_{n+1}q_n + q_{n-1}$ beats are a_{n+1} copies of the first q_n beats followed by the first q_{n-1} beats. By hypothesis these are s_n and s_{n-1}, so the first q_{n+1} beats are $s_n^{a_{n+1}} s_{n-1} = s_{n+1}$. Since $q_n \geq 2$ for all $n \geq 2$, the induction runs forever. □

We also confirmed the theorem by computer: for 200 random term sequences (terms drawn from $\{1, 2, 3, 4, 5, 7, 12\}$, words up to several thousand beats) and for $\pi - 3$ over the first 140,000 beats, the standard words agree letter for letter with the floor rule (3.1), and every block has q_n beats and p_n hits.

Remark 3.6. The theorem is the precise form of the statement that "every term of the continued fraction is a repeat count." The term a_{n+1} says how many times the level-n block repeats before the level-$(n-1)$ block closes the level-$(n+1)$ block. Big terms mean long runs of identical blocks; a term equal to 1 means the level adds a single block. For the golden ratio every term is 1, so every level is "one long block, then one short block," and the levels are the Fibonacci numbers.

3.4 Pi's levels, and the beat where the surprise lands

Table 3.1 lists the first levels of the rhythm of $\pi - 3$, and Fig. 3.1 draws them.

The two artifacts on which the companion site is based disagreed about where the first short phrase of level 4 "lands": one said beat 32,996, the other 33,102. Both are right about different things. Number the beats from 0, as the site does. Level 4 is $s_4 = s_3^{292} s_2$: 292 phrases of 113 beats followed by one phrase of 106. The 292 long phrases fill beats 0 through $292 \times 113 - 1 = 32{,}995$, so the short phrase *begins* at beat 32,996; it fills beats 32,996 to 33,101, so the level-4 block *ends* after beat 33,101, and the next block begins at beat $33{,}102 = q_4$. Our

Table 3.1. The levels of the rhythm of $\pi - 3 = [0; 7, 15, 1, 292, 1, 1, \dots]$. Each block is (term) copies of the block above it in the table, then the block two rows up. The fraction is hits over beats, the nth convergent of $\pi - 3$; adding 3 gives the convergent of π.

level n	term a_n	block s_n	beats q_n	hits p_n	convergent of π
1	7	$.^{6}\mathtt{x}$	7	1	22/7
2	15	$s_1^{15}\, s_0$	106	15	333/106
3	1	$s_2\, s_1$	113	16	355/113
4	292	$s_3^{292}\, s_2$	33,102	4,687	103993/33102
5	1	$s_4\, s_3$	33,215	4,703	104348/33215

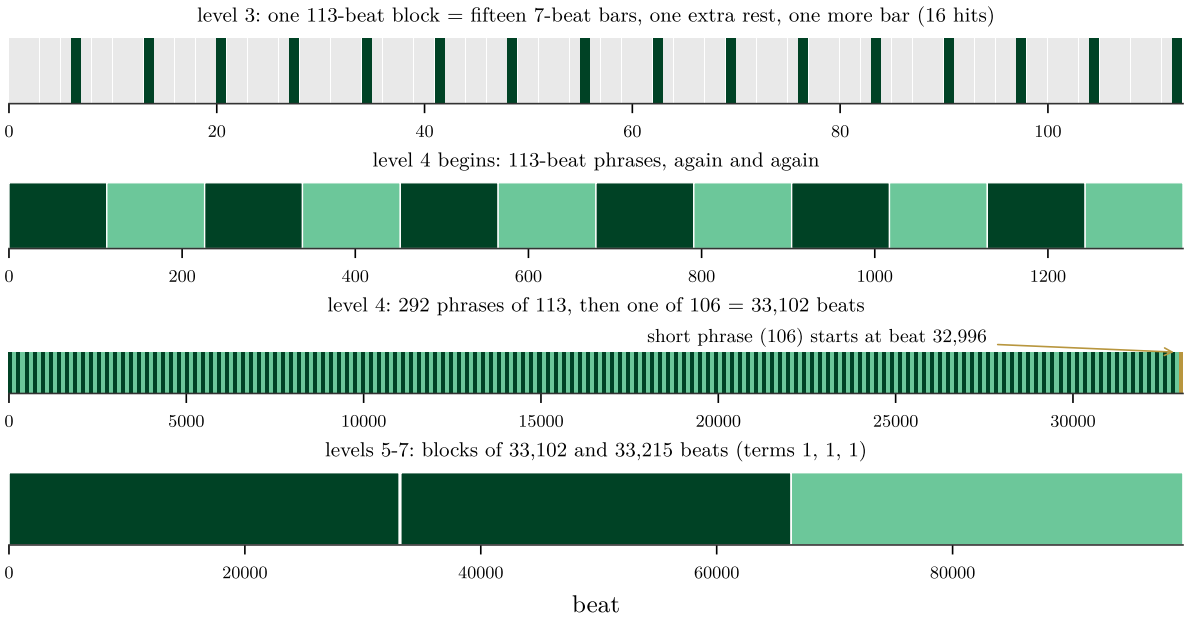


Figure 3.1. The nested blocks of π's rhythm, drawn to scale at four zoom levels. Top: one 113-beat block (hits in green). Second row: the start of level 4, a run of 113-beat phrases. Third row: all of level 4, 292 phrases of 113 beats and then the single 106-beat phrase (gold), which begins at beat 32,996 and closes the block at 33,102. Bottom: levels 5 to 7, where each term is 1.

script locates every short phrase in the first 140,000 beats: they begin at beats 32,996, 66,211, 99,313, 132,528. In the first 80,000 beats there are 706 complete phrases of 113 beats and 2 of 106.

When each beat is one bar of music, as in the fifth and sixth movements of the Suite (Chapter 11), bars are numbered from 1, and the same moment is the bar line after bar 32,996: the first bar of the short cycle is printed as bar 32,997, and the short cycle ends with bar 33,102. At a tempo of one bar every 2.5 seconds, reaching it takes 22.9 hours of continuous playing.

3.5 From levels to musical form

Theorem 3.5 turns any continued fraction into a hierarchy of blocks, and a hierarchy of blocks is what musicians call *form*: notes group into bars, bars into phrases, phrases into sections,

sections into movements. The reading proposed on the companion site, and used by Teddy Rook for the Suite, is simply to name the levels: the shortest block is a bar, the next a phrase, the next a section, and so on. Each term then says how many of one unit make the next, and the second, shorter block at each level is a contrasting unit, a "B" section that the number brings in on its own schedule.

Two small choices make the reading musical. First, a level that adds a single unit to the one below (a term equal to 1 at a low level, or the "extra rest" in $s_2 = s_1^{15} s_0$) can be absorbed: in the Composer on the site, π's bars are 7 beats, its phrases 113, and its sections 33,102. Second, the rhythm itself can be kept inside each bar, so that the accents of a piece follow the number at the smallest scale while its form follows the number at the largest. Chapter 11 shows both choices at work.

3.6 Three gaps

The rhythm of α has only two kinds of gaps between hits, $\lfloor 1/\alpha \rfloor$ and $\lfloor 1/\alpha \rfloor + 1$ (7 and 8 for $\pi - 3$), because it is balanced. A related classical fact concerns the points $\{\alpha\}, \{2\alpha\}, \ldots, \{N\alpha\}$ placed on a circle of circumference 1.

Theorem 3.7 (three-gap theorem; Sós, Świerczkowski)**.** *For any irrational α and any N, the points $0, \{\alpha\}, \ldots, \{N\alpha\}$ cut the circle into arcs of at most three different lengths, and when there are three, the largest is the sum of the other two.*

The statement was conjectured by Steinhaus. Proofs and the history are in Sós [90], Świerczkowski [92] and the survey of Alessandri and Berthé [1], which also explains the link with Sturmian words. The gap lengths are built from the distances $\|q_k \alpha\|$ of the multiples $q_k \alpha$ to the nearest integer, so the continued fraction controls them too. Figure 3.2 shows the distinct gap lengths for $\pi - 3$ and $N \le 120$; our script checks the theorem for every $N \le 1{,}200$. At $N = 100$ the three lengths are $0.0089, 0.0177, 0.0265$. In Chapter 9 this rigidity, two gaps in the rhythm and at most three on the circle, puts π's rhythm at the "quasicrystal" end of a scale whose other end is pure randomness.

For the video. Play the pi rhythm slowly and count the 7s. Then speed up and show Fig. 3.1 zooming out level by level, ending on the single gold phrase after 292 long ones. Explain "beat 32,996 vs. 33,102" with a ruler: one is where the short phrase starts, the other where the block ends. Instruments: "The levels" and the Composer's "Jump to the rarest block." About 10 minutes.

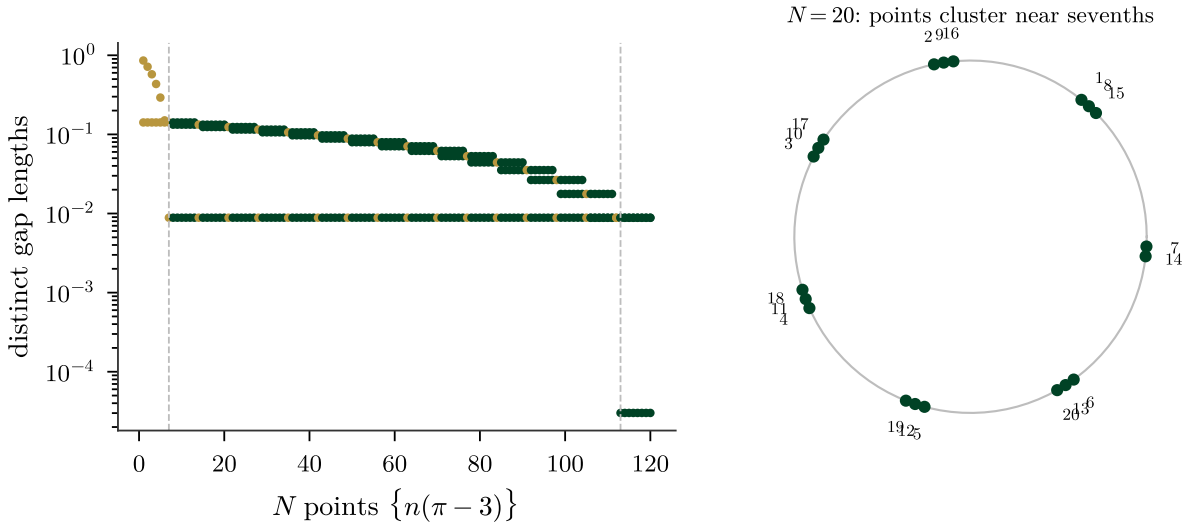


Figure 3.2. The three-gap theorem for $\alpha = \pi - 3$. Left: the distinct arc lengths among the points $0, \{\alpha\}, \ldots, \{N\alpha\}$ for $N = 1, \ldots, 120$ (green: three lengths; gold: two; dashed lines at the denominators $N = 7$ and $N = 113$). The count is two whenever the number of points, $N + 1$, is a denominator such as 7, 106 or 113. Right: the first 20 points, which cluster near multiples of $1/7$ because 7α is close to 1.

CHAPTER 4

Music from the same ladder

The continued fraction is not only a rhythm generator. The same ladder of fractions decides how musical scales are tuned, the same algorithm spreads drum hits around the world's traditional rhythms, and the same fractions are where coupled oscillators lock together. This chapter retells these classical connections in the language of Chapters 2 and 3.

4.1 Tuning: the fifth and its convergents

Two notes sound smooth together when their frequencies are in a simple ratio. The octave is $2:1$ and the fifth is $3:2$. Measured in octaves, a fifth is

$$\log_2 \tfrac{3}{2} = 0.584963\ldots = [0; 1, 1, 2, 2, 3, 1, 5, 2, 23, 2, 2, 1, 1, 55, \ldots].$$

A scale that divides the octave into n equal steps (an *n-note equal temperament*, or n-EDO) contains a good fifth when some k/n is close to $\log_2(3/2)$. The convergents of $\log_2(3/2)$ are

$$1/2, 3/5, 7/12, 24/41, 31/53, 179/306, \ldots$$

and they are the classical answers: $7/12$ is "a fifth is 7 of 12 half steps," and 41 and 53 are the next good stops, long known to theorists [7]. In cents (hundredths of an equal-tempered half step, so 1200 cents per octave), the 12-note fifth is 1.955 cents flat; the 41-note fifth misses by 0.484 cents and the 53-note fifth by 0.068.

Figure 4.1 shows the error of the best fifth in every n-note scale up to $n = 72$. The scales that set a new record for the fifth are $n = 1, 2, 3, 5, 7, 12, 29, 41, 53$. All are convergent denominators except 29, whose fifth $17/29$ is an *intermediate fraction* between the convergents $7/12$ and $24/41$. That is exactly what the theory of best approximations predicts: the record-setters for $|nx - k|$ are the convergents, and the record-setters for $|x - k/n|$ also include some intermediate fractions [48, Ch. II].

The Pythagorean comma. Twelve pure fifths overshoot seven octaves by a ratio, $3^{12}/2^{19} = 531441/524288$, called the Pythagorean comma, which is 23.46 cents. Equal temperament shares the comma out equally: each of its twelve fifths is $23.46/12 = 1.955$ cents flat, which our script confirms to nine decimals. The comma exists because $\log_2(3/2)$ is irrational: no number of pure fifths can ever equal a whole number of octaves.

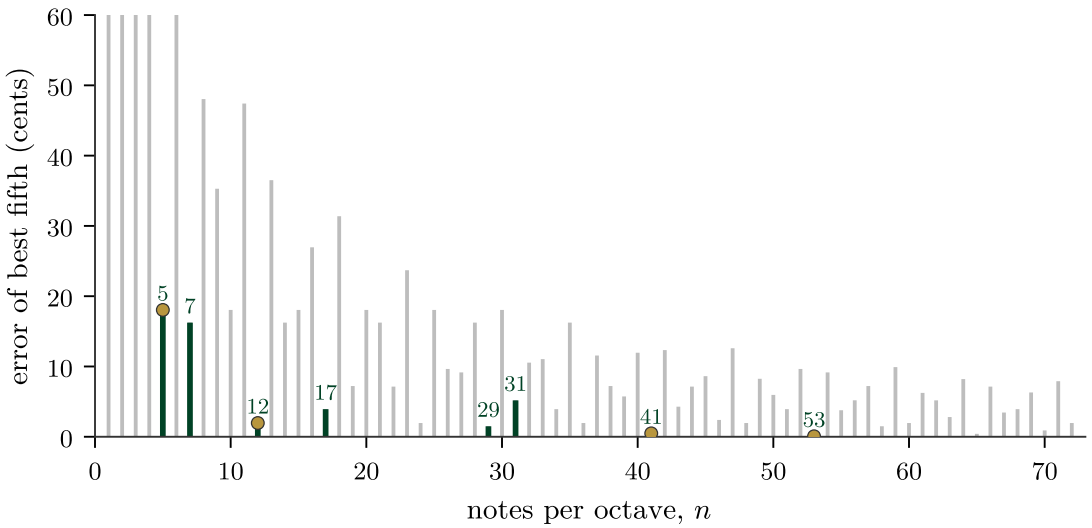


Figure 4.1. The error of the best fifth in every n-note equal temperament, $n = 1, \ldots, 72$, in cents. Green bars and labels mark well-known scales; gold dots mark the convergent denominators of $\log_2(3/2)$: 2, 5, 12, 41, 53.

Thirds, and a prime number of notes. The same comparison for the major third $5 : 4$ gives a 12-note third 13.69 cents sharp, while the 31-note scale, which Christiaan Huygens advocated [7], has a third within 0.78 cents of pure. Stepping around an n-note circle by s notes visits every note exactly when $\gcd(s, n) = 1$, so the number of step sizes that visit all notes is Euler's $\varphi(n)$. The circle of fifths visits all 12 notes because $\gcd(7, 12) = 1$. In a scale with a prime number of notes, such as 7, 31 or 53, every step size visits every note.

4.2 Euclidean rhythms

Suppose we want k drum hits in a cycle of n steps, spread as evenly as possible. In 2003 Eric Bjorklund described an algorithm for exactly this problem, for spacing pulses in the timing system of the Spallation Neutron Source accelerator; in 2005 Godfried Toussaint observed that its outputs include many traditional rhythms from around the world and named them *Euclidean rhythms*, because the algorithm has the structure of Euclid's algorithm [96]. Bjorklund's method starts with k groups `[x]` and $n - k$ groups `[.]`, and repeatedly appends one leftover group to each of the others until at most one leftover group remains. The group counts follow Euclid's algorithm on (n, k).

Proposition 4.1. *For $1 \le k < n$, the Euclidean rhythm $E(k, n)$ is a rotation of the lower Christoffel word of slope k/n, the period of the rhythm of k/n in the sense of* (3.1).

This is known; Gómez-Martín, Taslakian and Toussaint relate Euclidean rhythms to the maximally even sets of Clough and Douthett and to other even-spacing constructions [22, 37]. Our script confirms it for all 780 pairs with $n \le 40$. Two consequences connect the drum machine to π. Running Euclid's algorithm on 355 and 113 produces the quotients $3, 7, 16$, the beginning of π's continued fraction. And $E(16, 113)$ is a rotation of the level-3 block of π's rhythm (Table 3.1): the drum machine with 16 hits in 113 steps plays π at level 3. Figure 4.2 draws some of these rhythms as necklaces.

Table 4.1. Euclidean rhythms $E(k, n)$ from Bjorklund's algorithm, as computed by our script, with the gaps between hits. The names follow Toussaint [96]. The son clave (gaps $3, 3, 4, 2, 4$) is not a rotation of $E(5, 16)$, so it is not Euclidean.

name	rhythm	pattern	gaps
tresillo	$E(3, 8)$	`x..x..x.`	3, 3, 2
cinquillo	$E(5, 8)$	`x.xx.xx.`	2, 1, 2, 1, 2
West African bell	$E(7, 12)$	`x.xx.x.xx.x.`	2, 1, 2, 2, 1, 2, 2
ruchenitza	$E(3, 7)$	`x.x.x..`	2, 2, 3
khafif-e-ramal	$E(2, 5)$	`x.x..`	2, 3
Venda clapping	$E(5, 12)$	`x..x.x..x.x.`	3, 2, 3, 2, 2
samba	$E(7, 16)$	`x..x.x.x..x.x.x.`	3, 2, 2, 3, 2, 2, 2

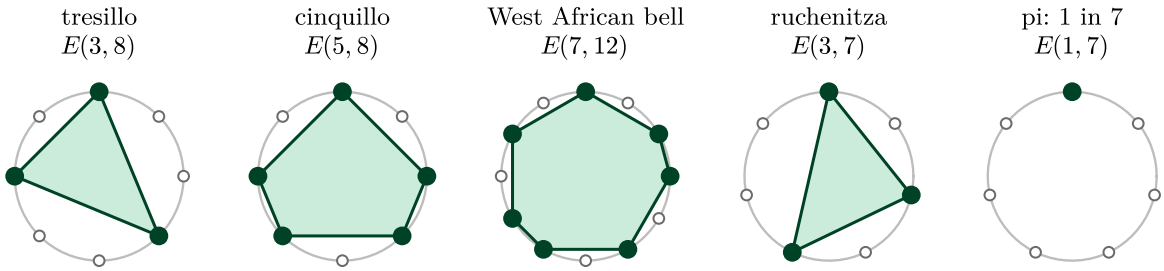


Figure 4.2. Euclidean rhythms drawn as necklaces: steps run clockwise from the top, hits are filled. The hits of each rhythm form a polygon that is as close to regular as the steps allow. The last necklace is the first level of π's rhythm, one hit in seven.

4.3 Polyrhythm and coprimality

Two pulse trains, one with a hit every a steps and one every b steps, coincide every $\operatorname{lcm}(a, b)$ steps. When a and b are coprime, they meet only once per cycle of ab steps: three against four meet only on the downbeat of a 12-step cycle. When they share a factor, they meet more often; four against six meet every 12 steps, twice per cycle of 24. The Polyrhythm layerer on the site stacks the rhythms of several convergents as concentric rings; for π, the rings $1/7$, $15/106$ and $16/113$ play three versions of one beat that drift apart and realign. Because $\gcd(7, 113) = 1$, the 7-ring and 113-ring meet only at the start of their common cycle. Chapter 5 turns "meeting less often" into an exact correlation.

4.4 Mode locking and the circle map

In 1665 Christiaan Huygens noticed that two pendulum clocks hanging from the same beam fell into step; a modern experimental study is Bennett, Schatz, Rockwood and Wiesenfeld [6]. Coupled oscillators tend to lock at simple frequency ratios, and the simplest model of this is the *sine circle map*

$$\theta_{t+1} = \theta_t + \Omega - \frac{K}{2\pi} \sin(2\pi\theta_t), \tag{4.1}$$

where Ω is the natural rotation per step and $K \geq 0$ the strength of the coupling. The *rotation number* $W = \lim(\theta_t - \theta_0)/t$ is the average rotation actually achieved. For $K = 0$, $W = \Omega$. For

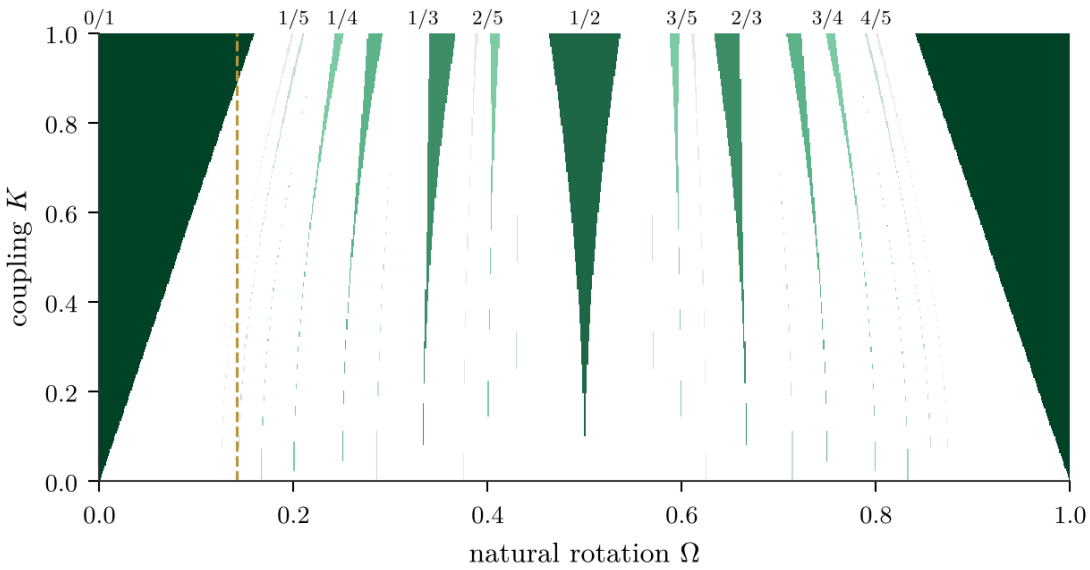


Figure 4.3. Arnold tongues of the sine circle map (4.1). Shaded points (Ω, K) have rotation number p/q with $q \le 8$ (darker: smaller q), computed from 400 iterations after a transient. The dashed gold line is $\Omega = \pi - 3$, which enters the 0/1 tongue at $K = 2\pi(\pi - 3) \approx 0.8897$.

$0 < K \le 1$ the map is a homeomorphism of the circle and W exists; it is locked to a fraction p/q on an interval of Ω around p/q, and these intervals widen with K into the *Arnold tongues* of Fig. 4.3. At $K = 1$ the locked intervals fill the whole line except a Cantor set of measure zero, a "complete devil's staircase" [47], which Chapter 10 explores.

The 0/1 tongue can be computed exactly. A fixed point of (4.1) (modulo 1) needs $\Omega = (K/2\pi)\sin(2\pi\theta)$, which has a solution if and only if $|\Omega| \le K/(2\pi)$. So at $K = 1$ the 0/1 step extends $1/(2\pi) = 0.1592$ on each side of 0; our numerical staircase gives 0.1592. The number $\pi - 3 = 0.14159\ldots$ is inside it, so a system whose natural rotation is $\pi - 3$ is captured into the 0/1 lock once $K \ge 2\pi(\pi - 3) = 0.8897$. Numerically, at $K = 1$ the 1/2 step is 0.074 wide and the 1/3 step 0.031, and the steps with denominators up to 10 already cover a fraction 0.598 of the interval $[0, 1]$.

Counting the tongues. Each tongue belongs to a fraction p/q in lowest terms, so the number of tongues with $0 \le p/q \le 1$ and $q \le N$ is $1 + \sum_{q \le N} \varphi(q)$, the length of the Farey sequence of order N. A classical theorem [42, Thm. 330] gives $\sum_{q \le N} \varphi(q) \sim 3N^2/\pi^2$. For $N = 50$, 100 and 200 our counts are 775, 3,045 and 12,233, against $3N^2/\pi^2 \approx 760$, 3040 and 12159. Here π appears through $6/\pi^2$, the subject of the next chapter.

For the video. Play a pure fifth against a 12-tone fifth and let the beating be heard. Show Fig. 4.1 with the convergent denominators lighting up. Then the drum machine: tresillo, cinquillo, and "pi: 16 in 113." Finish on the circle map with the tongues growing as K rises and the pi line being captured. Instruments: "Circle of fifths," "Euclid & Bjorklund," "Mode locking," the Polyrhythm layerer and the drum machine. About 10 minutes.

CHAPTER 5

Coprimality and $6/\pi^2$

Every level of a rhythm is a coprime pair: by Theorem 3.5(ii) a block of q_n beats holds p_n hits with $\gcd(p_n, q_n) = 1$. This chapter is about coprimality itself: how common it is, how it appears as statistical independence, and where π enters through $6/\pi^2$.

5.1 The orchard

Stand at the origin of the integer lattice and look out over an orchard with a tree at every point (a, b) with $a, b \geq 1$. The tree at (a, b) is visible exactly when no other tree stands on the segment from the origin to it, that is, when $\gcd(a, b) = 1$. Figure 5.1 shows the visible trees in a 40×40 corner, 0.6119 of the grid. For a 1000×1000 corner the share is 0.608383.

Theorem 5.1 (42, Thm. 332)**.** *The proportion of pairs* (a, b) *with* $1 \leq a, b \leq N$ *and* $\gcd(a, b) = 1$ *tends to* $6/\pi^2 = 0.607927\ldots$ *as* $N \to \infty$.

The heuristic behind it is short. A prime p divides both a and b with probability $1/p^2$, and divisibility by different primes behaves independently, so the probability that no prime divides both is

$$\prod_{p} \left(1 - \frac{1}{p^2}\right) = \frac{1}{\zeta(2)}, \qquad \zeta(2) = \sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}. \tag{5.1}$$

The first equality is Euler's product formula [31]: expanding each factor $(1 - p^{-2})^{-1} = 1 + p^{-2} + p^{-4} + \cdots$ and multiplying out, unique factorization produces each $1/n^2$ exactly once. The second is Euler's solution of the Basel problem [30]. The theorem makes the heuristic rigorous by counting with the Möbius function. Numerically, the product over the primes below one million is 0.60792714, against $6/\pi^2 = 0.60792710$.

So π, a number about circles, measures how often two whole numbers share no prime. The convergents of $\pi - 3$, read as pairs (beats, hits) such as $(7, 1)$, $(106, 15)$ and $(113, 16)$, are all visible trees.

5.2 Coprime means uncorrelated

The orchard looks like a matrix, and it is one. Treat "a hit every ath step" as a signal $x_t^{(a)} = 1$ if $a \mid t$ and 0 otherwise, and compare two such signals over one common period.

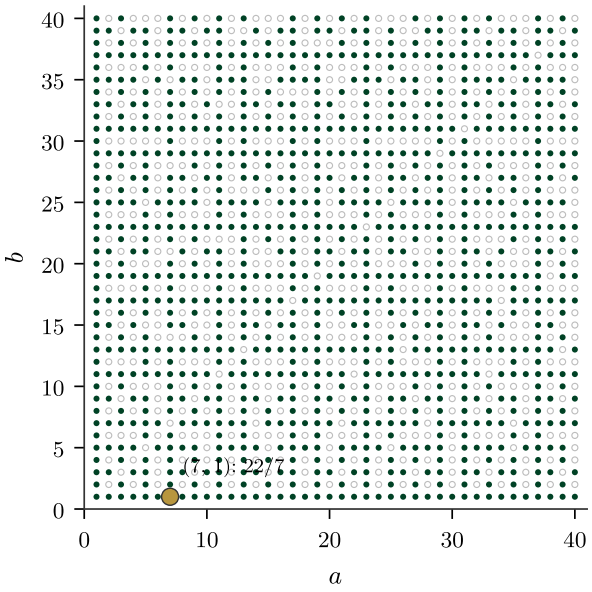


Figure 5.1. The orchard: lattice points (a, b) with $1 \le a, b \le 40$. Filled points are visible from the origin ($\gcd(a, b) = 1$); open circles are hidden. The gold point $(7, 1)$ is the first level of π's rhythm, read as (beats, hits).

Proposition 5.2 (the correlation of two pulse trains)**.** *For integers $a, b \ge 2$, the Pearson correlation of $x^{(a)}$ and $x^{(b)}$ over a full period of $\operatorname{lcm}(a, b)$ steps is*

$$\operatorname{corr}(a, b) = \frac{\gcd(a, b) - 1}{\sqrt{(a - 1)(b - 1)}}. \tag{5.2}$$

In particular the correlation is zero exactly when a and b are coprime, and it is 1 when $a = b$.

Proof. Over a period, $x^{(a)}$ has mean $1/a$ and, being 0 or 1, variance $\frac{1}{a}(1 - \frac{1}{a}) = (a - 1)/a^2$. The product $x_t^{(a)} x_t^{(b)}$ is 1 exactly when t is a multiple of $\operatorname{lcm}(a, b) = ab/\gcd(a, b)$, so its mean is $\gcd(a, b)/(ab)$. The covariance is therefore

$$\frac{\gcd(a, b)}{ab} - \frac{1}{a} \cdot \frac{1}{b} = \frac{\gcd(a, b) - 1}{ab},$$

and dividing by $\sqrt{(a - 1)(b - 1)}/(ab)$, the product of the standard deviations, gives (5.2). □

Our script checks (5.2) against a direct computation for every pair $2 \le a \ne b \le 30$. For example $\operatorname{corr}(4, 6) = 1/\sqrt{15} = 0.258$, which is why four against six "keeps colliding," while three against four, being coprime, is uncorrelated. In the 39×39 matrix of pulse trains $2, \ldots, 40$ (Fig. 5.2), the share of off-diagonal zeros is 0.607, close to $6/\pi^2$ for the same reason as the orchard. A prime's row is zero except at its multiples: primes are the "independent" pulse trains of arithmetic.

For periods 2 to 40 the matrix is positive definite: its eigenvalues run from 0.072 to 5.38. Random-matrix theory is used in finance to separate real correlations among stock returns from noise [54, 82]; how the spectrum of this arithmetic correlation matrix compares with such random benchmarks is a question we leave open (Chapter 14). Formula (5.2) itself is an elementary exercise; we did not find it stated in this form in the literature we searched, but we expect it is known, and we claim no novelty for it (Chapter 13).

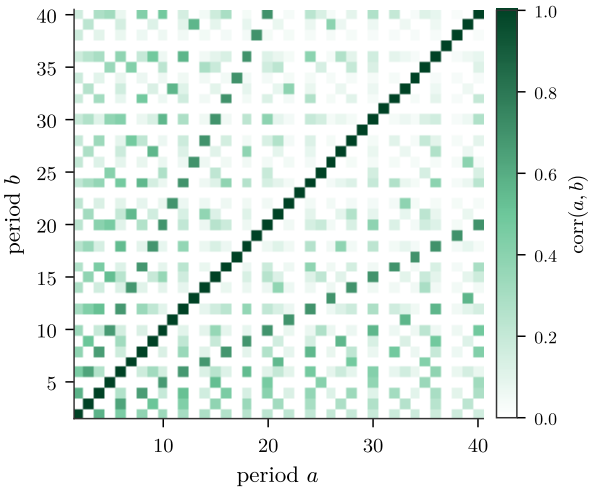


Figure 5.2. The correlation matrix (5.2) of pulse trains with periods 2 to 40. White cells are zero, exactly the coprime pairs; the pattern is the orchard of Fig. 5.1 seen as a matrix.

5.3 Hydrogen's $1/n^2$: a shared formula, not shared physics

Hydrogen's energy levels are proportional to $-1/n^2$, and its visible spectral lines, the Balmer series, come from the differences $1/4 - 1/n^2$. In wavelengths,

$$\lambda_n = B\,\frac{n^2}{n^2-4}, \qquad B = 364.56 \text{ nm}, \tag{5.3}$$

which is the formula Johann Balmer found in 1885 [5]. It gives $\lambda_3 = 656.2\,\text{nm}$ (the red H-alpha line) and $\lambda_4 = 486.1\,\text{nm}$.

Add up the reciprocal squares that set the levels, $1 + \frac{1}{4} + \frac{1}{9} + \cdots$, and the answer is $\zeta(2) = \pi^2/6$, the reciprocal of the orchard's $6/\pi^2$, and also, by (5.1), a product over the primes (Fig. 5.3). It is tempting to say that "hydrogen knows about the primes." It does not. The match is a match of formulas: the sum of $1/n^2$ is the same series wherever it appears. No physical quantity of hydrogen is the sum $\sum 1/n^2$, and nothing in the atom depends on prime factorization. We include it because it is a good illustration of how one series can sit inside several subjects, which is the theme of Chapter 12.

For the video. Walk into the orchard and count visible trees; show the share settling near 61%. Derive (5.2) on screen in four lines. Then 3 against 4 versus 4 against 6, heard and seen in the matrix. End with the Balmer lines and the careful sentence "same formula, not same physics." Instruments: "The orchard," "The correlation matrix," "Chemistry: hydrogen." About 8 minutes.

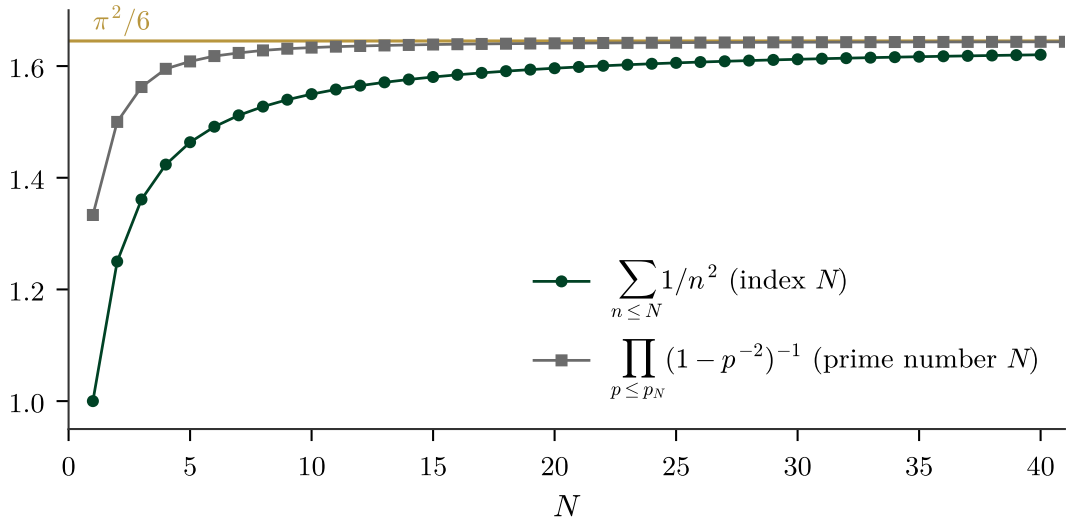


Figure 5.3. Two roads to $\pi^2/6$: the partial sums $\sum_{n \le N} 1/n^2$ (green, the reciprocal squares behind hydrogen's levels) and the partial Euler products $\prod (1-p^{-2})^{-1}$ over the first N primes (grey). The product converges much faster.

CHAPTER 6

Primes in sound and space

This chapter collects the places on the companion site where the primes are heard or seen: as notes, as arms of a spiral, as columns of a wheel, as peaks in a diffraction pattern and as a sum of waves. Each is a classical or recent published result; our contribution is to compute each one again and to tie it to the rhythm of π.

6.1 Primes as notes, and the bias of neighbors

Number the twelve notes of the octave $0, \dots, 11$. Every prime above 3 leaves remainder 1, 5, 7 or 11 on division by 12, because the other remainders share a factor with 12; these are the $\varphi(12) = 4$ notes C♯, F, G and B. Playing the primes in order as notes, with each gap to the next prime as a duration, gives the "prime music" instrument on the site.

If the remainders of consecutive primes were independent and equally likely, a prime would repeat the remainder of the one before it one time in four. They do not. Lemke Oliver and Soundararajan [57] showed numerically that consecutive primes avoid repeating their remainder, and explained the bias by a conjecture of Hardy–Littlewood type that predicts it fades only like $1/\ln x$. Our counts agree (Fig. 6.1). Modulo 10, among the primes from 7 up to 10^5, 10^6 and 10^7, the share of repeats is 13.9%, 15.4% and 16.3%, against 25% for independence; modulo 12 it is 12.8%, 15.0% and 16.4%. The share creeps up toward 25% as the range grows, slowly, as predicted.

6.2 A sunflower seeded with primes

Place seed number k at angle $2\pi k\alpha$ and distance $\sqrt{k}$ from the center. With $\alpha = \varphi - 1$ (the golden angle) this is the familiar model of a sunflower head; with $\alpha = \pi - 3$ the seeds fall into 7 spokes that curl into 113 arms (Fig. 6.2). The reason is the rhythm again: $113\alpha = 113\pi - 339$ differs from the integer 16 by only 3.0×10^{-5}, so seeds k and $k + 113$ sit at almost the same angle, and arm j of the 113 holds the seeds $k \equiv j \pmod{113}$.

Now light only the prime seeds. A prime larger than q can lie in arm j of q only if $\gcd(j, q) = 1$, each of those $\varphi(q)$ arms holds infinitely many primes by Dirichlet's theorem of 1837 [42, Thm. 15], and by the prime number theorem for arithmetic progressions they share the primes equally in the long run. Since 113 is prime, primes avoid only the arm $j = 0$; among

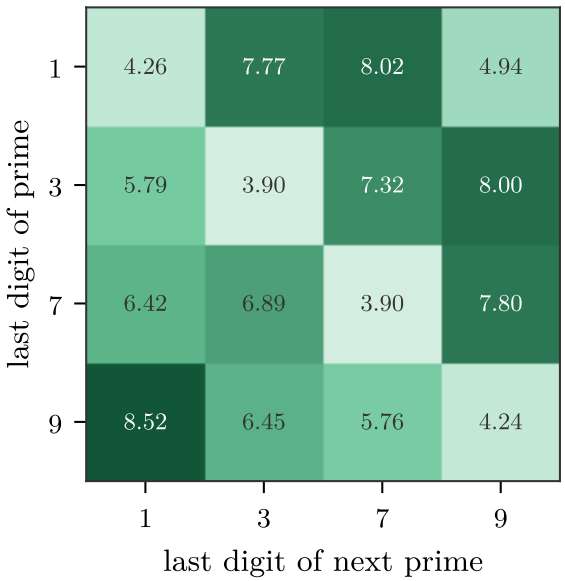


Figure 6.1. Last digits of consecutive primes up to 10^7 (664,576 primes from 7 on). Each cell is the percentage of pairs (prime, next prime) with the given last digits; independence would give 6.25% everywhere. The diagonal, where the digit repeats, is depleted.

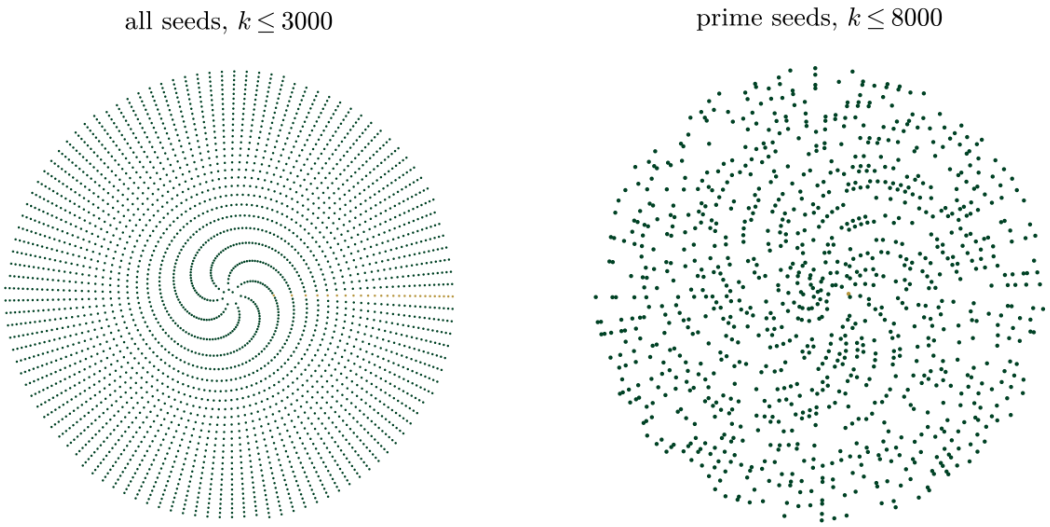


Figure 6.2. Pi's sunflower: seed k at angle $2\pi k(\pi - 3)$ and radius $\sqrt{k}$. Left, all seeds up to 3,000 form 7 spokes that curl into 113 arms (seeds divisible by 113 in gold). Right, only the prime seeds up to 8,000: they fill the arms evenly.

the primes up to 8,000 they occupy 112 of the 113 arms. With 106 arms, which is 2×53, primes occupy only $52 = \varphi(106)$ arms, and half the spiral goes dark.

Wheels. The same fact, straightened out, is *wheel factorization*: lay the integers in rows of width w, and the primes beyond w fall only in the $\varphi(w)$ columns coprime to w. With $w = 30$ a sieve can skip 73% of all numbers without testing them, and with $w = 210$, 77.1%. Pi's

rhythm, laid in rows of width 113, forms 16 nearly straight columns, one for each hit in a 113-beat block.

6.3 Diffraction: a quasicrystal and the primes

Crystallographers read the order of a material from its diffraction pattern, which is (up to constants) the squared modulus of the Fourier transform of the atom positions. For N points $x_1, \dots, x_N$ on a line, define the structure factor

$$S(k) = \frac{1}{N} \left| \sum_{j=1}^{N} e^{2\pi i k x_j} \right|^2 . \tag{6.1}$$

A random set gives $S(k) \approx 1$ for all $k \neq 0$; a periodic crystal gives sharp peaks of height about N at the multiples of its reciprocal lattice. In 1982 Dan Shechtman observed sharp diffraction from a metal alloy whose symmetry no periodic crystal can have [88]; such ordered but aperiodic materials are *quasicrystals*, and he received the 2011 Nobel Prize in Chemistry.

Pi's rhythm is a one-dimensional quasicrystal. Put an atom at each hit of π's rhythm. Sturmian point sets are standard examples of one-dimensional quasicrystals: their diffraction is pure point, with Bragg peaks at the points $m\alpha + n$ of the "Fourier module" $\mathbb{Z} + \alpha\mathbb{Z}$ [4]. In Fig. 6.3 (top) the peaks sit at $\{m(\pi - 3)\}$, folded into $[0, \frac{1}{2}]$; the peak at $k = \pi - 3$ has height 397 for the hits among the first 3,000 beats.

The primes are "effectively limit-periodic." Torquato, Zhang and de Courcy-Ireland [95] computed the structure factor of the primes in long intervals and found dense Bragg-like peaks at rational wavenumbers $k = a/q$ with *square-free* q, and none at $1/4$ or $1/9$; they called this order "effectively limit-periodic." The rational positions and the square-free rule have a short explanation. If the primes in a window are equidistributed among the $\varphi(q)$ residue classes coprime to q, then for $\gcd(a, q) = 1$

$$\sum_{p} e^{2\pi i a p/q} \approx \frac{N}{\varphi(q)} \sum_{\substack{r \bmod q \\ \gcd(r,q)=1}} e^{2\pi i a r/q} = \frac{N\,\mu(q)}{\varphi(q)},$$

because the inner sum is a Ramanujan sum, which equals the Möbius function $\mu(q)$ [42, §16.6]. So $S(a/q) \approx N\mu(q)^2/\varphi(q)^2$, which vanishes unless q is square-free. For the 1,472 primes in $[10^6, 10^6 + 2 \cdot 10^4)$ our computation gives $S(1/2) = 1472$ (all are odd), $S(1/3) = 369$ (the prediction $N/4$ is 368), but only $S(1/4) = 1.20$ and $S(1/9) = 0.57$, at the level of the random background.

Freeman Dyson suggested that the zeros of the zeta function, if the Riemann hypothesis is true, form a one-dimensional quasicrystal in a suitable sense, and that classifying such quasicrystals might lead somewhere [26]. Chapter 9 takes up that remark.

6.4 Primes as a sum of waves

Let $\psi(x) = \sum_{p^m \le x} \ln p$, a staircase that climbs by $\ln p$ at every prime power p^m. Riemann's 1859 paper [83] introduced the idea, and von Mangoldt proved [98], that for $x > 1$ not a prime

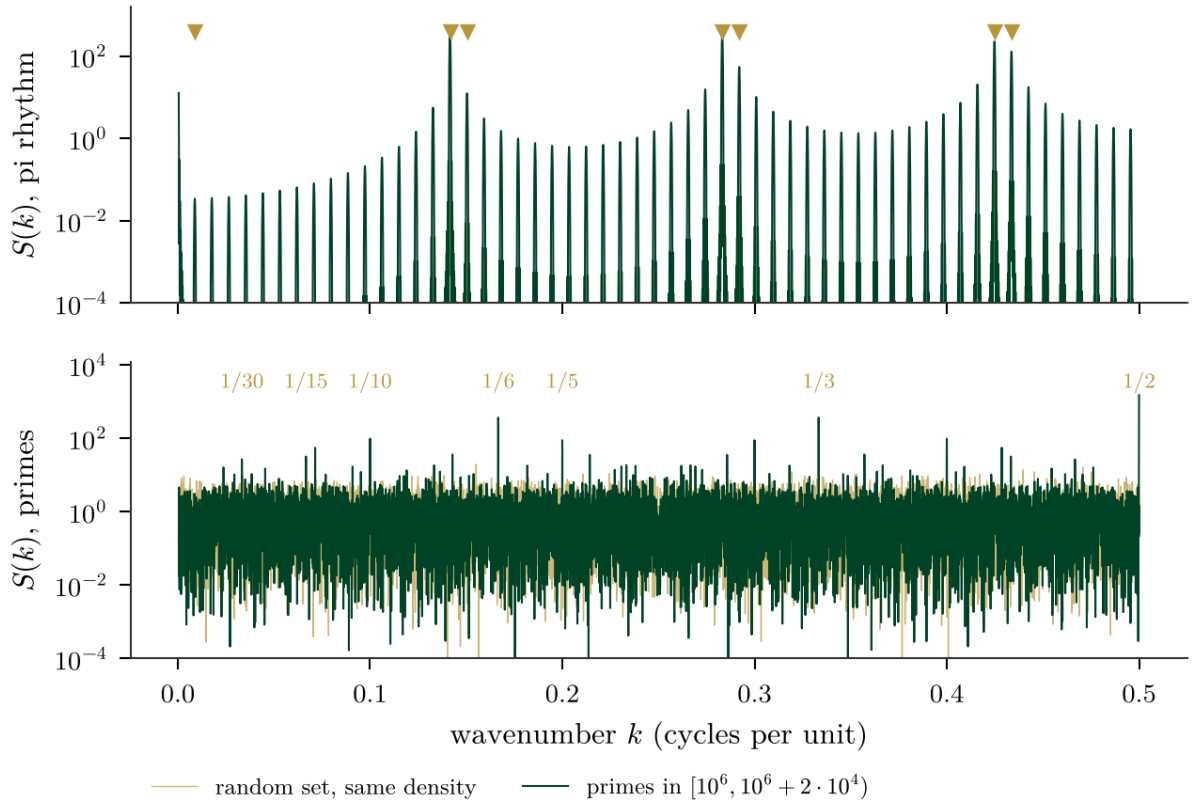


Figure 6.3. Structure factors (6.1) on a logarithmic scale (computed with a smooth taper to suppress side lobes). Top: atoms at the hits of π's rhythm in the first 3,000 beats; gold triangles mark $\{m(\pi-3)\}$ for $m = 1, \dots, 7$, folded into $[0, \frac{1}{2}]$. Bottom: atoms at the primes in $[10^6, 10^6 + 2 \cdot 10^4)$ (green) and at a random set of the same size (gold); the prime peaks sit at fractions with square-free denominators.

power

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \ln(2\pi) - \tfrac{1}{2} \ln\bigl(1 - x^{-2}\bigr)\,, \tag{6.2}$$

where ρ runs over the nontrivial zeros of the zeta function, summed symmetrically in pairs $\rho, \bar{\rho}$. Writing $\rho = \frac{1}{2} + i\gamma$ (as all computed zeros are), each pair contributes a wave $-2\,\mathrm{Re}(x^{\rho}/\rho)$ with amplitude about $\sqrt{x}/\gamma$ and frequency γ in the variable $\ln x$. The first zeros have $\gamma = 14.13473$, 21.02204 and 25.01086. Figure 6.4 rebuilds ψ from 3, 20 and 100 pairs of zeros: the waves add up to the staircase. The $\ln 2\pi$ in (6.2) is one more place where π sits next to the primes.

The Montgomery–Dyson observation, stated carefully. In 1972 Hugh Montgomery, studying the spacings between the γ's, derived a pair-correlation function $1 - (\sin \pi u / \pi u)^2$ for suitably normalized zeros; he proved it for a restricted class of test functions assuming the Riemann hypothesis, and conjectured it in general [70]. Freeman Dyson recognized the function as the pair correlation of eigenvalues of large random Hermitian matrices from the Gaussian Unitary Ensemble (GUE), a model of the energy levels of heavy nuclei [27, 66]. Odlyzko's large-scale computations showed excellent agreement [73]. What is known is the agreement of statistics; there is no known Hermitian operator whose eigenvalues are the zeros, and no proof that the statistics must agree. Chapter 9 computes these statistics for ourselves.

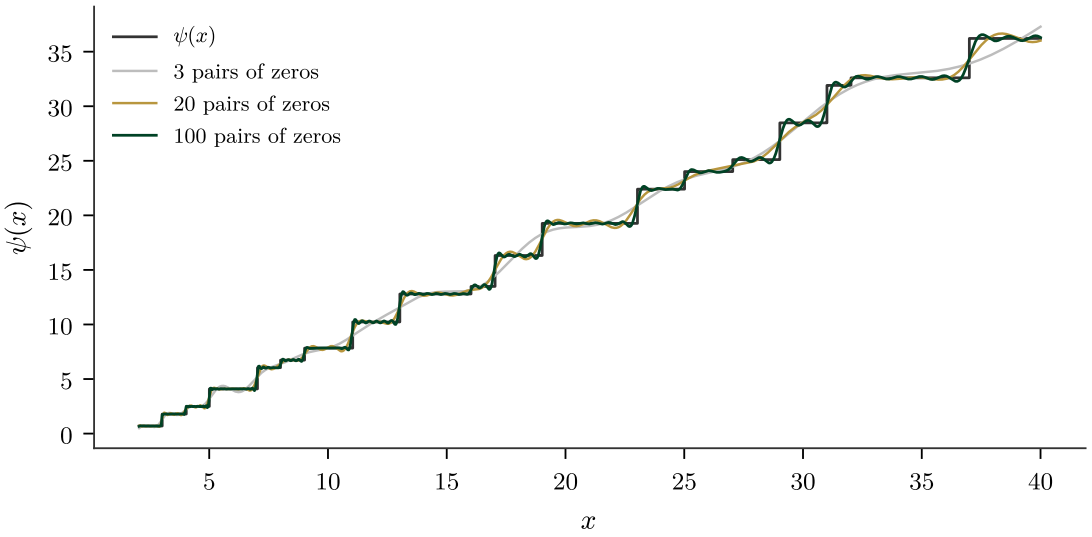


Figure 6.4. The explicit formula (6.2): the staircase $\psi(x)$ (black) and its reconstruction from the first 3, 20 and 100 pairs of zeta zeros.

For the video. Play the primes as notes and show the diagonal of Fig. 6.1 fading. Grow the sunflower seed by seed, then switch to primes only. Show both diffraction patterns, and point to the missing peaks at 1/4 and 1/9. End with Fig. 6.4 adding waves one by one, and say clearly what Montgomery and Dyson found and what is still unknown. Instruments: "Prime music," "Pi's sunflower," "Sieves & periods," "Quasicrystals," "Zeta zeros & nuclei." About 11 minutes.

CHAPTER 7

Tools that hunt primes

The continued fraction found 355/113. Can the same machine find primes? This chapter tries three tools from the companion site on the primes: a factoring method that is built from continued fractions, a picture of Fermat's little theorem made of rhythms, and an experiment that reads rhythms as binary numbers and tests them for primality. The first two are classical; the third is our experiment, and its result is a clear null.

7.1 Factoring with the continued fraction of $\sqrt{N}$

To factor an odd number N it is enough to find x and y with $x^2 \equiv y^2 \pmod N$ and $x \not\equiv \pm y$, because then $\gcd(x-y, N)$ is a proper factor. Continued fractions produce many congruences of the form $x^2 \equiv$ (small number) $\pmod N$. Expand $\sqrt{N}$ (or $\sqrt{kN}$ for a small multiplier k) with the integer recurrence

$$P_{n+1} = a_n Q_n - P_n, \qquad Q_{n+1} = \frac{kN - P_{n+1}^2}{Q_n}, \qquad a_{n+1} = \left\lfloor \frac{\lfloor \sqrt{kN} \rfloor + P_{n+1}}{Q_{n+1}} \right\rfloor, \tag{7.1}$$

starting from $P_0 = 0$, $Q_0 = 1$, $a_0 = \lfloor \sqrt{kN} \rfloor$. If A_n are the numerators of the convergents, reduced modulo N, then

$$A_{n-1}^2 \equiv (-1)^n Q_n \pmod N, \qquad 0 < Q_n < 2\sqrt{kN}. \tag{7.2}$$

These are standard facts about the continued fraction of a square root. Using them to factor, by searching the Q_n for squares, was described by D. H. Lehmer and R. E. Powers in 1931 [56]. In 1975 Michael Morrison and John Brillhart made it practical by combining many Q_n that factor over a small set of primes, using linear algebra modulo 2 to find a product that is a square, and with it factored the seventh Fermat number $F_7 = 2^{128} + 1$ [72].

Our script implements the simple Lehmer–Powers search. It is instructive to run it on $F_5 = 2^{32} + 1$, which Euler showed is divisible by 641 [42, §2.5]. Because $F_5 = 65536^2 + 1$, the continued fraction of $\sqrt{F_5}$ is $[65536; 131072, 131072, \dots]$ with every $Q_n = 1$, a square that gives only the trivial factorization. With multiplier $k = 2$ the search succeeds: at step $n = 340$ it finds a square Q_n, and the gcd gives $F_5 = 641 \times 6{,}700{,}417$. Table 7.1 shows the first steps.

Table 7.1. The first steps of the continued fraction of $\sqrt{2(2^{32}+1)}$ in the recurrence (7.1). By (7.2) each Q_n is, up to sign, a square modulo N; the search waits for one that is a perfect square at an even step.

n	a_n	Q_n	square?
1	1	166,833	
2	9	18,530	
3	289	639	
4	1	181,815	
5	62	2,930	
6	3	49,383	
7	1	109,871	
8	2	52,014	

A product of two six-digit primes, $999{,}983 \times 1{,}000{,}003$, falls after 2 steps, because two factors this close make N nearly a square.

This is the clearest case of "pi's machine doing prime work": the algorithm that, run on π, finds 22/7 and 355/113 is the algorithm that, run on $\sqrt{N}$, finds factors. It does not use any property of π, of course.

7.2 Mersenne necklaces and Fermat's little theorem

Take $m = 2^p - 1$ points on a circle and join each point x to $2x \bmod m$. Write x in binary with p digits. Since $2^p \equiv 1 \pmod{m}$, doubling modulo m simply rotates the digits by one place. So every orbit of the doubling map is a binary word of length p together with all its rotations: a *necklace*, or in our language a p-beat rhythm and all its rotations.

Proposition 7.1 (Fermat's little theorem for base 2, by necklaces)**.** *If p is prime, then p divides $2^p - 2$.*

Proof. Exclude the two constant words (all rests and all hits). Every other binary word of prime length p has exactly p distinct rotations, since a word equal to a nontrivial rotation of itself has a period dividing p, hence period 1. So the $2^p - 2$ nonconstant words split into orbits of size exactly p. □

The same argument with an alphabet of a letters gives $p \mid a^p - a$; this combinatorial proof is due to Golomb [36]. For $p = 7$ there are $(2^7 - 2)/7 = 18$ necklaces, each a 7-beat rhythm. For the composite length $p = 6$ the orbits have mixed sizes: 1 of length 2, 2 of length 3, 9 of length 6. The picture does *not* make $2^p - 1$ prime: $2^{11} - 1 = 2047 = 23 \times 89$. The test that does is the Lucas–Lehmer test, and with it the Great Internet Mersenne Prime Search found the largest known prime, $2^{136{,}279{,}841} - 1$, with 41,024,320 digits, in October 2024 [41].

7.3 The prime hunt

Here is an experiment from the companion site. Read the first n beats of a number's rhythm as a binary number, a hit as 1 and a rest as 0, and test it for primality as n grows. Does the rhythm of π attract primes?

Table 7.2. The prime hunt: probable primes among the binary prefixes of the first 1,500 beats of each rhythm. "Expected" is $\sum 2/\ln v_n$ over odd prefixes (the site's simpler $\sum 2/(n \ln 2)$ in parentheses). The last two columns are Poisson probabilities of a count at least as low, or at least as high, as the one found.

number	found	expected	$P(X \le \text{found})$	$P(X \ge \text{found})$
$\pi - 3$	1	2.4 (2.4)	0.30	0.91
$e - 2$	14	14.8 (14.3)	0.48	0.62
$\sqrt{2} - 1$	11	8.3 (8.8)	0.86	0.22
$\varphi - 1$	13	12.3 (11.8)	0.64	0.46

Method. For each of $\alpha = \pi - 3$, $e - 2$, $\sqrt{2} - 1$ and $\varphi - 1$ we built the first 1,500 beats with Theorem 3.5, formed the prefixes v_n (the binary number of the first n beats) for $n = 2, \ldots, 1500$, and tested every odd prefix with the Baillie–PSW probable-prime test (Appendix B). A prefix is odd exactly when beat n is a hit. For a random odd number of size v, the prime number theorem gives a probability of about $2/\ln v$ of being prime, so the expected count is $\sum 2/\ln v_n$ over the odd prefixes. The site uses the slightly cruder $\sum 2/(n \ln 2)$, which treats every prefix as an n-digit number; both are reported.

Results. Table 7.2 and Fig. 7.1 give the counts. The rhythm of $\pi - 3$ produced a single probable prime, at beat 49, where about 2.4 were expected; its rhythm is mostly rests, so its prefixes are odd only once every seven beats or so, and the expected count is small. The other three constants found close to their expected counts. In total the four rhythms gave 39 probable primes against 37.9 expected. No count is unusual: every Poisson tail probability in the table is above 0.2.

Interpretation. The rhythm of π does not attract primes, and neither does any other rhythm we tried. This is a null result, and we report it as one. The comparison is with Mersenne numbers $2^p - 1$, whose binary form is the all-hit rhythm: they *are* prime far more often than random numbers of their size, because every prime factor of $2^p - 1$ must be $\equiv 1 \pmod{2p}$, which removes all the small potential divisors. A construction that clearly beat its chance count in this test would deserve a closer look; none did. The same verdict, by a different route, comes from the statistics of π's terms in Chapter 10 and the prime convergents in Chapter 8.

For the video. Factor $2^{32} + 1$ live: show the Q_n scrolling past until a square appears and 641 drops out. Draw the doubling circle for $p = 7$ and let each necklace play as a rhythm. Run the prime hunt for π and watch it find one prime, then for e and watch it keep pace with chance. Say plainly: no attraction. Instruments: "Factoring," "Mersenne circles," "Rhythm → number," "Prime hunt." About 10 minutes.

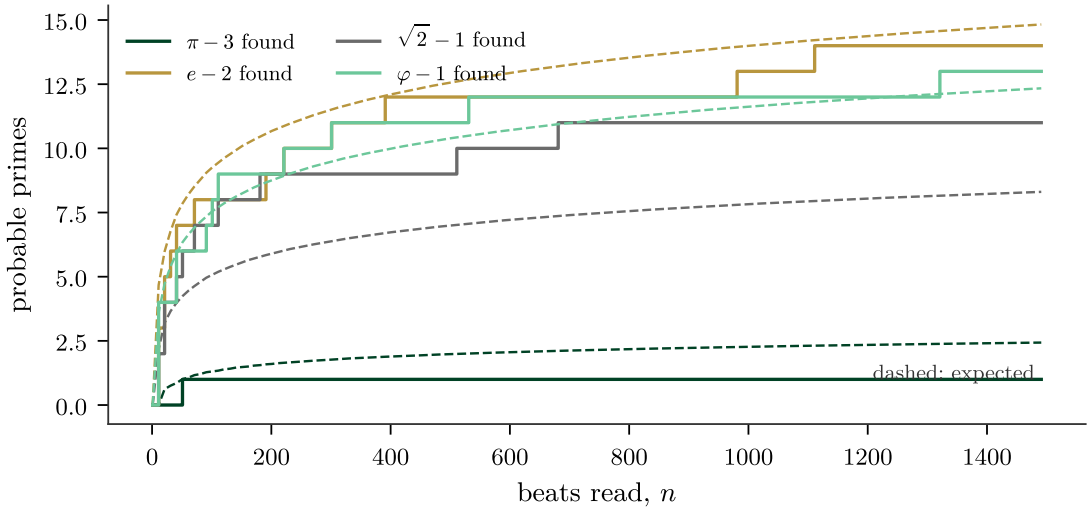


Figure 7.1. The prime hunt: cumulative probable primes among binary prefixes of each rhythm (steps) against the expected count $\sum 2/\ln v_n$ (dashed), for the first 1,500 beats.

CHAPTER 8

Thread (a): prime convergents

8.1 The question

The three classical fractions for π, 22/7, 223/71 and 355/113, all have prime denominators, and two of them are convergents. Among the convergents p_n/q_n of π, the prime denominators begin

$$q_1 = 7, \quad q_3 = 113, \quad q_8 = 265{,}381, \quad q_{32} = 842{,}468{,}587{,}426{,}513{,}207,$$

a list recorded in the OEIS [78]; the prime numerators begin $3, 103993, 833719, 4272943, \ldots$ [77]. Does the list go on forever? Nobody knows. The same question for the golden ratio asks whether there are infinitely many Fibonacci primes, and for $\sqrt{2}$ whether there are infinitely many Pell primes; both are famous open problems.

We cannot answer the question. What we can do is the next best thing: derive a heuristic prediction, with an explicit constant, for how many prime denominators a *typical* real number should have among its first N convergents; test the prediction on random numbers; and see where π, e, $\sqrt{2}$, φ and $\log_2(3/2)$ fall. This chapter does all three.

8.2 A heuristic with an explicit constant

Step 1: size. By Lévy's theorem (Theorem 2.7), for almost every x the denominators grow like

$$\ln q_n \sim Ln, \qquad L = \frac{\pi^2}{12 \ln 2} = 1.18657\ldots \tag{8.1}$$

Step 2: the naive guess. By the prime number theorem, a "random" integer near q is prime with probability about $1/\ln q$. If the q_n behaved like random integers of their size, the expected number of prime q_n with $n \le N$ would be about $\sum_{n \le N} 1/(Ln) \approx (1/L) \ln N$.

Step 3: local corrections. The q_n are not random integers. Consecutive denominators are coprime by (2.5), so, for example, two consecutive q_n are never both even. The right description of their local behavior is the following equidistribution theorem.

Theorem 8.1 (Szüsz; Moeckel; Jager and Liardet)**.** *Let $m \ge 2$. For almost every x, the pairs (q_{n-1}, q_n) reduced modulo m are equidistributed among the pairs (u, v) of residues with* $\gcd(u, v, m) = 1$.

Table 8.1. The share of convergent denominators q_n divisible by a prime ℓ: for π ($n \le 10{,}000$), for 200 random reals ($n \le 2{,}000$ each), the prediction $1/(\ell+1)$ of Corollary 8.2, and $1/\ell$ for random integers.

ℓ	π	random reals	$1/(\ell+1)$	$1/\ell$
2	0.3344	0.3331	0.3333	0.5000
3	0.2510	0.2499	0.2500	0.3333
5	0.1620	0.1666	0.1667	0.2000
7	0.1284	0.1243	0.1250	0.1429
11	0.0820	0.0833	0.0833	0.0909
13	0.0680	0.0708	0.0714	0.0769

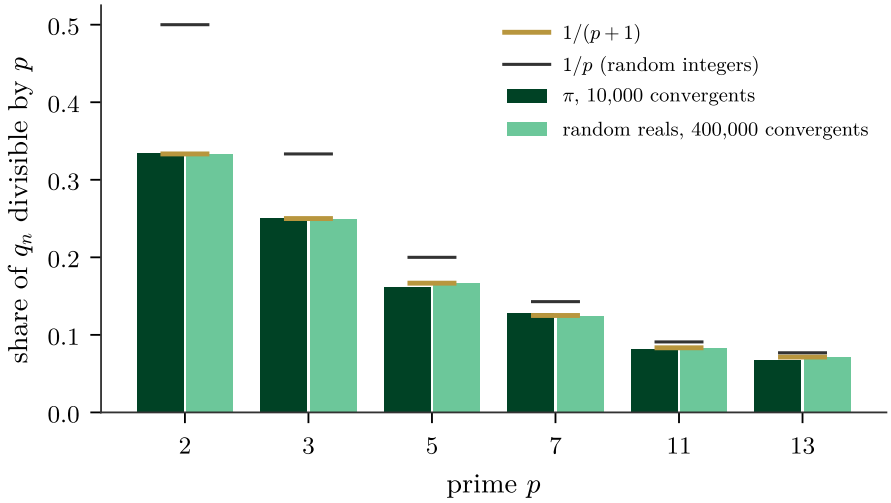


Figure 8.1. Convergent denominators avoid small prime factors: the share of q_n divisible by ℓ follows $1/(\ell+1)$ (gold), not $1/\ell$ (black), both for π and for random reals.

Szüsz proved the statement for pairs of denominators in 1962 [93, Satz 3.2]; Moeckel obtained it up to sign through the geodesic flow on modular surfaces [68]; Jager and Liardet proved equidistribution of the whole matrix $\left(\begin{smallmatrix} p_n & p_{n-1} \\ q_n & q_{n-1} \end{smallmatrix}\right)$ modulo m [44]. Borda's recent paper surveys this history and extends it [12]. The case $m = 2$ says that, for almost every x, exactly one third of the denominators are even.

Corollary 8.2. *For almost every x and every prime ℓ, the proportion of n with $\ell \mid q_n$ is $1/(\ell+1)$, not the $1/\ell$ of random integers.*

Proof. Modulo a prime ℓ, the admissible pairs are the $\ell^2 - 1$ nonzero pairs (u, v). Those with $v = 0$ are the $\ell - 1$ pairs $(u, 0)$, $u \neq 0$. By Theorem 8.1 the proportion is $(\ell-1)/(\ell^2-1) = 1/(\ell+1)$. □

Table 8.1 and Fig. 8.1 check the corollary on the first 10,000 convergents of π and on 400,000 convergents of 200 random reals. Both match $1/(\ell+1)$ to within sampling error; π is typical here too.

Step 4: the correction factor is $\zeta(2)$. In the usual way of refining the prime number theorem for structured sequences, the probability that q_n is prime is $1/\ln q_n$ multiplied, for

each prime ℓ, by the ratio of the chance that $\ell \nmid q_n$ to the chance that ℓ does not divide a random integer:

$$\mathfrak{S} = \prod_{\ell} \frac{1 - \frac{1}{\ell+1}}{1 - \frac{1}{\ell}} = \prod_{\ell} \frac{\ell}{\ell+1} \cdot \frac{\ell}{\ell-1} = \prod_{\ell} \frac{1}{1-\ell^{-2}} = \zeta(2) = \frac{\pi^2}{6}. \tag{8.2}$$

The product converges absolutely, so the correction is a single constant. It is a pleasant surprise that the constant is $\pi^2/6$, the same number as in Chapter 5, and for a related reason: the correction measures how much more often than random a denominator avoids each prime, and the reason it does is the coprimality of neighbors.

Step 5: the count. Putting the steps together, for a typical x,

$$\mathbb{E}\,\#\{n \le N : q_n \text{ prime}\} \approx \sum_{n \le N} \frac{\zeta(2)}{\ln q_n} \approx \frac{\zeta(2)}{L} \ln N = \frac{\pi^2}{6} \cdot \frac{12 \ln 2}{\pi^2} \ln N = 2 \ln 2 \cdot \ln N. \tag{8.3}$$

Heuristic 8.3. For almost every real x, the number of prime denominators among the first N convergents grows like $c \ln N$ with

$$c = 2 \ln 2 = 1.3863\ldots$$

The same holds for prime numerators, because Theorem 8.1 in the matrix form of Jager and Liardet gives the numerators the same local behavior.

Three remarks. First, the constant is independent of everything except the two laws used, which is why it is so clean: $\zeta(2)$ from the arithmetic and $\pi^2/(12 \ln 2)$ from the dynamics, and the two π^2 cancel. Second, a count growing like $\ln N$ is very slow: it predicts infinitely many prime denominators for a typical number, but only about one new one each time N is multiplied by $e^{1/c} \approx 2.06$, roughly one per doubling. Third, this is a heuristic. It treats primality of different q_n as independent events with the stated probabilities. It says nothing about any particular number such as π unless π is typical, which is unproved.

8.3 The test on random reals

To test Heuristic 8.3 we need many truly random real numbers with known continued fractions. We draw a random integer r with $D = 1{,}620$ digits and expand $r/10^D$ by exact integer arithmetic. Every real number in the interval $[r/10^D, (r+1)/10^D)$ shares the first terms, and since the error of the nth convergent shrinks like $q_n^{-2} \approx e^{-2Ln}$, about $D \ln 10/(2L) \approx 0.97D$ terms are shared, far more than the 1,000 we use. So the first 1,000 terms are those of a uniformly random real. We drew 400 such reals and counted probable primes among $q_1, \ldots, q_{1000}$ and $p_1, \ldots, p_{1000}$ (Appendix B).

Results. The mean number of prime denominators was 9.90 (standard deviation 2.87), and of prime numerators 9.68 (2.99). The growth is the test of the constant: between $n = 100$ and $n = 1000$ the mean count rose by 3.29, against the predicted $2 \ln 2 \cdot \ln 10 = 3.19$; between $n = 200$ and $n = 1000$ it rose by 2.22 against 2.23. A least-squares fit of the mean count against $\ln n$ over $50 \le n \le 1000$ gives slope 1.45 ± 0.04, slightly above $2 \ln 2$ because the earliest convergents, where q_n is small, still carry some weight. The absolute level has an additive constant that the heuristic does not fix: the pathwise sum $\sum \zeta(2)/\ln q_n$ averages 10.78 at $N = 1000$, a little above the observed mean, because $1/\ln q$ is a poor guide for the first few, very small denominators. The heuristic's *slope* is what matters, and it fits.

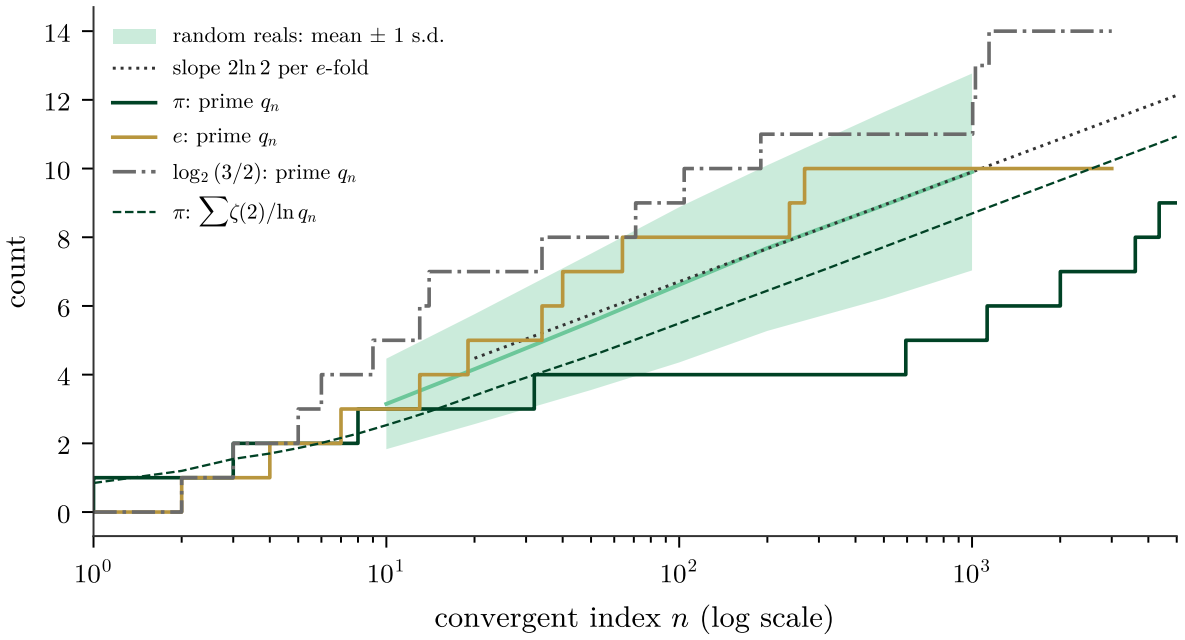


Figure 8.2. Prime convergent denominators, counted cumulatively against the index n (log scale). The shaded band is the mean ± one standard deviation over 400 random reals; the dotted line has the predicted slope $2\ln 2$ per e-fold. Steps: π (to $n = 5{,}000$), e and $\log_2(3/2)$ (to 3,000); dashed: π's own expected count $\sum\zeta(2)/\ln q_n$.

Table 8.2. Probable primes among convergents. Columns: the number of convergents N examined; prime denominators q_n, $n \le N$, and the heuristic expectation $\sum_{n\le N}\zeta(2)/\ln q_n$; prime numerators; the counts of prime q_n and p_n for $n \le 1000$ with the expectation; and the percentile of the q_n count at $n \le 1000$ among 400 random reals (share of random reals with at most as many). For $\sqrt{2}$ and φ the typical-number heuristic does not apply.

		$n \le N$			$n \le 1000$			
number	N	q_n	exp.	p_n	q_n	p_n	exp.	pct.
π	5,000	9	10.9	13	5	10	8.7	6
e	3,000	10	12.4	10	10	10	11.6	61
$\log_2(3/2)$	3,000	14	13.9	11	11	10	12.4	74
$\sqrt{2}$	3,000	23	16.8	18	17	16	14.7	–
φ	3,000	22	27.7	22	21	21	23.9	–

8.4 Is pi typical?

Table 8.2 gives the counts for the five constants of the Suite, and Figs. 8.2 and 8.3 compare them with the random sample.

Pi. Among its first 5,000 convergents, π has 9 prime denominators, at $n = 1$, 3, 8, 32, 594, 1126, 2002, 3610, 4355, against 10.9 expected; the last has 2,239 digits. It has 13 prime numerators, at $n = 4$, 8, 10, 15, 86, 229, 333, 593, 839, 852, 1148, 2725, 3787 (the largest has 1,949 digits). The first four denominators and the first numerators agree with the OEIS

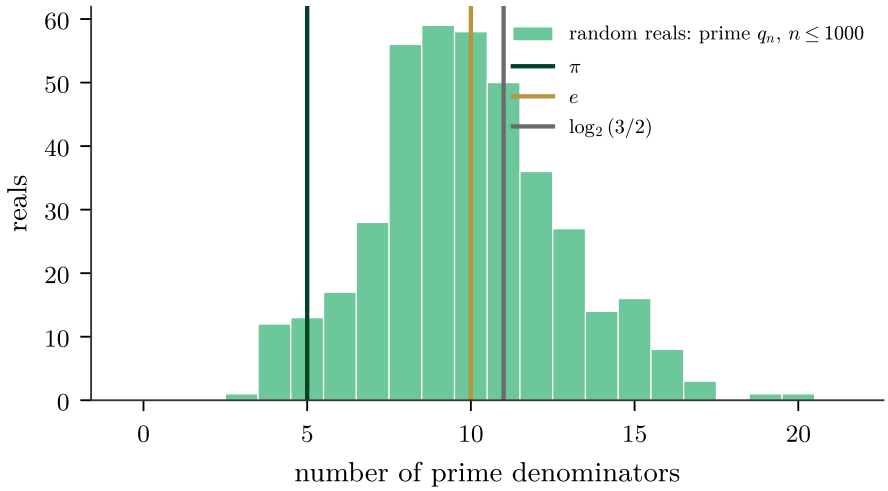


Figure 8.3. The number of prime denominators among the first 1,000 convergents, for 400 random reals (histogram), with π, e and $\log_2(3/2)$ marked.

lists [77, 78]. At $n \le 1000$, π's count of 5 prime denominators sits at the 6th percentile of the random sample: on the low side, but well inside the range of ordinary numbers, and its count of prime numerators is about average. Its even denominators make up 0.3330 of the first 10,000 (prediction 1/3) and those divisible by 3 make up 0.2484 (prediction 1/4). By every measure here, π looks like a typical real number. Heuristic 8.3 predicts that it has infinitely many prime denominators, with the next few spread out roughly one per doubling of n.

The fifth and e. The number $\log_2(3/2)$ has 14 prime denominators up to $n = 3{,}000$ against 13.9 expected; e has 10 against 12.4. Both are unremarkable, although e is not a typical number (its terms grow, and its denominators are even exactly one third of the time because of its pattern, not because of Theorem 8.1).

Quadratic irrationals. For φ, q_n is the Fibonacci number F_{n+1}; for $\sqrt{2}$ it is the Pell number P_{n+1}. Our counts recover the known lists of indices of Fibonacci and Pell primes [75, 79] up to index 3,001: the Fibonacci indices are 3, 4, 5, 7, 11, 13, 17, 23, 29, 43, 47, 83, 131, 137, 359, 431, 433, 449, 509, 569, 571, 2971, and the Pell indices are 2, 3, 5, 11, 13, 29, 41, 53, 59, 89, 97, 101, 167, 181, 191, 523, 929, 1217, 1301, 1361, 2087, 2273, 2393. These sequences break the typical-number heuristic in a structured way. They are divisibility sequences ($F_m \mid F_n$ whenever $m \mid n$), so F_n can be prime only when n is prime (or $n = 4$), and their denominators are not equidistributed modulo m: half of the Pell denominators are even, not a third. Grantham and Granville [40] develop heuristics for primes in such linear recurrences; for Fibonacci numbers they predict about $e^{\gamma} \log_{\varphi} N$ primes F_n with $n \le N$, which is 29.6 for $N = 3{,}001$, against the 22 found (they note that the Fibonacci data run below the prediction). Whether either list is infinite is open.

8.5 Do the site's tools help?

The owner of the companion site asked whether the tools assembled there help with this question. Here is a frank assessment.

- *Rhythms and Sturmian words.* The denominators are block lengths (Theorem 3.5), which makes the question easy to state musically: does π's form contain infinitely many sections whose length in bars is prime? But the rhythm adds no information about primality beyond the recurrence.
- *Coprimality and* $6/\pi^2$. This is the tool that does real work: the coprimality of neighboring blocks is exactly what produces the correction factor $\zeta(2)$ in (8.2).
- *Necklaces, diffraction, the prime hunt.* These do not bear on the question. The prime hunt is a different experiment with the same verdict: rhythms of π neither attract nor repel primes.
- *Self-similarity.* For quadratic irrationals, exact self-similarity means a linear recurrence, and the recurrence's divisibility structure is exactly what governs Fibonacci and Pell primes. For typical numbers, self-similarity in rule but not in shape is what makes the random model plausible.

In short, the tools organize the heuristic and explain its constant. They do not bring a proof closer. A proof that π has infinitely many prime denominators would require, at the least, control of π's terms that nobody has: it is not even known that they are unbounded.

8.6 What is known, and what is not

Known: the equidistribution of convergents modulo m for almost every x (Theorem 8.1); Lévy's theorem; the published lists of prime numerators and denominators for π and of Fibonacci and Pell primes; the Grantham–Granville heuristics for linear recurrences. Open: whether π, e, φ or $\sqrt{2}$ has infinitely many prime convergent denominators. We did not find in the literature we searched the explicit heuristic constant $2\ln 2$ for typical reals, or a test of it on a large random sample; both are offered here as a heuristic and an experiment, not a theorem.

For the video. State the question with π's four prime denominators on screen. Build the heuristic in five steps, ending with the two π^2's cancelling to leave $2\ln 2$. Show Fig. 8.2 and the histogram with π marked. Close with the frank assessment: the tools explain the constant but do not prove anything. Instrument: "The ladder," with the bar switched between π, φ and $\sqrt{2}$. About 12 minutes.

CHAPTER 9

Thread (b): one spectrum, from crystal to randomness

9.1 The question

Chapter 6 ended with the Montgomery–Dyson observation: the zeros of the zeta function seem to be spaced like the eigenvalues of large random Hermitian matrices, the model physicists use for energy levels of heavy nuclei. The companion site lists this as an open thread and asks whether the tools assembled here help. They cannot prove anything about the zeros. What they can do is supply a scale on which to place the zeros: at one end a perfect crystal, next the rhythm of π (a quasicrystal), then the zeros and random matrices, and at the other end pure randomness. This chapter computes the statistics that define that scale, for every point set on it, with one set of tools.

9.2 The point sets

All point sets are *unfolded*, that is, rescaled so that the mean spacing is 1, before any statistic is computed.

- *Crystal:* the integers.
- *Quasicrystal:* the hits of π's rhythm in the first 33,102 beats (Theorem 3.5), multiplied by $\pi - 3$. The spacings take two values, $7(\pi - 3) = 0.9911$ and $8(\pi - 3) = 1.1327$. The three-gap points $\{n(\pi-3)\}$ of Theorem 3.7 belong here too; for $N = 1{,}000$ their spacings, in units of $1/N$, take the values $0.030, 8.610, 8.640$.
- *Zeta zeros:* the imaginary parts $\gamma_1 < \gamma_2 < \cdots$ of the first 5,000 nontrivial zeros, up to height 5447.86, computed with mpmath and checked against the zero-counting function. They are unfolded with the smooth part of the Riemann–von Mangoldt counting formula, $\bar{N}(T) = \frac{T}{2\pi}\ln\frac{T}{2\pi e} + \frac{7}{8}$; the unfolded spacings have mean 1.0000. Odlyzko's tables [73] reach far higher; we used only zeros we computed ourselves.
- *GUE:* the eigenvalues of 16 random 1000×1000 complex Hermitian matrices with independent Gaussian entries, from the central half of each spectrum, unfolded with the semicircle law [66]; mean spacing 0.9999.

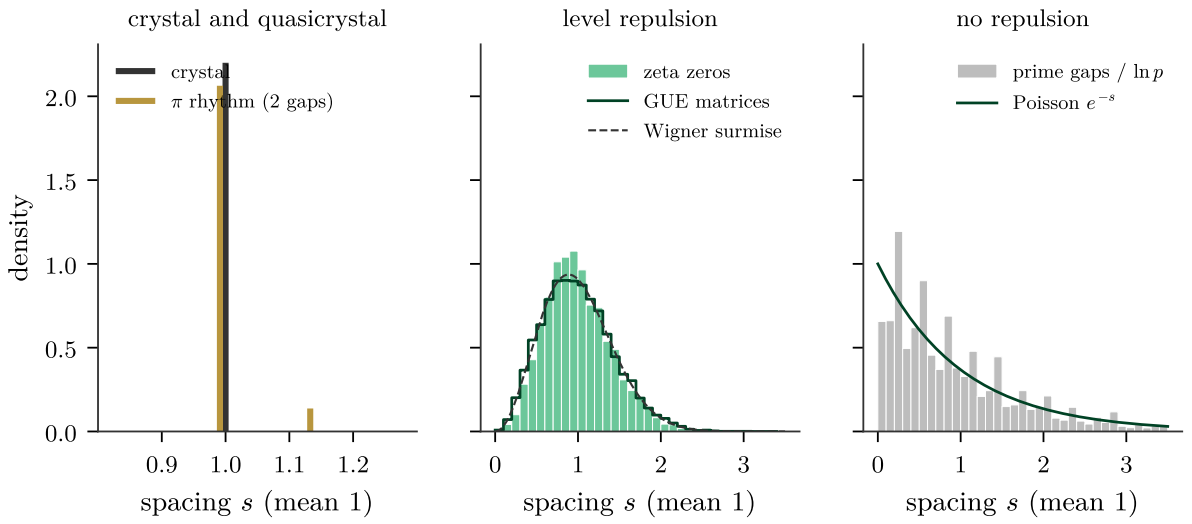


Figure 9.1. Nearest-neighbor spacing distributions, all with mean spacing 1. Left: a crystal (one spacing) and π's rhythm (two spacings, heights proportional to their frequencies; note the zoomed axis). Center: 5,000 zeta zeros (bars) and GUE eigenvalues (steps), with the Wigner surmise. Right: prime gaps divided by $\ln p$ near 10^9, with the Poisson law e^{-s}.

- *Primes:* the gaps between the 14,491 primes in $[10^9, 10^9 + 3 \cdot 10^5)$, divided by $\ln p$ and normalized to mean 1.
- *Poisson:* independent exponential spacings, the model of points with no correlation at all.

9.3 Nearest-neighbor spacings

Figure 9.1 shows the distribution of the spacings s between neighbors. The crystal has a single spacing and the quasicrystal two. The zeros and the GUE eigenvalues share a distribution that vanishes like s^2 at $s = 0$: close pairs are rare, a phenomenon called *level repulsion*. The curve drawn with them is the Wigner surmise $P(s) = \frac{32}{\pi^2} s^2 e^{-4s^2/\pi}$, a simple and very accurate approximation to the exact GUE spacing law [66]. The prime gaps, by contrast, look like the Poisson law e^{-s}, with the most likely spacing near zero.

A simple number captures the repulsion: the share of spacings below $1/4$. It is 0.009 for the zeros, 0.016 for the GUE matrices and 0.016 for the Wigner surmise, but 0.212 for the Poisson process and 0.129 for the normalized prime gaps (whose discreteness, all gaps being even, distorts the smallest bins). The Kolmogorov–Smirnov distance between the zero spacings and the GUE spacings is 0.036, against 0.305 between zeros and Poisson. The small remaining difference between zeros and GUE is expected: at the low heights we can reach, the zeros behave like random matrices of a modest effective size, a finite-height effect analyzed by Bogomolny, Bohigas, Leboeuf and Monastra [11]. Correspondingly, the variance of the zero spacings, 0.152, is a little below the GUE value 0.183.

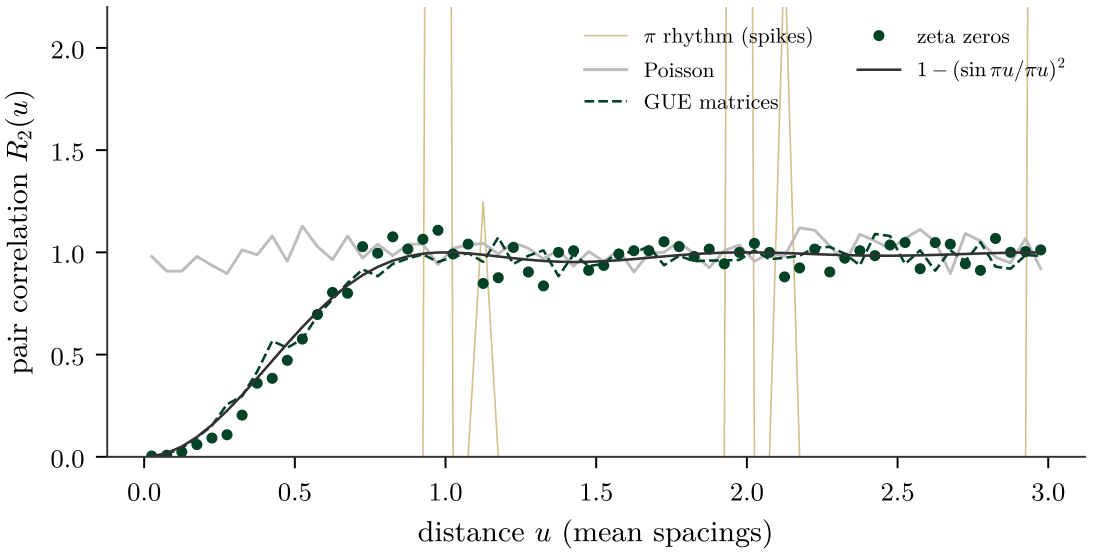


Figure 9.2. Pair correlation $R_2(u)$: 5,000 zeta zeros (dots), GUE eigenvalues (dashed), Poisson points (grey), π's rhythm (gold spikes), and Montgomery's function (9.1) (black).

9.4 Pair correlation

The pair correlation $R_2(u)$ measures how likely it is to find another point at distance u from a given point, relative to a uniform density. Montgomery's conjecture [70] is that for the unfolded zeros

$$R_2(u) = 1 - \left(\frac{\sin \pi u}{\pi u}\right)^2, \tag{9.1}$$

which is exactly the GUE pair correlation in the limit of large matrices [27, 66]. Figure 9.2 shows our estimate from the first 5,000 zeros, the GUE sample, a Poisson sample (flat at 1) and π's rhythm (sharp spikes at the sums of its two spacings). The root-mean-square difference between the zeros' histogram and (9.1) over $0 \le u \le 3$ is 0.063, comparable to the sampling noise of the histogram.

9.5 Number variance: a scale of rigidity

The most telling statistic is the *number variance* $\Sigma^2(L)$: drop a window of length L (in mean spacings) at a random place and count the points inside; $\Sigma^2(L)$ is the variance of that count. It measures how rigid the arrangement is over a range of L spacings.

- For a Poisson process, $\Sigma^2(L) = L$: the counts fluctuate as much as they can.
- For the GUE, $\Sigma^2(L) \approx \frac{1}{\pi^2}\left(\ln(2\pi L) + \gamma + 1\right)$ for large L, with γ Euler's constant [66]: the fluctuations grow only logarithmically.
- For a crystal, $\Sigma^2(L) = \{L\}(1-\{L\}) \le \frac{1}{4}$, bounded. Point sets with bounded number variance are called *hyperuniform* [94]; crystals and many quasicrystals, including Sturmian ones like π's rhythm, are of this kind.

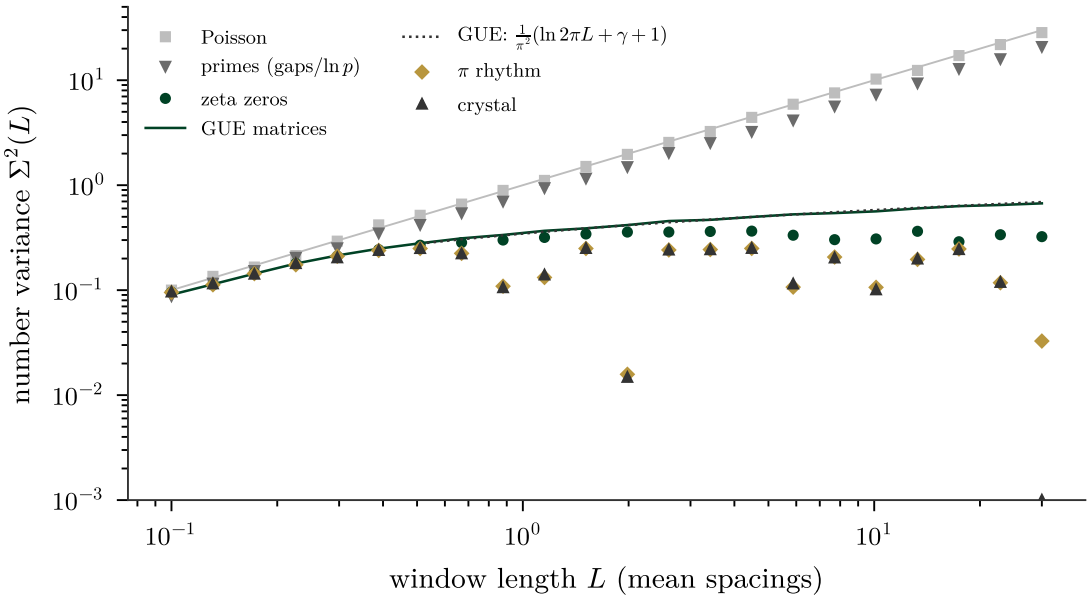


Figure 9.3. Number variance $\Sigma^2(L)$ on log–log axes for six point sets. From bottom to top: crystal and π's rhythm (bounded, with dips where L is close to a period), zeta zeros and GUE (logarithmic growth; dotted: the asymptotic GUE formula), normalized prime gaps and Poisson (linear growth).

For the zeros, Berry predicted that $\Sigma^2(L)$ follows the GUE law for small L and then saturates at a level set by the height of the zeros [9]. Figure 9.3 shows all six point sets on one logarithmic plot. At $L = 10.1$ we find $\Sigma^2 = 0.10$ for the crystal, 0.11 for π's rhythm, 0.31 for the zeros, 0.56 for the GUE sample (asymptotic formula 0.58), 7.36 for the normalized prime gaps and 10.26 for Poisson. The zeros lie below the GUE curve at this L, consistent with Berry's saturation at our low heights, while the GUE sample matches its formula.

9.6 The spectrum, and Dyson's bridge

The three statistics put the point sets in one order of decreasing rigidity:

$$\text{crystal} \;\to\; \text{quasicrystal } (\pi\text{'s rhythm}) \;\to\; \text{GUE and zeta zeros} \;\to\; \text{Poisson (and the primes at short range).}$$

The crystal has one spacing and bounded fluctuations. The rhythm of π has two spacings, at most three on the circle, and still bounded fluctuations: it is ordered without being periodic. The zeros have a continuous spacing law but repel each other and fluctuate only logarithmically. Poisson points do not interact at all, and the primes, seen through their gaps at short range, look Poisson-like, although Chapter 6 showed their long-range diffraction is far from random.

Where do the zeros belong, then? Freeman Dyson, in his lecture "Birds and Frogs," observed that if the Riemann hypothesis is true then the zeros form a one-dimensional quasicrystal in a precise sense, and suggested that classifying one-dimensional quasicrystals might lead to a proof [26]. His sense is about the Fourier transform of the zeros, not their spacings, and the explicit formula (6.2) explains it: the sum of the waves $\cos(\gamma t)$ over the zeros is concentrated at $t = \ln p^m$, the logarithms of prime powers, just as the diffraction of a quasicrystal is concentrated on a discrete set. Figure 9.4 shows this with our 5,000 zeros: the sum $\sum_\gamma \cos(\gamma t)$,

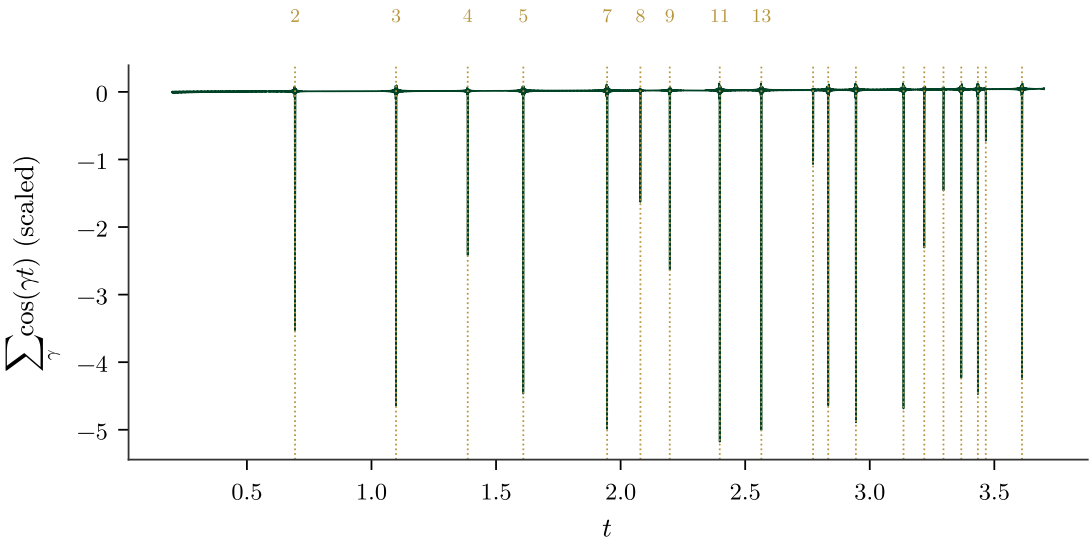


Figure 9.4. The zeros "diffract" at logarithms of prime powers: $\sum_\gamma \cos(\gamma t)$ over the first 5,000 zeros with a smooth cutoff, scaled by $1/\sqrt{\text{number of zeros}}$. Dotted gold lines mark $t = \ln n$ for the prime powers $n \le 40$; the dips sit exactly there, as the explicit formula predicts.

smoothly cut off, has sharp dips at $\ln 2$, $\ln 3$, $\ln 4$, $\ln 5$, $\ln 7$, … (depths -3.5 and -4.6 at $\ln 2$ and $\ln 3$ on the plotted scale, against a background fluctuation of about 0.01). So the zeros are GUE-like at short range and quasicrystal-like in their Fourier transform. The rhythm of π is quasicrystal-like in both. That is the bridge Dyson pointed to, and this picture organizes it; it does not cross it.

9.7 What this does and does not show

Everything in this chapter is a measurement of known phenomena with our own data, arranged so that a reader can see a crystal, π's rhythm, the zeros, random matrices and the primes on one scale. It illuminates the Montgomery–Dyson connection and makes Dyson's remark concrete. It proves nothing about the Riemann hypothesis, about the pair-correlation conjecture, or about any operator whose eigenvalues might be the zeros. Of the site's tools, the ones that help are the rhythm (as the quasicrystal end of the scale) and the diffraction picture (as the language for Dyson's remark); the others play no role.

For the video. Line up six strips of dots, from evenly spaced to random, and let the viewer guess which is which. Then show the three statistics, each with all six sets. End on Fig. 9.4: the zeros' waves spike at $\ln 2$, $\ln 3$, $\ln 5$; this is Dyson's quasicrystal remark made visible. Say clearly that this organizes the mystery and does not solve it. Instruments: "Zeta zeros & nuclei," "Quasicrystals." About 12 minutes.

CHAPTER 10

Self-similarity and fractals

10.1 Teddy Rook's picture

Teddy Rook pictured π as "a number, then a slightly smaller number after it, at least a decimal place over, then a smaller one after that, and so on." Does that idea fit with fractals? It does, and the precise version is one line long. By (2.1),

$$x = a_0 + \frac{1}{x_1}, \qquad x_1 = [a_1; a_2, a_3, \dots].$$

The "smaller number after it," the tail $1/x_1$, is built from a continued fraction made by exactly the same rule; so is its tail, and so on forever. An object that contains a copy of itself, made by the same rule, at every scale is *self-similar*, and self-similarity is the defining feature of a fractal. And each step really is smaller: by (2.7) the nth convergent misses by less than $1/q_n^2$. For π the complete quotients are $x_1, \dots, x_6 = 7.0625, 15.9966, 1.0034, 292.6346, 1.5758, 1.7367$ (Fig. 10.1).

Exact self-similarity, and self-similarity of rule. When the terms repeat, the tail is literally the number again: $\varphi = 1 + 1/\varphi$, so every complete quotient of φ is φ, and every complete quotient of $\sqrt{2} = [1; 2, 2, \dots]$ after the first is $1 + \sqrt{2}$. Every picture we draw from such a number repeats exactly under zoom. For π the terms $7, 15, 1, 292, 1, 1, 1, 2, \dots$ follow no known pattern, so the *rule* is the same at every level but the *shape* is new each time. That is not a flaw in the picture; it is one of the most interesting facts about π.

A twist: patterned fractions for π. If the numerators of a continued fraction are allowed to be numbers other than 1, π does have beautiful patterns, and then Teddy Rook's "never-ending string of continued fractions" holds with a clean rule all the way down. Brouncker found

$$\frac{4}{\pi} = 1 + \cfrac{1^2}{2 + \cfrac{3^2}{2 + \cfrac{5^2}{2 + \cdots}}} \tag{10.1}$$

in 1655, at the request of John Wallis, who published it in his *Arithmetica Infinitorum*; Khrushchev has reconstructed Brouncker's proof [50]. Lange found $\pi = 3 + 1^2/(6 + 3^2/(6 +$

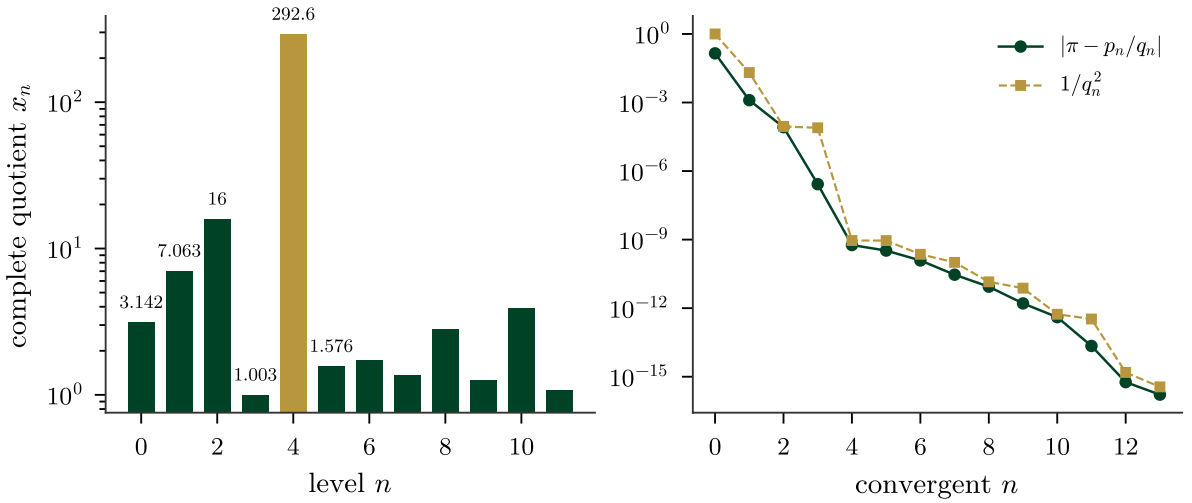


Figure 10.1. The nested tail of π. Left: the complete quotients x_n (log scale); each is the number "seen from" level n, and its whole part is the next term. The gold bar is $x_4 = 292.63$. Right: the error of each convergent (green) stays below $1/q_n^2$ (gold).

$5^2/(6+\cdots)))$ [55], and the continued fraction of the arctangent gives $\pi = 4/(1+1^2/(3+2^2/(5+3^2/(7+\cdots))))$. They converge at three very different speeds. After 10, 100 and 1000 levels Brouncker's fraction misses π by 0.0907, 0.0099 and 0.000999; its stopping points are exactly the partial sums of the Leibniz series $4(1-\frac{1}{3}+\frac{1}{5}-\cdots)$, as our script confirms with exact fractions. Lange's misses by 0.000186, $2.43e-7$ and $2.49e-10$ (its stopping points are the partial sums of Nilakantha's series). The arctangent fraction misses by 0.00013, $1.9e-8$ and $4.3e-16$ after 5, 10 and 20 levels; each level shrinks the error by a factor $0.1716 = (\sqrt{2}-1)^2$.

The rest of this chapter turns the rule "a number contains a copy of itself after every term" into pictures. Each is a classical fractal; for each we say who found it and draw it for π.

10.2 Ford circles

In 1938 Lester Ford drew, for every fraction p/q in lowest terms, a circle $C(p/q)$ tangent to the real axis at p/q with radius $1/(2q^2)$ [33].

Proposition 10.1 (Ford)**.** *Two Ford circles $C(p/q)$ and $C(r/s)$ never overlap. They are tangent if and only if $|ps-qr|=1$, that is, if p/q and r/s are neighbors in some Farey sequence; and between two tangent circles, the largest circle tangent to both is $C\big((p+r)/(q+s)\big)$.*

Proof of the first two statements. The squared distance between the centers minus the square of the sum of the radii is

$$\Big(\frac{p}{q}-\frac{r}{s}\Big)^2+\Big(\frac{1}{2q^2}-\frac{1}{2s^2}\Big)^2-\Big(\frac{1}{2q^2}+\frac{1}{2s^2}\Big)^2=\frac{(ps-qr)^2-1}{q^2s^2},$$

which is ≥ 0 for distinct fractions, with equality exactly when $|ps-qr|=1$. □

Zoom in anywhere and the same pattern of circles appears again; the picture is self-similar under the maps $x \mapsto (ax+b)/(cx+d)$ with integer entries and determinant ± 1. Ford circles turn Legendre's theorem into geometry. A vertical line through x passes through $C(p/q)$ exactly

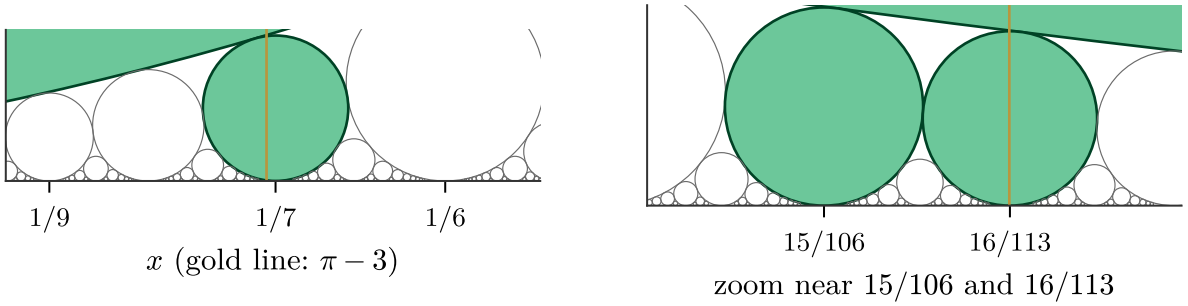


Figure 10.2. Ford circles near $\pi - 3$ (gold line), drawn to scale. Filled circles belong to convergents of $\pi - 3$. Left: the circle at $1/7$ with its neighbors. Right: a zoom by a factor of about 300, where the circles at $15/106$ and $16/113$ touch; the line passes through the second and misses the first.

when $|x - p/q| < 1/(2q^2)$, so by Theorem 2.4 every circle it pierces belongs to a convergent; and of any two consecutive convergents, at least one is pierced [48, Ch. II]. For $\pi - 3$ our script finds the pierced circles among the first convergents to be $0/1, 1/7, 16/113, 4703/33215, 14093/99532$, while $15/106$ and $4687/33102$ are convergents whose circles the line just misses (Fig. 10.2). Falling down the line toward $\pi - 3$, one drops through circle after circle, and after $16/113$ the fall is very long, because $355/113$ is so close: that is the 292 again.

10.3 The Stern–Brocot tree

Start with $0/1$ and $1/0$ and repeatedly insert the *mediant* $(p + r)/(q + s)$ between neighbors p/q and r/s. The result, described by Moritz Stern in 1858 [91] and independently by the clockmaker Achille Brocot in 1861 [14], who used it to design gear trains, is a binary tree that contains every positive fraction exactly once, in lowest terms [39, §4.5]. To locate a number, start at $1/1$ and go left (L) if the number is smaller and right (R) if larger. The path is a sequence of runs, and the run lengths are the terms of the continued fraction. For $355/113 = [3; 7, 16]$ the path is $R^3L^7R^{15}$ (the last run is one short because the path stops on the fraction itself), and for π our script finds $R^3L^7R^{15}LR^{292}LR\ldots$, with the 292 as a run of 292 Rs. The golden ratio zigzags $RLRL\ldots$ forever and $\sqrt{2}$ goes $R, LL, RR, LL, \ldots$. Every subtree is a copy of the whole tree with the fractions changed, which is self-similarity again (Fig. 10.3).

10.4 Minkowski's question-mark function

In 1904 Hermann Minkowski introduced a function that turns continued fractions into binary expansions [67]. For $x = [0; a_1, a_2, a_3, \ldots]$,

$$?(x) = 2\left(2^{-a_1} - 2^{-(a_1+a_2)} + 2^{-(a_1+a_2+a_3)} - \cdots\right), \tag{10.2}$$

which in binary is $a_1 - 1$ zeros, then a_2 ones, then a_3 zeros, and so on: each term becomes one run of binary digits. The function is continuous and strictly increasing from $?(0) = 0$ to $?(1) = 1$, maps rationals to dyadic rationals and quadratic irrationals to rationals, and yet its derivative is zero almost everywhere: it is a "singular" function, all of whose increase happens on a set of measure zero [23, 87]. It satisfies $?(x/(1 + x)) = ?(x)/2$, so the left half of

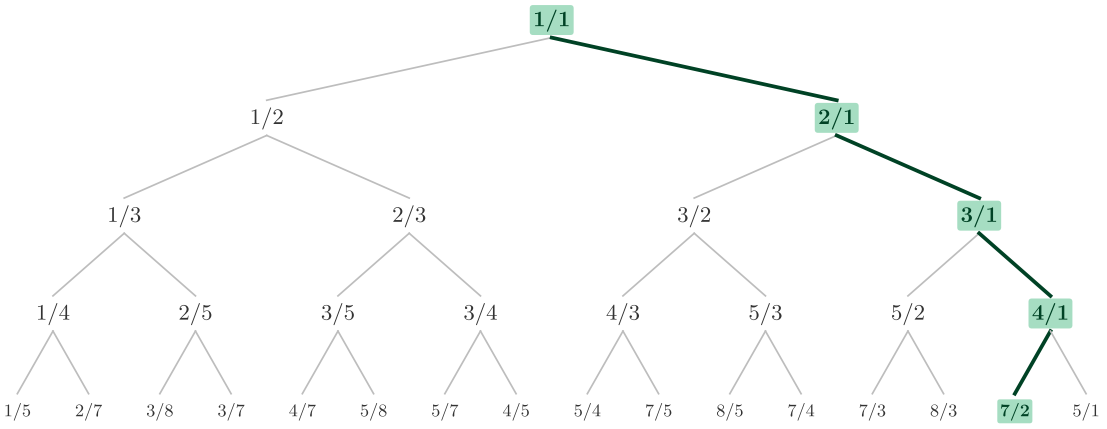


Figure 10.3. The first five levels of the Stern–Brocot tree, with the beginning $RRRL$ of the path to π highlighted: $1/1 \to 2/1 \to 3/1 \to 4/1 \to 7/2$. The path continues with six more Ls to $22/7$.

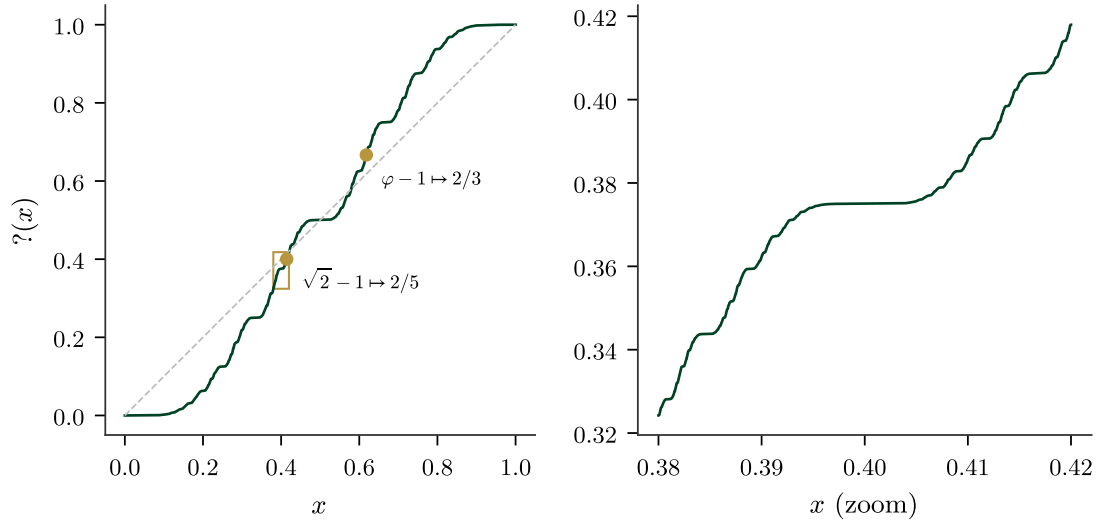


Figure 10.4. Minkowski's $?(x)$. Left: the whole graph with the diagonal dashed; quadratic irrationals land on fractions. Right: the boxed region magnified, showing flat stretches and bursts of growth at every scale.

its graph is the whole graph squashed by a factor 2. Our script confirms $?(\varphi - 1) = 2/3$ and $?(\sqrt{2} - 1) = 2/5$, and finds $?(\pi - 3) = 0.01562476\ldots$, just under $1/64$: six zeros, then fifteen ones (Fig. 10.4).

10.5 The devil's staircase

Return to the circle map (4.1) of Chapter 4, at full coupling $K = 1$. Plot the rotation number W against Ω: the result is a staircase with a flat step at every fraction (Fig. 10.5). At $K = 1$ the steps fill the whole interval except a Cantor set of measure zero, whose fractal dimension

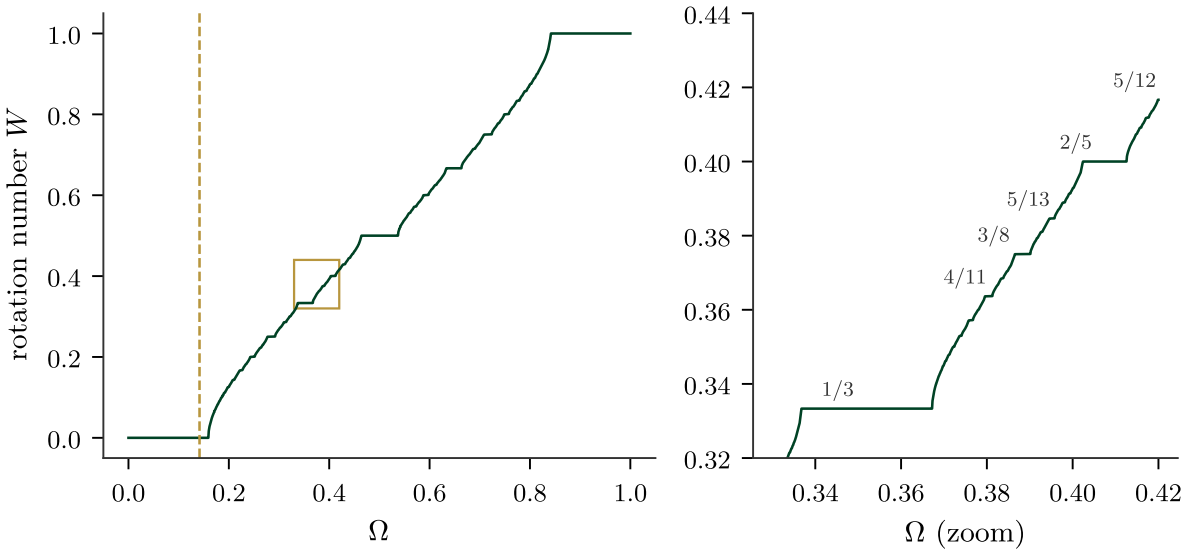


Figure 10.5. The devil's staircase: rotation number W of the sine circle map at $K = 1$ against Ω (2,520 iterations per point). The dashed line is $\Omega = \pi - 3$, inside the wide 0/1 step. Right: the boxed region, with steps labeled by their fractions; between 1/3 and 2/5 the widest step is 3/8, between 3/8 and 2/5 it is 5/13.

Jensen, Bak and Bohr measured as about 0.87 [47]. Between two adjacent steps p/q and r/s the widest step is the one at the mediant $(p+r)/(q+s)$, the Stern–Brocot rule once more, so zooming between two steps reproduces the whole staircase in miniature.

10.6 Cantor dust: numbers with small terms

Which numbers in $(0, 1)$ have continued fractions that use only the terms 1 and 2? Fixing the first term to be 1 or 2 keeps two intervals; fixing the second keeps two sub-intervals of each, and so on. What survives forever is a Cantor-like "dust" E_2 (Fig. 10.6). It has length zero, but more than a point's worth of size: its Hausdorff dimension is $\dim_H E_2 = 0.5312805062772051\ldots$. I. J. Good bounded it between 0.5306 and 0.5320 in 1941 [38]; Jenkinson and Pollicott computed it to 25 digits with a transfer-operator method [45] and later to 100 rigorously certified digits [46].

The method is elegant. The dimension of the set E_A of numbers whose terms all lie in a finite set A is the unique s for which the transfer operator

$$(\mathcal{L}_s f)(x) = \sum_{a \in A} (a + x)^{-2s} f\left(\frac{1}{a + x}\right)$$

has spectral radius 1. We discretize $\mathcal{L}_s$ on Chebyshev nodes and bisect on s (Appendix B). With 60 nodes our script gives $\dim_H E_{\{1,2\}} = 0.531280506277$, agreeing with the published value in all twelve printed digits. It also gives $\dim_H E_{\{1,2,3\}} = 0.70566$, $\dim_H E_{\{1,2,3,4\}} = 0.78895$, $\dim_H E_{\{1,\ldots,6\}} = 0.86762$ and $\dim_H E_{\{2,3\}} = 0.33744$. Since $\pi - 3$ starts with a 7 and contains a 292, it lies in none of these small-term dusts; the golden ratio's $\varphi - 1$ (all 1s) and $\sqrt{2} - 1$ (all 2s) are single points of $E_{\{1,2\}}$.

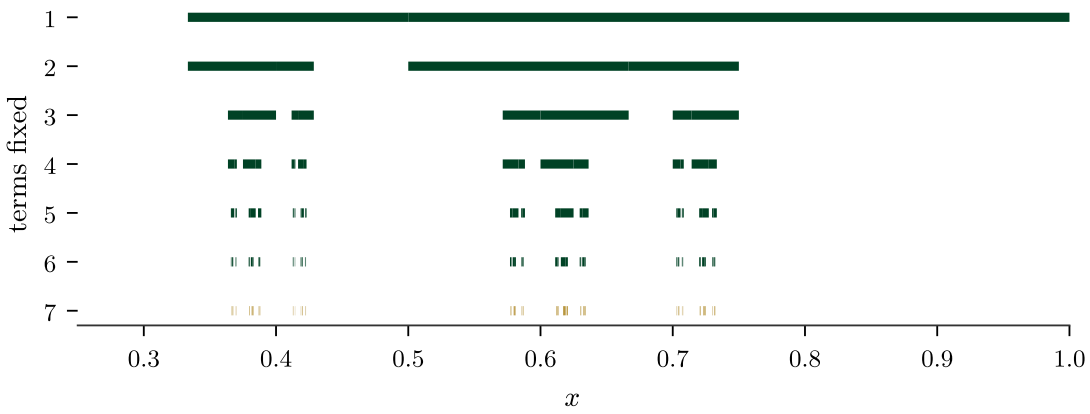


Figure 10.6. The numbers whose first k terms are all 1 or 2, for $k = 1, \ldots, 7$. Each row keeps two pieces of each piece of the row above; the limit is a Cantor set of dimension 0.5313.

10.7 The Mandelbrot set's Farey bulbs

The Mandelbrot set consists of the complex c for which $z \mapsto z^2 + c$, started at $z = 0$, stays bounded. Its heart-shaped main body is fringed with bulbs, and each bulb carries a fraction: the bulb attached at the point $c = e^{2\pi i p/q}/2 - e^{4\pi i p/q}/4$ of the main cardioid is the p/q bulb, and inside it the orbit settles on an attracting cycle of period q. Devaney explained how these fractions are arranged [24]: between the p/q and r/s bulbs, the largest bulb (measured by the share of the outside that leads into it) is the $(p + r)/(q + s)$ bulb, the mediant once more, which puts the Fibonacci fractions $1/2, 1/3, 2/5, 3/8, 5/13, \ldots$ on a path toward the golden-mean point. Our script places a test point just inside each bulb with $q \le 12$ and finds an attracting cycle of period exactly q in every case (0 exceptions). Figure 10.7 labels the larger bulbs. Walking $\pi - 3 = 0.14159$ of the way around the cardioid lands just short of the $1/7$ bulb, since $\pi - 3 < 1/7$, and π's convergents $15/106$ and $16/113$ are bulbs much smaller and much closer to that spot.

10.8 The Fibonacci word fractal

In 2009 Alexis Monnerot-Dumaine drew the Fibonacci word as a path [69]: take one unit step for each letter, and after the kth letter, if it is a 0, turn a quarter turn, left if k is even and right if k is odd. The Fibonacci word $0100101001001\ldots$ is exactly the rhythm of $\varphi - 1$ from Chapter 3 with hits and rests exchanged (our script checks this over 30,000 letters), so this is the golden rhythm, walked. The path is a fractal curve made of smaller copies of itself, of dimension $3 \ln \varphi / \ln(1 + \sqrt{2}) = 1.6379$ [69]. Figure 10.8 draws the same rule for three numbers: the golden rhythm makes the fractal, $\sqrt{2} - 1$ weaves a grid, and $e - 2$ draws a zigzag of growing arms. For $\pi - 3$ the turns nearly cancel within each 7- or 8-beat gap, and over the first 33,102 beats the path is a long, thin zigzag.

10.9 Six numbers, six pictures

The Continued-Fraction Suite of Chapter 11 uses five numbers. One construction draws them all on the same footing: cut a $1 \times x$ rectangle into as many squares as fit (that count is the first

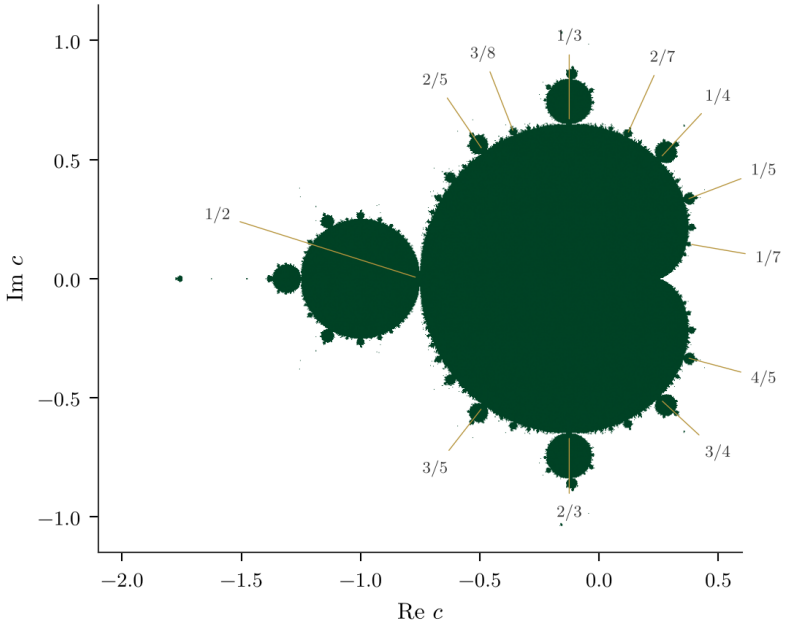


Figure 10.7. The Mandelbrot set with its main bulbs labeled p/q by the point where they attach to the cardioid. Inside the p/q bulb the orbit of 0 settles on a cycle of q points.

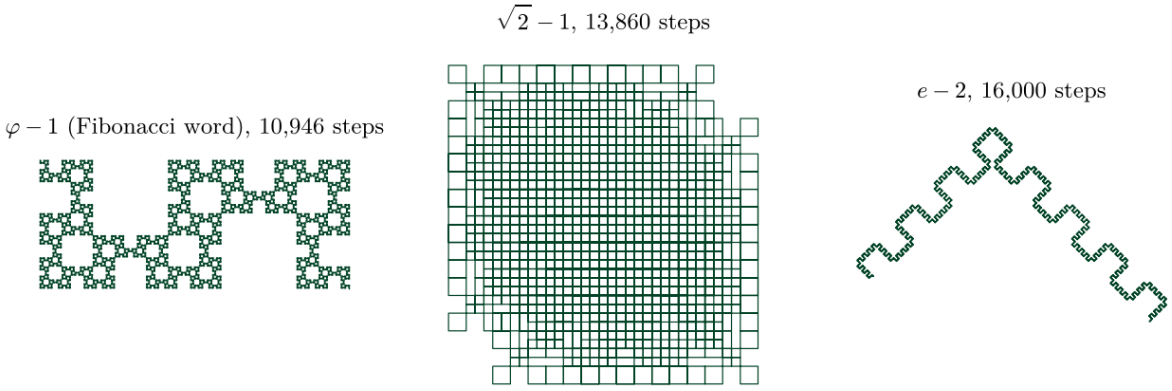


Figure 10.8. Rhythms walked as paths with Monnerot-Dumaine's rule. Left: the golden rhythm, the Fibonacci word fractal. Center: $\sqrt{2} - 1$. Right: $e - 2$.

term), turn the leftover a quarter turn, and repeat. Figure 10.9 shows the first six levels of each number, each level's rectangle stretched to the same height so that thin levels stay visible. The golden ratio cuts one square at every level and the silver ratio two: exactly self-similar. The fifth and e change their counts, e in its stretching pattern $1, 2, 1, 1, 4, 1, 1, 6$. Pi cuts 7, then 15, then 1, then 292 squares, the last a dense gold band. The sixth panel zooms to level 4 of π, the 292, which is where the sixth movement lives.

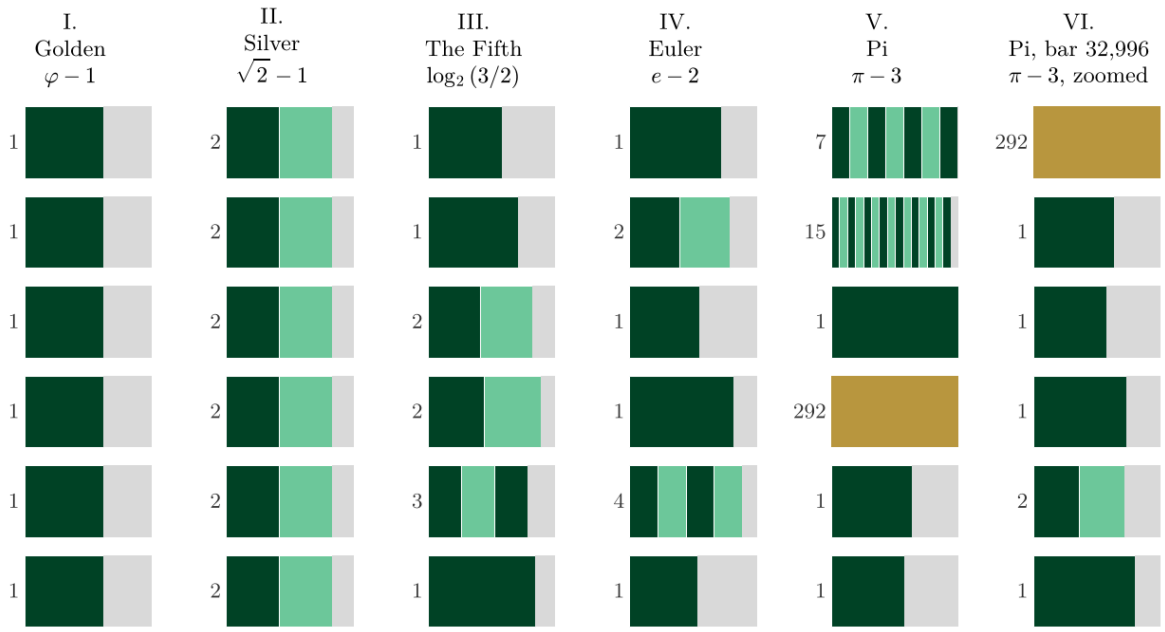


Figure 10.9. The Suite's numbers as square-cutting ladders. Each row is one level: the number of squares cut (printed at the left) is the term, and the grey sliver is what passes to the next level. Rows are stretched to equal height. A count above 50 is drawn as a gold band.

10.10 Is pi typical?

The Gauss map (2.2) reads off the terms, and the Gauss–Kuzmin law (Theorem 2.6) says what a typical number's terms look like. Here is a formal test of π's first 10,000 terms against it. Group the terms as $1, 2, \ldots, 12$ and 13+; the expected count in each group is 10,000 times its Gauss–Kuzmin probability. The chi-square statistic is $\chi^2 = 13.5$ on 12 degrees of freedom, with $p = 0.33$. With the finer grouping $1, \ldots, 20$ and 21+, $\chi^2 = 25.3$ on 20 degrees of freedom, $p = 0.19$. Either way the counts are fully consistent with the law (Fig. 10.10): 4,206 ones (42.1%, law 41.5%), 1,672 twos and 882 threes. The geometric mean of the terms is 2.6831 after 100 terms, 2.6627 after 1,000 and 2.6632 after 10,000, wandering around Khinchin's $K_0 = 2.685452001$ as a random number's would; and $\ln q_{10000}/10000 = 1.17941$ against Lévy's 1.18657 (Fig. 10.11). The largest term so far is 20,776, at position 431.

Nobody has proved that π obeys these laws, or even that its terms are unbounded, but by every count we can make it behaves like a typical number. This fits the null result of the prime hunt (Section 7.3) and the typical behavior of its prime convergents (Chapter 8). The number e, by contrast, fails the test on purpose: its terms $1, 2, 1, 1, 4, 1, 1, 6, \ldots$ have far too many ones and a geometric mean that grows without bound.

For the video. Begin with Teddy Rook's sentence and the one-line formula. Then six short visits: falling through Ford circles to π; walking the Stern–Brocot tree as L and R; the question-mark function heard as pitch; zooming into the devil's staircase; the Cantor dust building row by row; the Fibonacci word fractal drawing itself. Finish with Fig. 10.9 and the chi-square verdict: π looks typical. Instruments: all of Chapter VII on the fractals page. About 14 minutes.

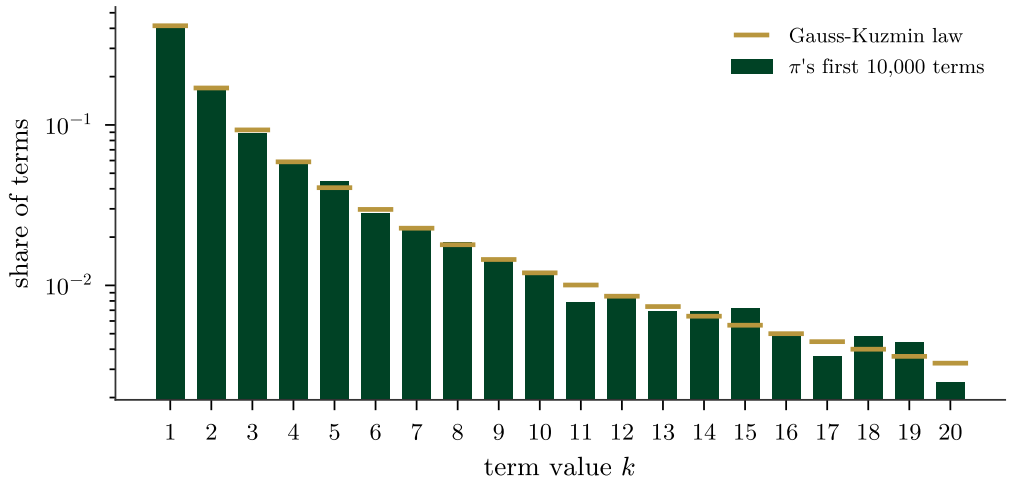


Figure 10.10. The shares of the values $1, \ldots, 20$ among π's first 10,000 terms (bars) against the Gauss–Kuzmin law (gold).

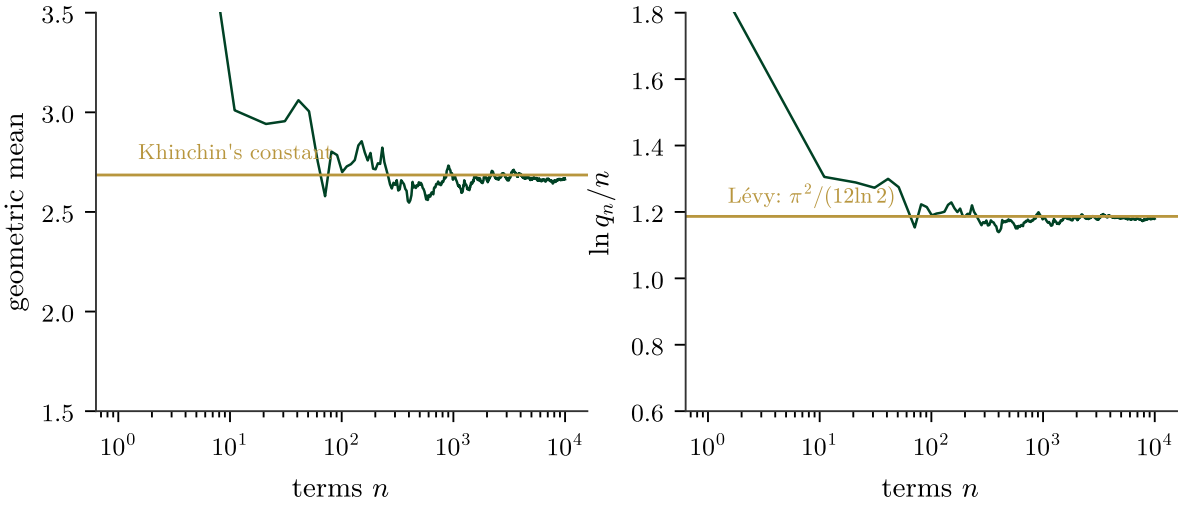


Figure 10.11. Running geometric mean of π's terms against Khinchin's constant (left) and $\ln q_n/n$ against Lévy's constant $\pi^2/(12 \ln 2)$ (right).

CHAPTER 11

Composition: the Continued-Fraction Suite as a method

11.1 The work

The *Continued-Fraction Suite* is an original composition by Teddy Rook: six short pieces for piano, one number each, engraved with LilyPond and published as a free score with a recorded performance [86]. It grew out of the Composer and "Compose a piece" instruments of the companion site, which play a number's rhythm with one voice per level. Whatever the prior art on its method (Chapter 13), the Suite is a new artwork. This chapter describes it as a method, in words and form diagrams; it does not reproduce the score. The musical facts below (keys, meters, tempos, bar counts, rehearsal marks) were read from the score by the site's builders and recorded in the site's data, and our script checks every bar count against the continued fractions.

11.2 The method

Each movement applies the same five decisions to its number.

1. *Choose the number and its levels.* Compute the terms $a_1, a_2, \ldots$ and the block lengths q_n (Theorem 3.5).

2. *Choose which level is a bar.* The block lengths then become lengths in bars of phrases, sections and larger units. Levels that are too short to be phrases are passed over: Golden starts its phrases at 3 and 5 bars, Silver at 2 and 5, Pi at 7.

3. *Lay out the form as the standard word.* At each level the long block appears a_{n+1} times and is closed by one short block. The score marks the long unit A, the short one B, and a unit that the number brings in only rarely as a "rare block."

4. *Put the rhythm inside each bar.* The left hand carries accents on the eighth notes where the number's own beat word has a hit (with α replaced by $1 - \alpha$ when $\alpha > \frac{1}{2}$, so the accents stay sparse). So the same number shapes the music at the smallest scale and the largest.

Table 11.1. The six movements of the Continued-Fraction Suite. "Bars" is the length of the movement as printed; Movement VI begins at bar 32,968.

movement	number	key	meter, tempo	bars
I. Golden	$\varphi - 1 = [0; 1, 1, 1, \dots]$	G major	3/4, Andante cantabile	89
II. Silver	$\sqrt{2} - 1 = [0; 2, 2, 2, \dots]$	D major	4/4, Adagio	70
III. The Fifth	$\log_2 \frac{3}{2} = [0; 1, 1, 2, 2, 3, \dots]$	F major	4/4, Allegretto	53
IV. Euler	$e - 2 = [0; 1, 2, 1, 1, 4, \dots]$	A minor	3/4, Tempo di valzer	71
V. Pi	$\pi - 3 = [0; 7, 15, 1, 292, \dots]$	C major	4/4, Moderato	113
VI. Pi, bar 32,996	$\pi - 3$, at level 4	C major	4/4, Moderato	50

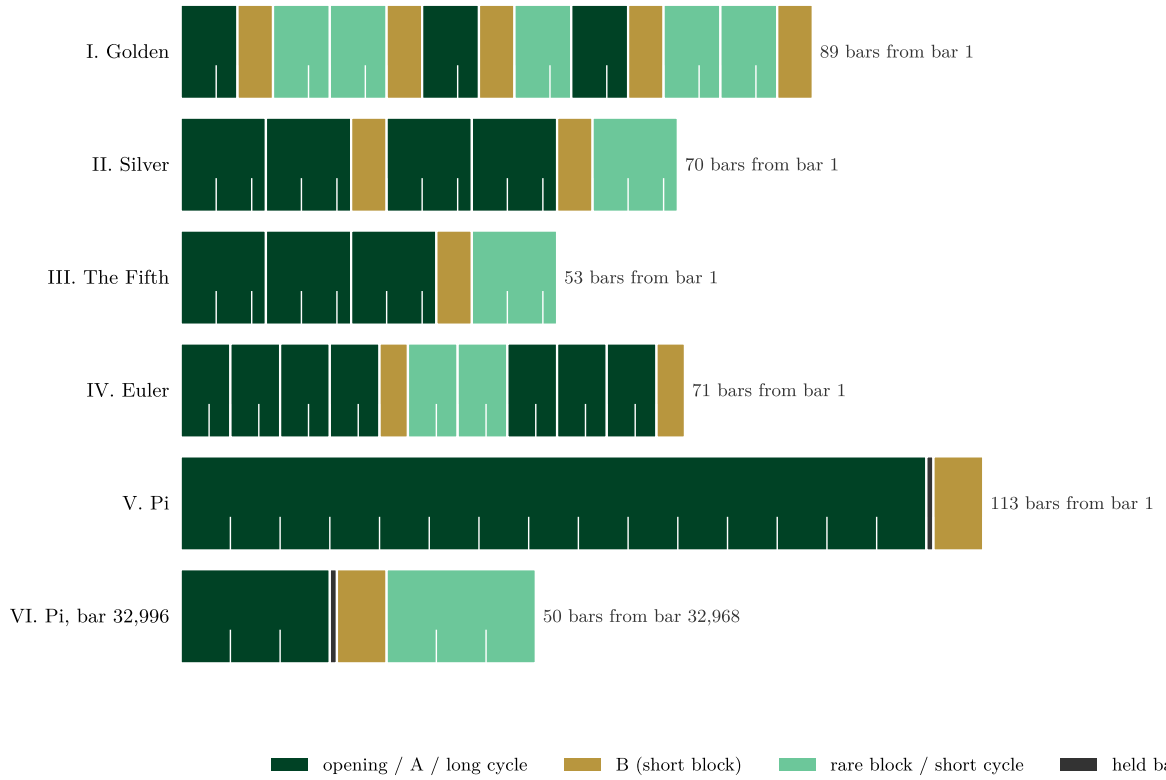


Figure 11.1. The forms of the six movements, drawn to scale (one unit per bar) from the section list read from the score. Colors follow the rehearsal marks: green for the opening and A sections (and Movement VI's long cycle), gold for B, mint for rare blocks and the short cycle, dark grey for a held bar. White ticks mark phrase boundaries.

5. *Let the melody walk the digits.* The score's preface says that the melody follows the number's decimal digits; this is a free compositional choice and carries no structural weight in what follows.

Figure 11.1 draws the six forms to scale, and the rest of this section reads each one.

I. Golden. Every term of $\varphi - 1$ is 1, so each block is the two blocks before it joined: the Fibonacci rule. Phrases are 5 and 3 bars, sections are $8 = 5 + 3$ and 5 bars, and the sections follow the Fibonacci word: $8, 5, 8, 8, 5, 8, 5, 8, 8, 5, 8, 8, 5$, which is eight 8s and five 5s, $64 + 25 =$

89 bars, itself a Fibonacci number. Our script checks that this sequence is the golden rhythm, letter for letter. Because every term is 1, no section ever comes three times in a row.

Golden also shows a subtlety of labeling that we call *two-level labels*. In the other movements the short block is truly rare. Here every level is "one long block, then one short block," and the sizes slide up the Fibonacci numbers: an 8-bar unit is the *long* block at the level of sections (8 against 5) and the *short* block at the level above ($13 = 8 + 5$ against 8). The score names each 8-bar unit by whichever level's boundary falls there, so the same 8 bars of music are marked A in some places and "rare block" in others. That is the self-similarity of the golden ratio, written into rehearsal marks.

II. Silver. Every term of $\sqrt{2} - 1$ is 2: "two of these, then one of those." Phrases are 5 and 2 bars, sections $12 = 5 + 5 + 2$ and 5 bars, and the whole is $29 + 29 + 12 = 70$ bars, where $29 = 12 + 12 + 5$. The numbers $2, 5, 12, 29, 70$ are the Pell numbers, the denominators of $\sqrt{2}$.

III. The Fifth. The terms of $\log_2(3/2)$ change: $1, 1, 2, 2, 3, 1, 5, \ldots$ Phrases are 5 and 2 bars and sections $12 = 5{+}5{+}2$; then the 3 takes over, three 12-bar sections and a 5 make 41, and one more 12 makes 53. The section sequence $12, 12, 12, 5, 12$ builds the denominators $5, 12, 41, 53$ of the best fractions for the fifth (Chapter 4) directly into the form.

IV. Euler. The terms of $e - 2$ grow in a pattern: $1, 2, 1, 1, 4, 1, 1, 6, \ldots$ Phrases are 4 and 3 bars, sections $7 = 4 + 3$ and 4; the 4 in the continued fraction makes a 32-bar block of four 7s and a 4; then $39 = 32 + 7$ and $71 = 39 + 32$. The denominators $7, 32, 39, 71$ are those of the convergents 5/7, 23/32, 28/39, 51/71 of $e - 2$.

V. Pi. The first three terms of $\pi - 3$ build a 113-bar cycle: fifteen 7-bar phrases, one held bar, and a last 7-bar phrase marked B, so $15 \times 7 + 1 + 7 = 113$. The held bar is the "one extra rest" of the standard word $s_2 = s_1^{15} s_0$, and the B phrase is the closing s_1 of $s_3 = s_2 s_1$.

VI. Pi, bar 32,996. The fourth term, 292, says that the 113-bar cycle repeats 292 times before the first short cycle of 106 bars. If Movement V never stopped, that moment would come after $292 \times 113 = 32{,}996$ bars, about 22.9 hours at the marked tempo (2.5 seconds per bar). The last movement jumps there. It plays bars 32,968 to 33,017: the end of the 292nd cycle (three phrases, the held bar and the B phrase, here in F major) and the opening three phrases of the short cycle, which begins at the printed bar 32,997. A short cycle is a long one without its closing phrase, so on paper it starts exactly like a long one; the composer marks the moment instead with a change of key and a hush. Section 3.4 explains why "beat 32,996" on the site and "bar 32,997" in the score name the same moment.

11.3 What the method gives a composer

The method gives proportions without dictating content. A number fixes how long each unit is and when contrast arrives; the composer still chooses every note. Quadratic irrationals give forms that repeat exactly at every level (Golden, Silver); numbers with changing terms give forms in which each level follows a different rule (The Fifth, Euler); and a number with one huge term gives a form with a surprise that is postponed almost indefinitely (Pi). In the language of Chapter 10, the first kind is self-similar in shape and the others only in rule.

For the video. Play eight bars of each movement while its strip from Fig. 11.1 scrolls underneath with a playhead. Pause on Golden's two-level labels. For Movement VI, show a counter running from bar 1 to 32,996 at high speed, then cut to the real music at bar 32,968. Instruments: the Composer's "Jump to the rarest block," "Compose a piece," and the six piano lessons in the Music section. About 10 minutes.

CHAPTER 12

Connections: music, patterns, math and chemistry

The Rhythm of Numbers is the deep dive of a larger page on the companion site, a four-way Venn diagram of music, patterns, mathematics and chemistry whose fifteen regions each show a place where the subjects meet [84]. This chapter places the work of the earlier chapters in that diagram, briefly and only with claims we have checked.

12.1 Music and math: ratios, clocks and rhythms

This region is the home of most of this work. Simple frequency ratios (2:1, 3:2, 4:3) sound smooth together; the twelve notes form a clock in which the step of a fifth, 7, is coprime to 12 and so visits every note; and the comma that keeps the clock from closing exactly is the irrationality of $\log_2(3/2)$ (Chapter 4). Euclidean rhythms, the drum machine and the Suite belong here too.

12.2 Music and chemistry: octaves of elements

In 1865 John Newlands arranged the known elements by atomic weight and noticed that chemically similar elements recurred every eighth place, which he compared with the octave of a musical scale and called the law of octaves [61]. He presented the arrangement to the Chemical Society of London on March 1, 1866, which declined to publish it [61]; the periodic table of Mendeleev, with gaps for undiscovered elements, followed in 1869. His "octave" was a way of counting, not a sound. Molecules, on the other hand, do have characteristic vibration frequencies, far above hearing; infrared spectra are their fingerprints.

12.3 Music and patterns: sand that draws sound

In 1787 Ernst Chladni bowed sand-covered metal plates and recorded the patterns the sand made when the plate rang [20]. The sand collects on the nodal lines of a standing wave. For a square plate a classical approximation to the modes is

$$z(x,y) = \cos(n\pi x)\cos(m\pi y) \pm \cos(m\pi x)\cos(n\pi y), \qquad 0 \le x, y \le 1,$$

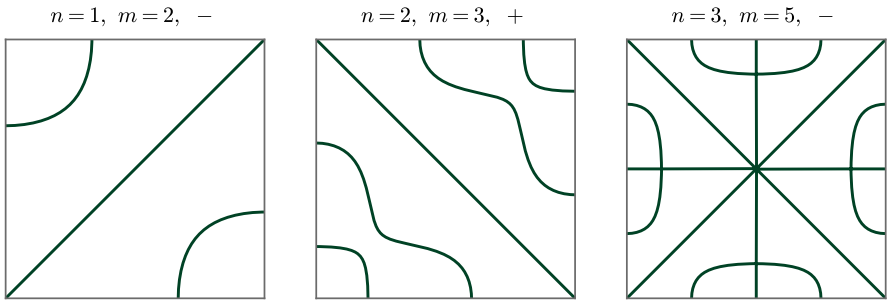


Figure 12.1. Chladni figures: the nodal curves $z = 0$ of the approximate square-plate modes for three choices of n, m and the sign.

with the sand on the curve $z = 0$ (Fig. 12.1). Real plates differ in detail near the edges. Melodies have symmetries of the same kind: transposition, inversion and retrograde are the translations and reflections of a pitch–time grid.

12.4 Math and patterns, patterns and chemistry

Periodic patterns in the plane and in space are classified by their symmetry groups, and crystals by the symmetries of their unit cells. Quasicrystals (Chapter 6) do not fit that classification: they are ordered, with sharp diffraction, but have no repeating cell [88]. On the site, the stained glass of cathedral windows shows the chemistry of color: dissolved metal ions absorb some wavelengths, and tiny metal particles scatter others.

12.5 Math and chemistry: why 2, 8, 18, 32

The electron shell n of an atom has subshells $l = 0, \dots, n-1$, each with $2l+1$ orbitals holding two electrons apiece. Since $1 + 3 + \cdots + (2n-1) = n^2$, shell n holds $2n^2$ electrons: $2, 8, 18, 32$. The squares here are the same squares whose reciprocals sum to $\pi^2/6$ in Chapter 5; as there, this is a shared formula, not shared physics.

12.6 All four: waves and symmetry

At the center of the diagram is the standing wave. A string's modes are whole-number multiples of its lowest frequency, which is why it has a clear pitch. A circular drum's modes are not: their frequencies are proportional to zeros of Bessel functions, and the lowest four are in the ratios $1 : 1.59 : 2.14 : 2.30$, which is why a drum thumps rather than sings. An electron in an atom is a standing wave in three dimensions; its orbitals sort by their number of nodal surfaces just as drum modes sort by nodal lines, and their counts, filled in order, give the shape of the periodic table.

12.7 Where the Rhythm of Numbers sits

The Rhythm of Numbers lives mainly in the music–math region and reaches into the others: tuning and rhythm (music and math), coprimality and diffraction (math and patterns, patterns

and chemistry), the $1/n^2$ of hydrogen and the n^2 of shells (math and chemistry), and self-similarity, which is the subject of patterns itself. What the diagram adds is a reminder that the same few structures (ratios, standing waves, symmetry, and the continued fraction that measures how far a ratio is from simple) appear in each subject for different reasons.

For the video. Show the four-way diagram, then light up each region for a few seconds with its one picture: the circle of fifths, Newlands' table, a Chladni plate, a quasicrystal pattern, the $2n^2$ squares, a drum and an orbital. Then zoom into the music–math region and say "this is where the rest of the film lives." Instruments: the Connections hub and its region pages. About 6 minutes.

CHAPTER 13

What is new here, and what is re-found

A work that walks through this much classical mathematics owes its readers a careful account of what is old. This chapter audits every idea in the work and in the companion site against the literature, and sorts it into four classes:

Classical: a standard result with a long history; we cite a primary source or a standard text.

Specialist: known in a research literature, though not in standard texts; we cite it.

New presentation: a new arrangement, picture, synthesis or computation of known results.

New: something we did not find anywhere: an original artifact, a computation, or a heuristic that we could not locate in the literature we searched.

The searches covered the standard texts cited throughout, the music-theory literature on maximally even sets, well-formed scales, Christoffel words and Sturmian rhythms (searching for, among others, "Sturmian words music," "continued fractions musical form," "Christoffel words rhythm"), the OEIS, and the number-theory literature on convergent denominators. No search of this kind is complete. When we say "we did not find prior work," we mean exactly that, and nothing stronger.

13.1 The audit

Table 13.1. The prior-art audit, idea by idea.

idea	class	notes and sources
The guess $\pi \approx 3+k/(7k+1)$ and $k = 16$	classical	The guess is the continued fraction of π stopped after two terms (Chapter 1); 355/113 is Zu Chongzhi's Milü [65]; 223/71 and 22/7 are Archimedes' bounds. Re-found independently by Teddy Rook.
Continued fractions, convergents, best approximation, Legendre, Lagrange	classical	Hardy and Wright [42], Khinchin [48].

idea	class	notes and sources
Gauss–Kuzmin, Khinchin, Lévy	classical	Khintchine [49], Kuzmin [53], Lévy [59, 60].
The "rhythm of a number"	classical	It is the characteristic Sturmian word of α, the cutting sequence of a line of slope α [2, 10, 63]. The rhythm of π in particular is the characteristic word of $\pi - 3$; we found no claim that it is new and make none.
Nested blocks: block lengths are the q_n	classical	Standard words; Brown [15], Fraenkel et al. [34]; our proof in Chapter 3 is a direct version for completeness.
The beats 32,996 and 33,102	new presentation	Elementary consequences of the standard word $s_4 = s_3^{292} s_2$; computing and reconciling the two counts is ours.
The continued fraction as musical *form* (bar, phrase, section, movement)	specialist / new presentation	Closely related work exists. Carey and Clampitt studied self-similar pitch structures and their rhythmic analogues [18, 19]; Clampitt and Noll use Sturmian morphisms and Christoffel words for modes [21, 25]; Callender built rhythmic canons from Sturmian words whose voices and relations are set by the continued fraction of the slope [16, 17]; Fibonacci proportions in form have a long analytic literature [58]. We did not find the specific reading of a number's *whole* continued fraction as a nested hierarchy of formal units, each term a repeat count with a closing contrasting unit, used to lay out complete pieces.
The Composer and "Compose a piece" instruments	new (artifact)	Software on the companion site.
Euclid's algorithm and Euclidean rhythms	specialist	Clough and Douthett [22], Gómez-Martín et al. [37], Toussaint [96]; Bjorklund's 2003 algorithm as reported by Toussaint. That $E(16, 113)$ is π's level-3 block is an observation that follows at once.
Tuning from $\log_2(3/2)$, the comma, 41 and 53	classical	Benson [7].
Circle map, Arnold tongues, devil's staircase	classical	Jensen et al. [47]; Huygens's clocks [6].
$6/\pi^2$, the Euler product, $\zeta(2)$	classical	Euler [30, 31], Hardy and Wright [42].
Correlation of pulse trains, $(\gcd - 1)/\sqrt{(a-1)(b-1)}$	new presentation	An elementary computation. We did not find it stated in this form, but expect it is known; the "orchard as correlation matrix" reading is a presentation, not a theorem.

idea	class	notes and sources
Hydrogen's $1/n^2$ and $\pi^2/6$	classical, with a caveat	Balmer [5]; a shared formula, not shared physics (Chapter 5).
Residue bias of consecutive primes	specialist	Lemke Oliver and Soundararajan [57]; our counts re-done.
The prime sunflower of π	new presentation	Dirichlet's theorem and wheel sieves are classical [42]; seeding the $\pi-3$ spiral with primes is a picture of them.
Diffraction of π's rhythm and of the primes	specialist	Sturmian diffraction [4]; primes [95]. The Ramanujan-sum explanation of the square-free rule is standard reasoning; our computation re-done.
Explicit formula and zeros	classical	Riemann [83], von Mangoldt [98].
Continued-fraction factoring	classical	Lehmer and Powers [56], Morrison and Brillhart [72].
Mersenne necklaces and Fermat's little theorem	classical	Golomb [36].
The prime hunt in rhythm prefixes	new (experiment)	We did not find prior work testing binary prefixes of Sturmian words for primality. The outcome is a null result (Chapter 7).
The prime-convergent heuristic with $c = 2\ln 2$	new (heuristic)	Built entirely from known theorems [44, 60, 68, 93]. We did not find the constant or the identification of the correction factor as $\zeta(2)$ in the literature we searched. Related heuristics for linear recurrences: Grantham and Granville [40].
The random-real test of the heuristic, and π's placement	new (computation)	Our computation (Chapter 8); the lists for π match the OEIS [77, 78].
The spacing spectrum (crystal $\to$ quasicrystal $\to$ GUE $\to$ Poisson)	new presentation	Every ingredient is known: Berry [9], Bogomolny et al. [11], Dyson [27], Mehta [66], Montgomery [70], Odlyzko [73]; ordering point patterns by number variance is the hyperuniformity program of Torquato and Stillinger [94]; Dyson's quasicrystal remark is in Dyson [26]. Putting π's rhythm, the zeros, GUE and the primes on one scale with one set of tools is our synthesis.
Teddy Rook's picture of the nested tail	classical	The self-similarity of continued fractions; Chapter 10.
Patterned continued fractions for π	classical	Brouncker [50]; Lange [55].

idea	class	notes and sources
Ford circles, Stern–Brocot, $?(x)$, Cantor dust, Farey bulbs, Fibonacci word fractal	classical / specialist	Brocot [14], Denjoy [23], Devaney [24], Ford [33], Good [38], Jenkinson and Pollicott [45, 46], Minkowski [67], Monnerot-Dumaine [69], Salem [87], Stern [91]. Our dimension computation re-does the published method.
The six-numbers pictures	new presentation	Square cutting is a classical picture of Euclid's algorithm; applying one construction to the Suite's numbers is ours.
Gauss–Kuzmin test of π's first 10,000 terms	new presentation	Statistical checks of π's terms are a known kind of computation; the specific statistics are ours.
The Continued-Fraction Suite	new (artwork)	An original composition by Teddy Rook [86].
Golden's two-level labels	new presentation	An observation about labeling form in a self-similar piece.
Twelve million certified terms of π and calibrated tests of the Gauss–Kuzmin, Khinchin, Lévy and Galambos laws	new presentation (computation)	Far longer expansions exist [74, 99]; our certified divide-and-conquer expansion, the simulation-calibrated p-values and the pair and block tests are ours (Chapter 15).
Prime convergents of π to $n = 20{,}000$	new (computation)	Denominators agree with OEIS A259490 [81]; the Poisson-binomial comparison is ours.
The exact expected number of n with $q_n(x) = q$, and the expected prime count $2\ln 2 \cdot \ln N + O(1)$	new (theorem)	Built on the classical cylinder lengths [48] and Mertens' theorem; we did not find either statement in the literature we searched.
Almost-everywhere bounds $\ln N$ and $2\ln N$; finitely many n with p_n, q_n both prime	new (corollaries)	Immediate from deep theorems of Erdős [28] and Koukoulopoulos et al. [52], with Legendre's theorem and Lévy's theorem; the last part is an elementary Borel–Cantelli argument.
Divisibility of e's denominators by small primes	new (observation)	Periodicity modulo ℓ is elementary; the densities and their correlation are our computation.
Structure of the correlation matrix (Gram form, determinant, mean and variance, top-eigenvalue bound)	new presentation / new	The Gram form and determinant are the pattern of Smith's determinant [89]; the moment and eigenvalue statements we did not find for this matrix. GCD matrices: Hong and Loewy [43], Lindqvist and Seip [62].
Ranking of equal temperaments, and prime sizes	new presentation	Measures from Erlich [29]; equidistribution over primes from Vinogradov [97].

13.2 The two questions the owner asked

Is the rhythm of π new? No. The rhythm of any number, in the sense of Chapter 3, is its characteristic Sturmian word, a classical object, and the nested-block structure is a classical theorem. The rhythm of π is one instance. What is less common is to *listen* to it and to name its levels as musical form; that reading has close relatives in the music-theory literature, notably the Sturmian canons of Callender, in which the continued fraction governs the relations between voices, and the work of Carey, Clampitt and Noll on Sturmian structures in scales and rhythms. We did not find the full form reading (every term a repeat count, from bar to movement) laid out as a method for composing complete pieces.

Is the new music new? Yes, as music. The Continued-Fraction Suite is an original work by Teddy Rook, and its originality as an artwork does not depend on the novelty of its method, any more than a fugue's does on the novelty of counterpoint. Of its method, the Fibonacci proportions of Golden have a long history in analysis and composition; Sturmian rhythms and continued-fraction-governed canons exist in the specialist literature. We did not find a prior composition whose whole form is laid out level by level from a number's continued fraction, nor one that sets out to reach the 292nd return of π's 113-bar cycle.

For the video. A brisk tour of Table 13.1, colored by class. Spend most of the time on the two questions: the rhythm of π is a classical object, heard in a new way; the Suite is new music. About 6 minutes.

CHAPTER 14

Discussion

14.1 What the work shows

One guess about π led, through one algorithm, to a surprising amount of mathematics and music. The guess $3+k/(7k+1)$ is a continued fraction; a continued fraction is a rhythm whose levels are its terms (Theorem 3.5); the same algorithm builds tunings, Euclidean rhythms, mode-locking staircases and a factoring method; coprimality, which every level of the rhythm obeys, is where $6/\pi^2$ enters; and the self-similarity of the continued fraction is the common root of a family of classical fractals. Along the way we found the precise meaning of two numbers that the site's sources disagreed about (32,996 and 33,102), and we checked every numerical claim by script.

The two research threads gave concrete answers. For prime convergents, a heuristic built from Lévy's theorem and the equidistribution of convergents modulo m predicts $2\ln 2 \cdot \ln N$ prime denominators among the first N convergents of a typical number, with the correction factor turning out to be $\zeta(2)$; a sample of random reals fits the predicted growth, and π is typical. For the Montgomery–Dyson connection, one set of statistics places a crystal, π's rhythm, the zeta zeros, random matrices, the primes and Poisson points on a single scale, and the explicit formula makes Dyson's quasicrystal remark visible. Neither thread proves a theorem about π or the zeta function, and we have been careful to say so.

14.2 Limitations

- *Heuristics are not theorems.* Heuristic 8.3 treats primality of different denominators as independent events; the agreement with data supports it but does not prove it. Chapter 15 proves it in expectation, not for almost every number.
- *Finite samples.* The prime-convergent test uses 400 random reals and 1,000 convergents each; the slope fit has a standard error of 0.04 and is slightly above the prediction because of early terms. A larger sample, and longer expansions, would sharpen the test.
- *Low zeros.* We used the first 5,000 zeros, which we computed ourselves. At these heights finite-size corrections are visible [9, 11]; Odlyzko's tables of zeros at much greater heights [73] would show closer agreement with the GUE.

- *Probable primes.* Large "primes" are Baillie–PSW probable primes. No composite is known to pass that test, but it is not a proof of primality.
- *Score facts.* The facts about the Suite come from the site's data, which its builders read from the engraved score; we checked the bar arithmetic but not each note.
- *Literature search.* The audit of Chapter 13 is as careful as we could make it, but the music-theory literature in particular is large, and related work may exist that we did not find.

14.3 Open questions

Six questions were left open when this work was first written. Chapter 15 reports what further work on them produced; the status after that work is given in brackets.

1. Does π have infinitely many prime convergent denominators? Infinitely many prime numerators? [Open. We tested q_n and p_n to $n = 20{,}000$ using 12 million certified terms; the counts stay inside the heuristic's 95% range (Section 15.2).]
2. Are there infinitely many Fibonacci primes? Pell primes? [Open. The known lists agree reasonably with the Grantham–Granville heuristic (Section 15.3).]
3. Can Heuristic 8.3 be proved in a model? For example, for almost every x, is the number of $n \le N$ with q_n prime asymptotic to $2\ln 2 \cdot \ln N$? Even an upper bound of that order for almost every x would be interesting. [Partly answered. For a random x the expected count is $2\ln 2 \cdot \ln N + O(1)$, a theorem (Corollary 15.4); for almost every x the count lies between $\ln N$ and $2\ln N$ asymptotically, using recent work on the Duffin–Schaeffer conjecture (Theorem 15.5). The almost-everywhere constant $2\ln 2$ is still a conjecture.]
4. Are π's terms unbounded? Do they obey the Gauss–Kuzmin law? [Open. Calibrated tests on 12 million terms are all consistent with typical behavior (Section 15.5).]
5. How does the spectrum of the arithmetic correlation matrix of Section 5.2 compare with random-matrix benchmarks, and does $6/\pi^2$ appear in its bulk? [Partly answered. Its eigenvalues have mean 1 and variance tending to 2, and its determinant and geometric mean are explicit (proved); $6/\pi^2$ appears in a lower bound $(2/\pi^2)\ln^2 m$ for the top eigenvalue, not in the first two moments; its spacings look like the GOE's (Section 15.6).]
6. Rank all equal temperaments up to 200 notes by how well they approximate several just intervals at once. Do prime sizes stand out? [Answered: no, by eight measures and by an equidistribution theorem (Section 15.7).]

14.4 Future work

On the mathematical side: find the second-moment estimate that would prove Heuristic 8.3 for almost every x (Section 15.4); find the exact growth of the top eigenvalue of the correlation matrix; and repeat the spacing statistics with zeros at large height. On the musical side: the method of Chapter 11 invites pieces in tunings whose number of notes is itself a convergent denominator (a pi movement in a 113-note tuning, for instance), and pieces built from numbers with chosen terms, whose form a composer could design by picking the continued fraction. The

companion site's instruments already do much of this; the score of each new piece is a continued fraction made audible.

For the video. Close the film: return to the scrap of paper with $3+k/(7k+1)$, list the open questions on screen, and play the last bars of Movement VI. About 4 minutes.

CHAPTER 15

Working on the open questions

Section 14.3 ended with six open questions. This chapter reports what happened when we went back to them. Most of them are famous problems, and we did not solve them. Whether π has infinitely many prime convergent denominators, whether there are infinitely many Fibonacci or Pell primes, and whether π's terms obey the Gauss–Kuzmin law are all still open. What we can offer is new computation, sharper heuristics, and some results that we can actually prove.

How to read the labels. In this chapter, a *Theorem* or *Proposition* with no citation in its heading is proved here, in full, from results that are cited. One with a citation is quoted from the literature. When a proof leans on a deep cited theorem, we say so in the statement. A *Heuristic* predicts without proving. Numbers from computation are *numerical evidence*: they can support a conjecture, never prove it. "Prime" for a large number means a Baillie–PSW probable prime (Appendix B); no composite number is known to pass that test, but passing it is not a proof.

15.1 Twelve million terms of pi

Every question about π starts with its terms. Chapter 2 used the first 10,000. Here we use the first 12,000,000, enough to reach the term 878,783,625 at position 11,504,930.

Method. We take π from mpmath, accurate to the last binary place, as an interval $[(m-2)/2^b, (m+2)/2^b]$ with b about 43 million bits. A divide-and-conquer version of Euclid's algorithm then keeps exactly the terms that every real number in the interval shares. It expands the interval cut to its top half, which gives a prefix of the terms and the matrix $\left(\begin{smallmatrix} p_k & p_{k-1} \\ q_k & q_{k-1} \end{smallmatrix}\right)$ of (2.4); it applies the inverse of that matrix to the full interval; and it repeats on what is left. Since the set of reals that share a given prefix is an interval, a term shared by both endpoints is a term of π. The computation is done twice, at two precisions 4,096 bits apart, and the two lists must agree. It takes under two minutes on one core.

Checks. The first 10,000 terms agree with those of Chapter 2, which were found by a different method. The record terms of our list, and their positions (292 at position 4, then 436, 20,776, 78,629, 179,136, 528,210, 12,996,958 and 878,783,625 at positions 307, 431, 28,421, 156,381,

Table 15.1. Probable primes among π's convergents, $n \le N$: the observed count, the expected count $\sum \zeta(2)/\ln$ of Heuristic 8.3, the central 95% range of the Poisson-binomial law with those chances, and the share of that law at or below the observed count.

	denominators q_n				numerators p_n			
N	obs.	exp.	95% range	share	obs.	exp.	95% range	share
1,000	5	8.7	4–14	0.11	10	8.0	3–14	0.82
5,000	9	10.9	5–17	0.34	13	10.3	5–17	0.85
20,000	10	12.9	7–20	0.25	15	12.2	6–19	0.83

267,313, 453,293 and 11,504,930), agree with the OEIS tables of record terms [76], which come from much longer computations. The script stops if any of these checks fails.

Prior computations. We are far from the first. Gosper computed about 17 million terms in 1977; Havermann reached 20 million in 1999 and 180 million in 2002; Bickford 458 million in 2010; Weisstein 15 billion by 2013; Fahad 30 billion in 2021; and Frank 653.52 billion in 2025 [74, 99]. Our twelve million are a small, independent, self-checking sample, enough for the statistics below and for every prime-convergent search we can afford.

15.2 Question 1: the prime convergents of pi

The search. We tested q_n and p_n for every $n \le 20{,}000$; the largest of these numbers have about 10,268 digits. Each number first goes through trial division by the primes below 10^5 and then the Baillie–PSW test. The prime denominators occur at

$$n = 1, 3, 8, 32, 594, 1{,}126, 2{,}002, 3{,}610, 4{,}355, 6{,}925,$$

so 10 in all, and none between $n = 6{,}926$ and $n = 20{,}000$. This agrees with OEIS A259490, whose index k is our $n + 1$ and which records that there are no more below $k = 25{,}000$ [81]. The prime numerators occur at

$$n = 4, 8, 10, 15, 86, 229, 333, 593, 839, 852, 1{,}148, 2{,}725, 3{,}787, 7{,}441, 8{,}750,$$

15 in all, and none between $n = 8{,}751$ and $n = 20{,}000$. The first thirteen are the ones of Chapter 8; the two beyond $n = 5{,}000$ are at $n = 7{,}441, 8{,}750$. The OEIS list of prime numerators, A086785, holds 15 terms [77], the same number as ours; we compared the first thirteen values.

The comparison. Heuristic 8.3 gives each q_n a chance $\zeta(2)/\ln q_n$ of being prime, and each p_n a chance $\zeta(2)/\ln p_n$. If these were independent events, the count would follow a Poisson-binomial law, which we compute exactly. Table 15.1 and Fig. 15.1 compare. Both counts sit inside the central 95% range of the heuristic at every checkpoint. The denominators run on the low side (the share of the law at or below the observed count is between 0.1 and 0.35), and the numerators on the high side (about 0.83). A low count for one and a high count for the other is what chance alone often produces, and nothing in the table is unusual.

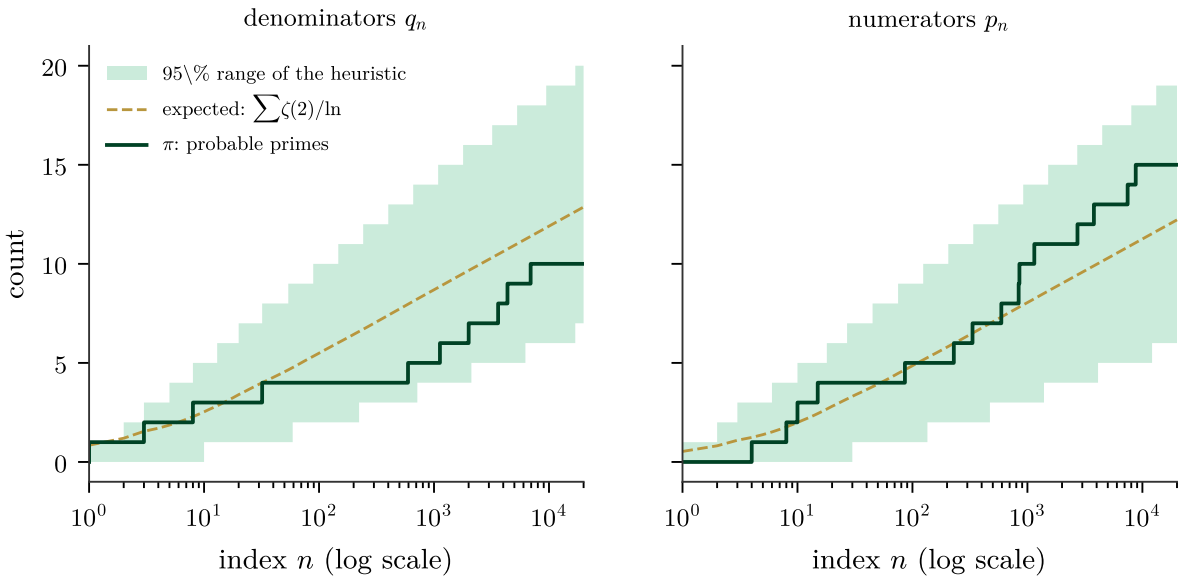


Figure 15.1. Probable-prime denominators (left) and numerators (right) of π's convergents up to $n = 20{,}000$ (green steps), the expected count of Heuristic 8.3 (gold, dashed) and the central 95% range of the heuristic (shaded).

Both at once. Only $n = 8$ gives a convergent, 833,719/265,381, whose numerator and denominator are both prime. For a typical number this is expected to happen only finitely often; Theorem 15.5(iv) below proves it for almost every real number. The same local reasoning as in Heuristic 8.3 gives a correction factor $\zeta(2)$ for the pair (the pair (p_n, q_n) modulo a prime ℓ is equidistributed over the $\ell^2 - 1$ nonzero pairs, by the theorem of Jager and Liardet [44]), and the expected number of further cases beyond $n = 8$ is then only about 0.11.

The number e. The earlier count for e (Chapter 8) was 10 prime denominators up to $n = 3{,}000$, the last at $n = 268$. The terms of e are not typical, so Heuristic 8.3 does not apply. What replaces it? One piece can be proved.

Proposition 15.1. *Let q_n be the convergent denominators of e and ℓ a prime. The sequence $q_n \bmod \ell$ is purely periodic. Hence the share f_ℓ of n with $\ell \mid q_n$ exists and equals the share of zeros in one period.*

Proof. The terms of e are $a_{3k-2} = a_{3k} = 1$ and $a_{3k-1} = 2k$, so $a_n \bmod \ell$ depends only on $n \bmod 3\ell$. The state ($n \bmod 3\ell$, $q_{n-1} \bmod \ell$, $q_n \bmod \ell$) therefore determines the next state, by $q_{n+1} = a_{n+1}q_n + q_{n-1}$. This map is a bijection of a finite set (it can be run backward, since $q_{n-1} = q_{n+1} - a_{n+1}q_n$), so every orbit is a cycle. □

For every prime $\ell < 3{,}000$ our script finds that the period divides 6ℓ (we have not proved this in general), and computes f_ℓ exactly: $f_2 = 1/3$, $f_3 = 1/3$, $f_5 = 1/5$, $f_7 = 5/21$, $f_{11} = 5/33$, $f_{13} = 7/39$, $f_{17} = 1/17$. A typical number would have $f_\ell = 1/(\ell + 1)$ (Corollary 8.2). For e, the average of ℓf_ℓ over the odd primes below 3,000 is 2.00: its denominators are divisible by a given small prime about twice as often as random integers. If the events "$\ell \mid q_n$" were independent across ℓ, the correction factor would be $\prod_\ell (1 - f_\ell)/(1 - 1/\ell)$, which falls from 0.66 (primes below 100) to 0.45 (below 1,000) and 0.38 (below 3,000) and does not settle.

But they are far from independent. Among e's first 3,000 denominators, the share with no prime factor below 1,000 is 0.093, against 0.036 if the events were independent and 0.081 for random integers (below 100: 0.160, 0.080 and 0.120). The same denominators are divisible by many small primes at once, and the rest are almost as free of small factors as random integers. Using the observed ratio 1.14 as the correction gives 8.6 expected prime denominators up to $n = 3{,}000$, against 10 found; the typical-number factor $\zeta(2)$ gives 12.4, and the independent product gives 5.0. We offer the first as a rough guide, not a heuristic with a constant.

The growth of e's denominators is easy to pin down: since $\prod_{k\le n} a_k \le q_n \le \prod_{k\le n}(a_k+1)$, we get $\ln q_n = \frac{n}{3}\ln n\,(1+o(1))$, much faster than Lévy's linear rate. Any heuristic of the form "prime with chance $c/\ln q_n$" then predicts about $3c\ln\ln N$ prime denominators up to N. So e should have infinitely many, but very few: with $c = 1.14$, about two more between $n = 3{,}000$ and $n = 10^6$.

Status. Open. The new computation quadruples the range searched; everything still looks typical.

15.3 Question 2: Fibonacci and Pell primes

For $\varphi = [1;1,1,\dots]$ the convergent denominators are the Fibonacci numbers, $q_n = F_{n+1}$; for $\sqrt{2} = [1;2,2,\dots]$ they are the Pell numbers, $q_n = P_{n+1}$ ($P_1 = 1$, $P_2 = 2$, $P_{n+1} = 2P_n + P_{n-1}$). Whether either sequence has infinitely many primes is a famous open problem.

Proposition 15.2 (classical)**.** *If F_n is prime then $n = 4$ or n is prime. If P_n is prime then n is prime.*

Proof. Both sequences ($u_n = F_n$ or $u_n = P_n$, with $u_0 = 0$, $u_1 = 1$) satisfy $u_{m+k} = u_m u_{k+1} + u_{m-1}u_k$ for $m \ge 1$, $k \ge 0$: it holds for $k = 0$ and $k = 1$, and both sides obey the sequence's recurrence in k. Taking $k = m$ and m replaced by jm gives $u_{(j+1)m} = u_{jm}u_{m+1} + u_{jm-1}u_m$, so by induction on j, $u_m \mid u_{jm}$. If $n = jm$ with $1 < m < n$, then u_m divides u_n. For Fibonacci numbers $1 < F_m < F_n$ unless $m \le 2$; a composite $n \ne 4$ has a divisor m with $3 \le m < n$, so F_n is composite. For Pell numbers $P_m > 1$ for every $m \ge 2$, and every composite n has such a divisor. □

The proposition is why the typical-number heuristic fails for these two numbers: their denominators are divisibility sequences, not random-looking integers. Grantham and Granville give the heuristic that replaces it. For a Lucas sequence $u_n = (\alpha^n - \beta^n)/(\alpha - \beta)$ with $\alpha > |\beta|$, they predict

$$\#\{n \le N : u_n \text{ prime}\} \sim e^{\gamma}\log_\alpha N \tag{15.1}$$

[40]. The Fibonacci numbers have $\alpha = \varphi$. The Pell numbers are the Lucas sequence with $\alpha = 1+\sqrt{2}$, so the same formula applies to them, though Grantham and Granville do not discuss them by name.

What is known. OEIS A001605 lists 51 indices of Fibonacci primes up to $n = 3{,}340{,}367$; the first 37 (up to $n = 201{,}107$) are proven primes and the rest probable primes, and later searches added five more and then $n = 11{,}964{,}299$ in 2025 [75]. OEIS A096650 lists 31 indices of Pell primes below 10^5, and records that there are no others below 10^5; those above 523 are probable primes [79]. Our own lists, up to index 3,001, agree with both (Chapter 8). Table 15.2 compares the counts with (15.1).

Table 15.2. The number of Fibonacci primes F_n and Pell primes P_n with $n \le N$, from the OEIS lists, against the Grantham–Granville prediction $e^{\gamma}\log_{\alpha} N$ with $\alpha = \varphi$ and $\alpha = 1+\sqrt{2}$. The Pell list is complete only below 10^5.

	Fibonacci		Pell	
N	found	predicted	found	predicted
100	12	17.0	11	9.3
1,000	21	25.6	17	14.0
10,000	26	34.1	24	18.6
100,000	33	42.6	31	23.3
1,000,000	43	51.1	–	–
3,340,367	51	55.6	–	–

The Fibonacci counts run steadily below the prediction, by about a sixth at the larger N, as Grantham and Granville note themselves; the Pell counts run above it, by about a third at 10^5. For counts this small and this slowly growing, such gaps are not alarming, and the constant in such heuristics is the least certain part. Both lists are consistent with infinitely many primes and a count that grows like $\log N$; that is as far as the data go.

Status. Open, and not touched by anything in this work; the heuristic and the data agree to within the accuracy one can expect.

15.4 Question 3: a theorem in the model

Here we can prove something. Pick x at random, uniformly in $(0,1)$, and write $q_n = q_n(x)$. Recall that the set of x whose first n terms are $a_1,\dots,a_n$ is an interval (a *cylinder*) of length

$$\frac{1}{q_n(q_n+q_{n-1})} \tag{15.2}$$

[48, Ch. III], and that, by (2.4), $q_n/q_{n-1} = [a_n; a_{n-1},\dots,a_1]$ (the "mirror formula": $q_n/q_{n-1} = a_n + q_{n-2}/q_{n-1}$, and induct).

Theorem 15.3 (the exact expected count)**.** *For every integer $q \ge 2$,*

$$\mathbb{E}\,\#\{n \ge 1 : q_n(x) = q\} = \frac{2}{q}\sum_{\substack{1\le r<q\\ \gcd(r,q)=1}} \frac{1}{q+r}. \tag{15.3}$$

For a prime p this is $\frac{2}{p}(H_{2p-1}-H_p)$, where $H_m = 1+\frac12+\cdots+\frac1m$. Consequently

$$\mathbb{E}\,\#\{n \ge 1 : q_n(x) \text{ is prime and } q_n(x) \le X\} = \sum_{p\le X}\frac{2}{p}\,(H_{2p-1}-H_p) = 2\ln 2\cdot\ln\ln X + B_0 + o(1) \tag{15.4}$$

as $X\to\infty$, with $B_0 = 2\ln 2\cdot M - C_1 = -0.2945\ldots$, where $M = 0.2615\ldots$ is the Meissel–Mertens constant and $C_1 = \sum_p \frac{2}{p}(\ln 2 - H_{2p-1} + H_p) = 0.6570\ldots$

Proof. Sequences $(a_1, \dots, a_n)$ of positive integers correspond to pairs (q_{n-1}, q_n). By the mirror formula, the sequence read backward is a continued fraction expansion of the rational number q_n/q_{n-1}, all of whose terms are at least 1. Conversely, every rational q/r with $q \geq 2$, $1 \leq r < q$ and $\gcd(r, q) = 1$ has exactly two such expansions, one ending in a term at least 2 and one ending in 1 (for instance $7/2 = [3; 2] = [3; 1, 1]$), and each, read backward, is a sequence $(a_1, \dots, a_n)$ with $q_{n-1} = r$ and $q_n = q$ (the two lengths differ by one). So the cylinders with $q_n = q$, for all n together, are exactly two for each r, each of length $1/(q(q + r))$ by (15.2). Summing the lengths over n gives the expected number of n with $q_n = q$, which is (15.3). For a prime p every r from 1 to $p - 1$ is coprime to p, and $\sum_{r=1}^{p-1} 1/(p + r) = H_{2p-1} - H_p$.

For the asymptotic, $H_{2p-1} - H_p = \ln 2 + O(1/p)$, so the series defining C_1 converges absolutely, and $\sum_{p \leq X} \frac{2}{p}(H_{2p-1} - H_p) = 2 \ln 2 \sum_{p \leq X} \frac{1}{p} - C_1 + O(1/X)$. Mertens' theorem $\sum_{p \leq X} 1/p = \ln \ln X + M + o(1)$ [42, Thm. 427] finishes the proof. □

Our script checks (15.3) against a direct search over all sequences with $q_n = q$ for every $q < 40$, in exact rational arithmetic, and computes C_1 from the primes below 10^7 (the omitted tail is below 10^{-7}).

Where $\zeta(2)$ comes from. Theorem 15.3 shows the correction factor of Chapter 8 at work, now as an identity. A prime q gets weight about $2 \ln 2/q$, because all $q - 1$ numbers $r < q$ are coprime to it. A general q gets about $\frac{\varphi(q)}{q} \cdot \frac{2 \ln 2}{q}$, and $\varphi(q)/q$ averages $6/\pi^2$. Relative to all integers, primes are favored by exactly $\pi^2/6 = \zeta(2)$. (Summed over $q \leq X$, the general weight gives about $\frac{12 \ln 2}{\pi^2} \ln X = (\ln X)/L$ convergents up to X, which matches Lévy's theorem.)

Corollary 15.4. *For x uniform in $(0, 1)$,*

$$\mathbb{E}\, \#\{n \leq N : q_n(x) \text{ prime}\} = 2 \ln 2 \cdot \ln N + O(1).$$

Proof. Lower bound. Since $q_n \geq F_{n+1} \geq \varphi^{n-1}$, any n with $q_n \leq \varphi^{N/2}$ has $n \leq N/2 + 1 \leq N$ (for $N \geq 2$). So the count is at least the number of n with q_n prime and at most $X = \varphi^{N/2}$, and by (15.4) its expectation is at least $2 \ln 2 \cdot \ln(N \ln \varphi/2) + B_0 + o(1)$.

Upper bound. Let $C = 2 \ln \mu + 2 = 4.92\ldots$, where $\mu = \sum_{k \geq 1} 2/(k\sqrt{k+1}) = 4.312\ldots$. If $n \leq N$ and q_n is prime, then either $q_n \leq e^{CN}$, or $q_N \geq q_n > e^{CN}$. Hence the count is at most $\#\{n : q_n \text{ prime}, q_n \leq e^{CN}\} + N \cdot \mathbf{1}[q_N > e^{CN}]$. The first term has expectation $2 \ln 2 \cdot \ln(CN) + B_0 + o(1)$ by (15.4). For the second, the cylinder lengths give, for the conditional chance of the next term,

$$\Pr(a_{n+1} = k \mid a_1, \dots, a_n) = \frac{1 + s}{(k + s)(k + 1 + s)} \leq \frac{2}{k(k + 1)}, \qquad s = q_{n-1}/q_n \in [0, 1],$$

so $\mathbb{E}[(1 + a_{n+1})^{1/2} \mid a_1, \dots, a_n] \leq \mu$. Since $q_N \leq \prod_{k \leq N}(1 + a_k)$, conditioning one term at a time gives $\mathbb{E}\, q_N^{1/2} \leq \mu^N$, and Markov's inequality gives $\Pr(q_N > e^{CN}) \leq \mu^N e^{-CN/2} = e^{-N}$. So the second term has expectation at most Ne^{-N}. □

So, in expectation, Heuristic 8.3 is a theorem, with exactly the constant $2 \ln 2$. To the best of our search we did not find Theorem 15.3 or Corollary 15.4 stated in the literature, although the counting behind them is classical. What about almost every x, which is what the heuristic actually claims? Here deep recent theorems carry us part of the way. Write $\theta_n = q_n^2 \, |x - p_n/q_n|$; every convergent has $\theta_n < 1$, and by Legendre's theorem, any fraction a/q in lowest terms with $|x - a/q| < 1/(2q^2)$ is a convergent of x [42, Thm. 184].

Theorem 15.5 (almost every x; parts (i)–(iii) rest on cited theorems)**.** *For almost every real x:*

(i) infinitely many q_n are prime, and infinitely many p_n are prime;

(ii) the number of $n \le N$ with q_n prime and $\theta_n < 1/2$ is $(1+o(1))\ln N$;

(iii) hence $1 \le \liminf_{N\to\infty} \dfrac{\#\{n \le N : q_n \text{ prime}\}}{\ln N} \le \limsup_{N\to\infty} \dfrac{\#\{n \le N : q_n \text{ prime}\}}{\ln N} \le 2$;

(iv) only finitely many n have both p_n and q_n prime.

Proof. (i) Erdős proved that if $n_1 < n_2 < \cdots$ are integers with $\sum \varphi(n_i)/n_i^2 = \infty$, then for almost every x infinitely many n_i are convergent denominators of x [28]. For the primes, $\sum_p (p-1)/p^2 = \infty$. For numerators, note that the convergents of $1/x$ are the reciprocals of those of x (if $x > 1$, then $1/x = [0; a_0, a_1, \dots]$; if $0 < x < 1$, then $1/x = [a_1; a_2, \dots]$), so the numerators of x are, up to a shift of index, the denominators of $1/x$; and the map $x \mapsto 1/x$ sends sets of measure zero to sets of measure zero.

(ii) Koukoulopoulos, Maynard and Yang proved the following quantitative form of the Duffin–Schaeffer conjecture [52], extending [51]. Let ψ take the positive integers to $[0, 1/2]$, and suppose that

$$\Psi(Q) = \sum_{q \le Q} \frac{2\varphi(q)\psi(q)}{q} \to \infty.$$

Then for almost every x the number of coprime pairs (a, q) with $q \le Q$ and $|x - a/q| < \psi(q)/q$ is $\Psi(Q)(1 + o(1))$. Take $\psi(q) = 1/(2q)$ when q is prime and 0 otherwise; then $\Psi(Q) = \sum_{p \le Q} (p-1)/p^2 = \ln\ln Q + O(1)$. By Legendre's theorem the pairs counted are exactly the convergents with q_n prime, $q_n \le Q$ and $\theta_n < 1/2$, one for each such n. To pass from $q_n \le Q$ to $n \le N$, use Lévy's theorem: for almost every x and any $\varepsilon > 0$, $e^{(L-\varepsilon)n} \le q_n \le e^{(L+\varepsilon)n}$ for all large n, with $L = \pi^2/(12\ln 2)$. Then for large N, every n with $q_n \le e^{(L-\varepsilon)N}$ has $n \le N$, and every $n \le N$ has $q_n \le e^{(L+\varepsilon)N}$; and $\ln\ln e^{(L\pm\varepsilon)N} = \ln N + O(1)$.

(iii) The lower bound is (ii). For the upper bound, use the same theorem with $\psi(q) = 1/q$ for prime q (so $\Psi(Q) = 2\ln\ln Q + O(1)$): every convergent with q_n prime has $|x - p_n/q_n| < 1/q_n^2$, so it is one of the pairs counted. Convert from Q to N as before.

(iv) This part is elementary, and we give it for $0 < x < 1$ (for other x, replace the bound on $\pi(q)$ below by the Brun–Titchmarsh inequality [71]). If $p_n = a$ and $q_n = q$ are both prime, then x lies within $1/q^2$ of a/q with $a < q$ prime. For each prime q, the set of such x has measure at most $2\pi(q)/q^2 \le C'/(q\ln q)$ by Chebyshev's bound $\pi(q) = O(q/\ln q)$ [42, Thm. 7], and $\sum_p 1/(p\ln p) < \infty$. By the Borel–Cantelli lemma almost every x lies in only finitely many of these sets, so only finitely many primes q can occur, and each value q_n occurs for at most two n. □

Where the argument stops. The conjecture behind Question 3 sits strictly inside Theorem 15.5(iii): $2\ln 2 \approx 1.3863$ is between 1 and 2. And it matches the theorem in a precise way. For almost every x, the share of n with $\theta_n < 1/2$ is $1/(2\ln 2)$ (the Doeblin–Lenstra law, proved by Bosma, Jager and Wiedijk [13]), and $2\ln 2 \times 1/(2\ln 2) = 1$, which is exactly the constant in (ii). The missing piece is the convergents with $\theta_n \ge 1/2$. The set of x for which a given fraction a/q is a convergent is an interval around a/q whose two halves have lengths $1/(q(q+r))$ and $1/(q(2q-r))$, where r is determined by a ($ar \equiv \pm 1 \pmod q$). The Duffin–Schaeffer machinery counts approximations whose target interval depends only on q; here it

Table 15.3. Simulation of 20,000 uniform random reals. Top: the mean number of prime $q_n \le X$, with its standard error, against (15.4); and the same for prime $q_n \le X$ with $\theta_n < 1/2$, against $\sum_{p\le X}(p-1)/p^2$. Bottom: the mean number of prime q_n with $n \le N$, its standard error and variance, and $2\ln 2 \cdot \ln N$.

X	mean	s.e.	exact	mean, $\theta_n < \frac{1}{2}$	s.e.	exact
10^5	3.084	0.011	3.093	2.235	0.009	2.253
10^{10}	4.049	0.012	4.054	2.929	0.011	2.946
10^{20}	5.008	0.014	5.015	3.626	0.012	3.639
10^{50}	6.279	0.016	6.285	4.543	0.014	4.555
10^{100}	7.230	0.018	7.246	5.230	0.015	5.248
10^{200}	8.196	0.019	8.207	5.925	0.016	5.942
N	mean	s.e.	variance	$2\ln 2 \cdot \ln N$		
10	3.079	0.010	1.82	3.192		
20	4.063	0.012	2.78	4.153		
50	5.346	0.014	4.13	5.423		
100	6.311	0.016	5.15	6.384		
200	7.270	0.017	6.12	7.345		
400	8.237	0.019	7.17	8.306		

depends on a too, through a modular inverse. A second-moment bound for these events would give the full almost-everywhere statement, and that is what we cannot supply. Note also that Corollary 15.4 alone, with Markov's inequality and the Borel–Cantelli lemma along $N_k = e^{2^k}$, gives the weaker but elementary bound $O(\ln N\,(\ln\ln N)^{1+\varepsilon})$ for almost every x, without the deep theorem.

The numbers. Table 15.3 and Fig. 15.2 check Theorem 15.3 on 20,000 random reals, each given by a random fraction with 2,600 binary digits. The simulated means agree with the exact values within two standard errors at every X (the six values are cumulative, so their deviations are strongly correlated), for all prime denominators and for those with $\theta_n < 1/2$; the ratio of the two is close to $1/(2\ln 2) \approx 0.72$, as the Doeblin–Lenstra law predicts. The counts by index (second part of the table) grow by $2\ln 2$ per e-fold, as Corollary 15.4 says; their level lies a constant below $2\ln 2 \cdot \ln N$, as the $O(1)$ allows. The variance is a little below the mean, as for a sum of independent events with these chances (a Poisson-binomial count has variance below its mean).

Status. Partly answered. In expectation the heuristic's constant is now a theorem (Corollary 15.4). For almost every x the count is between $\ln N$ and $2\ln N$ asymptotically, using the recent quantitative Duffin–Schaeffer theorem; that it is $2\ln 2 \cdot \ln N$ remains a conjecture. Nothing here applies to π, which is one number and may or may not be typical.

15.5 Question 4: is pi typical?

The Gauss–Kuzmin law, Khinchin's constant and Lévy's constant are laws for almost every number. Whether π obeys them is open; it is not even known whether π's terms are bounded. All we can do is test, and every test has to be careful about two things. First, the terms of a typical number are weakly dependent, so textbook chi-square p-values, which assume

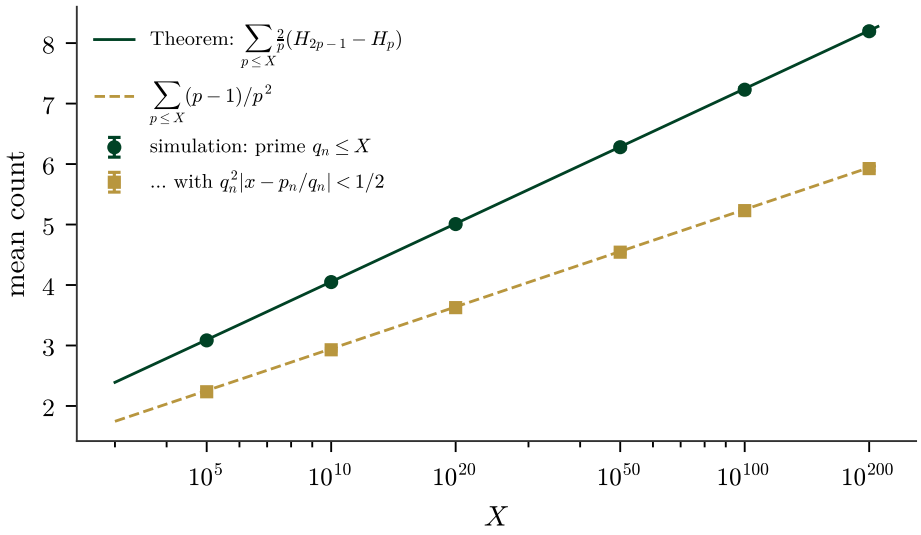


Figure 15.2. Theorem 15.3 against simulation: the mean number of prime convergent denominators up to X (green, with two standard errors) and of those with $\theta_n < 1/2$ (gold), for 20,000 random reals, with the exact expectations.

independence, are only approximate. We therefore calibrated every statistic on 1,000 random reals with 100,000 terms each, expanded exactly. Second, a p-value is not a probability that π is typical; a pass is evidence of the absence of a large deviation, not proof of typicality.

Results. On the first 12,000,000 terms (Fig. 15.3):

- *Values 1 to 100 and above 100* (101 cells): $\chi^2 = 100.9$ on 100 degrees of freedom, textbook $p = 0.45$, calibrated $p = 0.45$. The ones make up 41.489%, against 41.504% for the law.
- *The tail*, in the dyadic cells $[2^j, 2^{j+1})$, $j = 0, \dots, 19$, and $[2^{20}, \infty)$: $\chi^2 = 20.6$ on 20 degrees of freedom, $p = 0.42$.
- *Pairs* (a_n, a_{n+1}) with values $1, \dots, 5$ and "6 or more" (36 cells, probabilities from the Gauss measure of the two-term cylinders): $\chi^2 = 26.0$ on 35 degrees of freedom, textbook $p = 0.86$, calibrated $p = 0.82$. The calibration matters here: for random reals the textbook test rejects at the 5% level 0.077 of the time, more than it should (for single values, 0.044).
- *Blocks.* Cut into 120 blocks of 100,000 terms, the 120 block p-values look uniform (Kolmogorov–Smirnov $p = 0.30$), as they should if no stretch of π's expansion is unusual.
- *Khinchin.* The geometric mean of the terms is 2.68655, against $K_0 = 2.685452\dots$; measured in the standard deviation that random reals show at this length, the deviation is 1.29.
- *Lévy.* $\ln q_n/n = 1.186889$ at $n = 12$ million, against $\pi^2/(12\ln 2) = 1.186569\dots$, a deviation of 1.21 standard deviations. The two standardized curves in Fig. 15.4 move almost together, because $\ln q_n$ is $\sum_{k\le n}\ln a_k$ plus a sum of corrections $\ln(1 + q_{k-2}/(a_k q_{k-1}))$, each between 0 and $\ln 2$, so the two sums rise and fall with the same large terms.

Table 15.4. The largest of π's first n terms, its position, and its percentile under Galambos's law $e^{-n/(M_n \ln 2)}$.

n	largest term M_n	position	percentile
1,000	20,776	431	0.93
10,000	20,776	431	0.50
100,000	78,629	28,421	0.16
1,000,000	12,996,958	453,293	0.89
10,000,000	12,996,958	453,293	0.33
12,000,000	878,783,625	11,504,930	0.98

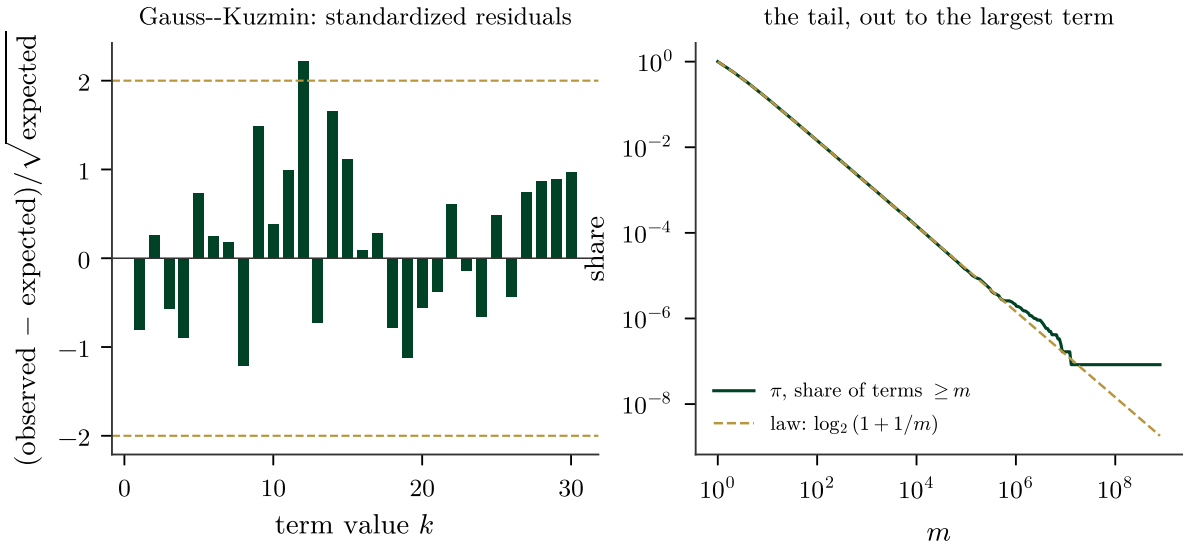


Figure 15.3. π's first 12,000,000 terms against the Gauss–Kuzmin law. Left: standardized differences between observed and expected counts for the values 1 to 30; under the law they behave roughly like standard normal variables. Right: the share of terms at least m, out to the largest term, against the law's $\log_2(1 + 1/m)$.

- *Largest terms.* Galambos proved that for almost every x, the largest of the first n terms, M_n, satisfies $\Pr(M_n \le nt/\ln 2) \to e^{-1/t}$ [35]. Table 15.4 gives the percentile $e^{-n/(M_n \ln 2)}$ at several n. They wander over the whole range, as they should; the term 878,783,625 at position 11,504,930 is large (98th percentile at $n = 12$ million), but not extreme.
- *Missing values.* The smallest positive integer that does not occur among the first 12 million terms is 2,150. A value near 2,000 is expected about $12 \cdot 10^6/(k^2 \ln 2) \approx 4$ times, so a first gap in this range is ordinary.

The much longer computations behind the record tables and MathWorld's plots of the running Khinchin and Lévy means [76, 99] show the same picture, and we know of no published statistic of π's terms that looks atypical.

Status. Open. All statistics we computed on 12 million terms are consistent with π being a typical number for the Gauss measure. That is evidence, and the evidence is good; it is not a proof, and no method we know of could turn it into one.

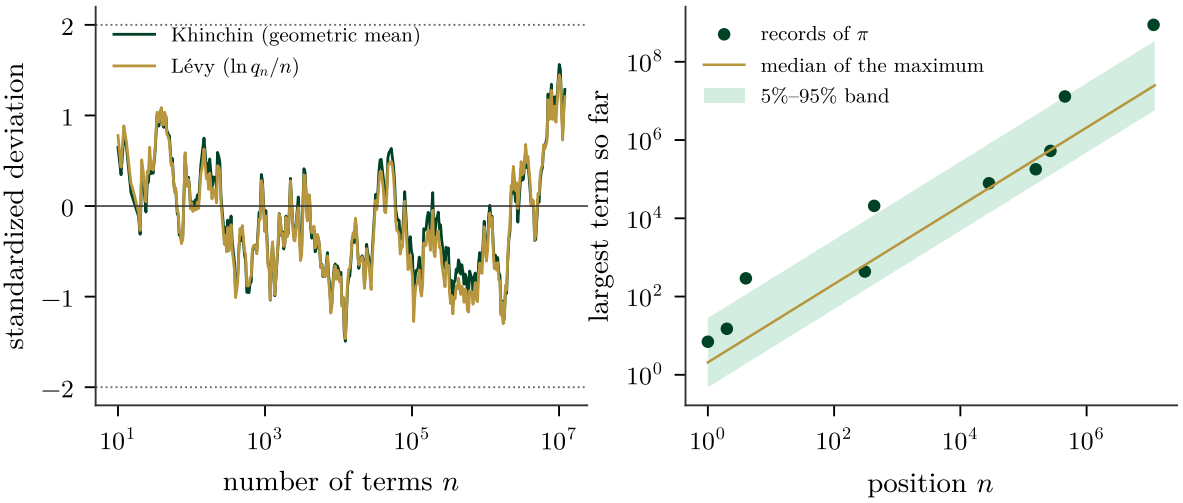


Figure 15.4. Left: the running deviation of the geometric mean of π's terms from Khinchin's constant, and of $\ln q_n/n$ from Lévy's constant, in units of the standard deviation that random reals show after n terms; the dotted lines are ± 2. Right: the record terms of π against the median and the 5%–95% band of Galambos's law for the largest term.

15.6 Question 5: the spectrum of the correlation matrix

The matrix of Section 5.2, for periods $2, \dots, m$, is

$$C_m = \left[\frac{\gcd(a,b)-1}{\sqrt{(a-1)(b-1)}} \right]_{a,b=2}^{m},$$

an $(m-1)\times(m-1)$ matrix with ones on the diagonal (Section 5.2 used $m = 40$). Its structure is cleaner than its look.

Proposition 15.6 (structure)**.** *Let V be the $(m-1)\times(m-1)$ matrix with $V_{ad} = 1$ if $d \mid a$ and 0 otherwise $(a, d = 2, \dots, m)$, $\Phi = \operatorname{diag}(\varphi(d))$ and $D = \operatorname{diag}(1/\sqrt{a-1})$. Then*

(i) $C_m = DV\Phi V^{\mathsf{T}}D$, so C_m is positive definite;

(ii) $\det C_m = \prod_{a=2}^{m} \varphi(a)/(a-1)$;

(iii) $(\det C_m)^{1/(m-1)}$, the geometric mean of the eigenvalues, tends to $\exp\bigl(\sum_p \frac{1}{p}\ln(1-\frac{1}{p})\bigr) = 0.5599\ldots$;

(iv) for every prime p with $m/2 < p \le m$, the unit vector e_p is an eigenvector with eigenvalue 1.

Proof. (i) Gauss's identity $\sum_{d\mid g}\varphi(d) = g$ gives $\gcd(a,b)-1 = \sum_{d\mid a,\ d\mid b,\ d\ge 2}\varphi(d) = (V\Phi V^{\mathsf{T}})_{ab}$. The matrix V is lower triangular with ones on the diagonal (a divisor d of a is at most a), so it is invertible, and Φ has positive entries; hence C_m is a Gram matrix of independent vectors. (ii) $\det V = 1$, so $\det C_m = \det(D)^2 \prod \varphi(d)$. This is the analogue of Smith's determinant $\det[\gcd(i,j)] = \prod \varphi(k)$ [89]. (iii) Since $\ln(\varphi(a)/a) = \sum_{p\mid a}\ln(1-1/p)$ and a prime p divides a share $1/p$ of integers, the average of $\ln(\varphi(a)/a)$ over $a \le m$ tends to $\sum_p p^{-1}\ln(1-1/p)$; the extra factor $\prod a/(a-1) = m$ contributes $\ln m/(m-1) \to 0$. (iv) For such p, the only multiple of p in range is p itself, so $\gcd(p,b) = 1$ for $b \ne p$, and row p of C_m is e_p. □

For $m = 40$, $\det C_{40} = 6.22 \times 10^{-9}$. Part (iv) gives at least $\pi(m) - \pi(m/2)$ eigenvalues equal to 1, and in every case we computed the multiplicity of the eigenvalue 1 is exactly that.

Theorem 15.7 (the bulk: mean 1 and variance 2)**.** $\operatorname{tr} C_m = m-1$*, and* $\operatorname{tr} C_m^2 = 3m+O(\ln^3 m)$*. So the eigenvalues of* C_m *have mean 1 and, in the limit, mean square 3 and variance 2.*

Proof. The trace is clear. For the square, let $T(m) = \sum_{a,b\le m} \gcd(a,b)^2/(ab)$. Writing $a = ga'$, $b = gb'$ with $\gcd(a',b') = 1$,

$$T(m) = \sum_{\gcd(a',b')=1} \frac{1}{a'b'} \left\lfloor \frac{m}{\max(a',b')} \right\rfloor = m \sum_{\substack{\gcd(a',b')=1 \\ \max(a',b')\le m}} \frac{1}{a'b' \max(a',b')} + O(\ln^2 m).$$

Over all pairs, $\sum_{a,b\ge 1} 1/(ab\max(a,b)) = \zeta(3) + 2\sum_a H_{a-1}/a^2 = \zeta(3) + 2\zeta(3) = 3\zeta(3)$, by Euler's identity $\sum_a H_a/a^2 = 2\zeta(3)$ [32]. The summand is homogeneous of degree -3, so restricting to coprime pairs divides by $\zeta(3)$ (Möbius inversion), giving 3; the pairs with $\max(a',b') > m$ contribute $O(\ln m/m)$. So $T(m) = 3m + O(\ln^2 m)$. Finally, $\operatorname{tr} C_m^2 = \sum_{a,b=2}^{m} (\gcd(a,b)-1)^2/((a-1)(b-1))$ differs from $T(m)$ by $O(\ln^3 m)$: expanding, the difference is bounded by a constant times $\sum \gcd(a,b)/(ab) + \sum 1/(ab) + \sum \gcd(a,b)^2(a+b)/(a^2b^2)$ over $a,b \le m$, and these sums are $O(\ln^3 m)$, $O(\ln^2 m)$ and $O(\ln^2 m)$ by the same substitution. □

Numerically $\operatorname{tr} C_m^2/(m-1)$ is 2.95 at $m = 6{,}000$ and 2.9783 at $m = 20{,}000$, closing in on 3 at the rate the error term allows. Note that $\zeta(3)$ appears and cancels; $6/\pi^2$ does not appear in these moments.

Theorem 15.8 (the top eigenvalue: a lower bound)**.** $\lambda_{\max}(C_m) \ge \dfrac{2}{\pi^2} \ln^2 m\,(1 + o(1)) = \dfrac{1}{3} \cdot \dfrac{6}{\pi^2} \ln^2 m\,(1 + o(1))$.

Proof. Take the test vector $x_a = 1/\sqrt{a-1}$. Then $x^{\mathsf{T}}x = H_{m-1} = \ln m + O(1)$, and

$$x^{\mathsf{T}} C_m x = \sum_{a,b=2}^{m} \frac{\gcd(a,b)-1}{(a-1)(b-1)} \ge \sum_{a,b\le m} \frac{\gcd(a,b)}{ab} - H_m^2 - 2H_m.$$

With $a = ga'$, $b = gb'$ as before, $\sum_{a,b\le m} \gcd(a,b)/(ab) = \sum_{g\le m} g^{-1} S(m/g)$, where, by Möbius inversion,

$$S(y) = \sum_{\substack{a',b'\le y \\ \gcd(a',b')=1}} \frac{1}{a'b'} = \sum_{d\le y} \frac{\mu(d)}{d^2} H_{\lfloor y/d \rfloor}^2 = \frac{6}{\pi^2} \ln^2 y + O(\ln y + 1).$$

Then $\sum_{g\le m} g^{-1} \frac{6}{\pi^2} \ln^2(m/g) = \frac{6}{\pi^2} \cdot \frac{\ln^3 m}{3} + O(\ln^2 m)$. Dividing by $x^{\mathsf{T}}x$ gives the bound, since $\lambda_{\max} \ge x^{\mathsf{T}} C_m x / x^{\mathsf{T}} x$. □

Comparison with random-matrix benchmarks. Tables 15.5 and 15.6 and Fig. 15.5 give the numbers.

- *Not Marchenko–Pastur.* A sample correlation matrix of independent noise follows the Marchenko–Pastur law [64], with bounded support. With the same mean 1 and variance 2, that law would put half of all eigenvalues at 0 and none above $(1+\sqrt{2})^2 \approx 5.8$. Our

Table 15.5. The spectrum of C_m (dense diagonalization): $\operatorname{tr} C_m^2/(m-1)$ (limit 3), the geometric mean of the eigenvalues (limit 0.5599), the smallest eigenvalue, the multiplicity of the eigenvalue 1 and the lower bound $\pi(m)-\pi(m/2)$, the largest eigenvalue, and the mean spacing ratio $\langle r\rangle$ in the central 80% of the spectrum (Poisson 0.386, GOE 0.531).

m	$\operatorname{tr} C^2/(m-1)$	geo. mean	$\lambda_{\min}$	mult. of 1	bound	$\lambda_{\max}$	$\langle r\rangle$
500	2.755	0.5672	0.0246	42	42	14.887	0.493
1,000	2.838	0.5640	0.0196	73	73	18.773	0.515
2,000	2.895	0.5622	0.0159	135	135	23.279	0.539
4,000	2.934	0.5611	0.0130	247	247	28.458	0.529
6,000	2.950	0.5607	0.0117	353	353	31.822	0.525

Table 15.6. The largest eigenvalue of C_m, computed by the Lanczos method with a matrix-vector product of cost $O(m \log m)$ from Proposition 15.6(i); the Rayleigh quotient of the test vector of Theorem 15.8; and $(2/\pi^2)\ln^2 m$.

m	$\lambda_{\max}$	test vector	$(2/\pi^2)\ln^2 m$
1,000	18.773	10.680	9.670
3,000	26.221	13.952	12.990
10,000	36.429	18.091	17.190
30,000	47.879	22.377	21.536
100,000	63.078	27.632	26.860
300,000	79.678	32.938	32.230
1,000,000	101.252	39.312	38.678

matrix has no eigenvalue at 0 (the smallest is about 0.01 at $m = 6{,}000$ and shrinks slowly), about two thirds of its eigenvalues below 1, the eigenvalue 1 exactly at the large primes, and a top eigenvalue that grows without bound. It is not a noisy correlation matrix, and the comparison mainly shows how much arithmetic structure it carries.

- *Not a semicircle.* The GOE's semicircle is symmetric about its center; this spectrum is positive and strongly skewed.
- *But GOE-like locally.* The mean ratio of consecutive spacings, $\langle r\rangle$, needs no unfolding and is about 0.386 for uncorrelated (Poisson) levels and 0.531 for the GOE [3]. In the bulk of our spectrum it is 0.51 to 0.54 for $m \geq 1{,}000$ (0.525 at $m = 6{,}000$), close to the GOE value: neighboring eigenvalues repel each other as in a random symmetric matrix. We report this as an observation; we have no explanation for it.
- *The top eigenvalue.* It grows faster than our lower bound: the ratio $\lambda_{\max}/\ln^2 m$ rises from about 0.39 at $m = 1{,}000$ to 0.53 at $m = 10^6$. The test vector's quotient tracks $(2/\pi^2)\ln^2 m$ closely, as the proof says, but the true top eigenvector puts extra weight on numbers with many small factors. GCD matrices of this kind are studied in the literature on GCD sums and on eigenvalues of GCD matrices [43, 62]; we did not find the asymptotic for this particular matrix, and we do not know it.

Status. Partly answered. The matrix is a Gram matrix with an exact determinant; its spectrum has mean 1 and variance tending to 2 (proved); its geometric mean converges to

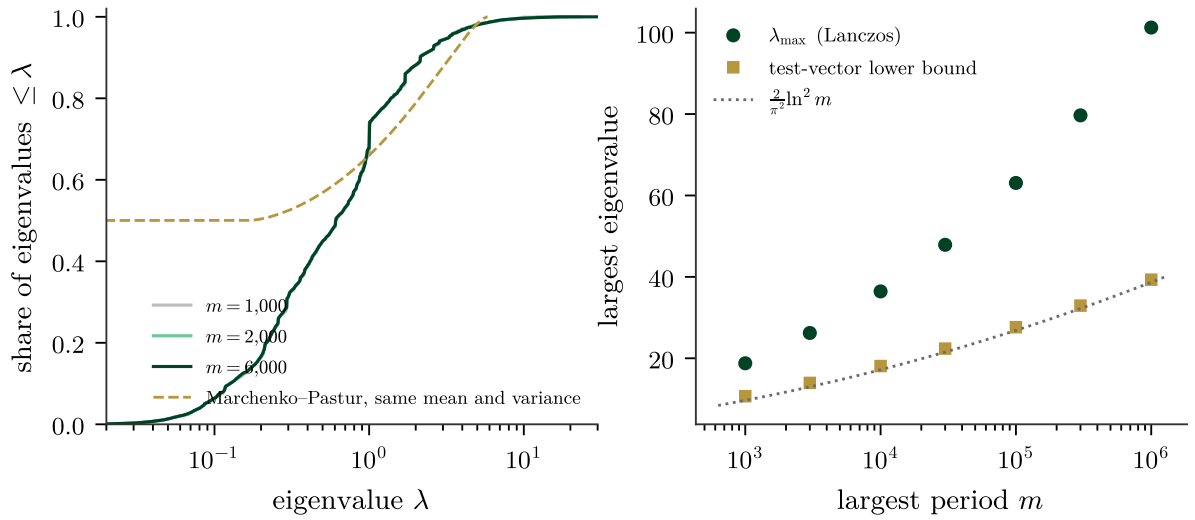


Figure 15.5. Left: the distribution of the eigenvalues of C_m for three sizes (the curves almost coincide), against the Marchenko–Pastur law with the same mean and variance (which puts half its mass at 0). Right: the largest eigenvalue up to $m = 10^6$, the proven lower bound from the test vector, and $(2/\pi^2)\ln^2 m$.

an explicit constant (proved); its top eigenvalue is at least $(2/\pi^2)\ln^2 m$, which is where $6/\pi^2$ appears (proved), but grows faster in our data. It is far from the Marchenko–Pastur and semicircle laws, yet its level spacings look like those of the GOE (numerical evidence). Whether $6/\pi^2$ appears in the limiting bulk distribution is open; it does not appear in its first two moments.

15.7 Question 6: ranking the equal temperaments

The measures. For each n from 5 to 200 we measured how well the n-note equal temperament approximates two sets of just intervals: the 5-limit consonances 3/2, 5/4, 6/5, and the 7-limit set that adds 7/4, 7/5, 7/6 (their inversions have the same errors). We used four measures:

RMS: the root-mean-square error in cents, each interval approximated by its nearest step;

relative: the RMS error in units of one step, that is RMS $\times\, n/1200$. Large n wins any absolute measure simply by having small steps; the relative error asks how good the temperament is for its size;

TOP: Erlich's Tenney-optimal error [29]: map each prime $2, 3, 5$ (and 7) to its nearest number of steps, weight the error at the prime p by $1/\log_2 p$, allow the octave to be stretched, and take the smallest possible maximum weighted error, in cents per octave;

TOP relative: TOP in units of one step.

The ranking. Table 15.7 lists the top 20 for each measure. The familiar names are all there: 171, 118, 99, 72, 53, 41, 31, 19 and 12, and 140 and 130. Of the 16 sizes between 5 and 200

Table 15.7. The top 20 equal temperaments between 5 and 200 notes for each measure, best first. Prime sizes are in bold.

measure	best 20
RMS, 5-limit	171, 118, 152, 183, 190, **137**, 186, **193**, 140, 164, **53**, 106, 159, 99, 198, **149**, 87, 174, 65, 130
RMS, 7-limit	171, 140, 99, 198, 130, **193**, 183, **167**, 152, 186, 190, 161, **181**, 175, **197**, 156, **103**, **149**, 187, 72
TOP, 5-limit	171, 118, 152, 183, 190, **53**, 106, 159, 164, 99, 198, **193**, **137**, 186, 140, 65, 130, 195, 178, **149**
TOP, 7-limit	171, 99, 198, **193**, 183, 152, 140, 130, 190, **149**, 159, 118, 200, **197**, 186, 176, **179**, 125, **181**, 72
relative, 5-limit	118, 171, **53**, 152, 65, 34, 87, 99, **19**, 183, 84, 106, **137**, **31**, 12, 140, 190, 72, 46, 186
relative, 7-limit	171, 99, **31**, 140, 72, 130, 68, **41**, **103**, 198, **53**, 152, 27, 118, 183, **167**, 84, **193**, 62, 46
TOP relative, 5-limit	171, 118, **53**, 65, 106, 152, 99, 183, 34, 12, 87, **19**, 72, 159, 46, 84, 164, **137**, 190, 130
TOP relative, 7-limit	171, 99, 72, 130, **31**, **41**, 140, 152, 118, **53**, 198, **149**, 22, 183, **193**, **19**, 12, 46, 159, 27

Table 15.8. Prime against composite sizes $n = 5, \ldots, 200$: mean score of each group (smaller is better), two-sided p-values of a permutation test (100,000 permutations) and of the Mann–Whitney test, the number of primes in the top 20, and the chance of at least that many in a random 20.

measure	primes	composites	p (perm.)	p (M–W)	top-20 primes	p
relative, 5-limit	0.283	0.273	0.51	0.46	4	0.70
relative, 7-limit	0.283	0.282	0.91	0.37	6	0.27
TOP relative, 5-limit	0.112	0.110	0.80	0.82	3	0.87
TOP relative, 7-limit	0.128	0.128	1.00	0.91	6	0.27
RMS, 5-limit	8.433	5.889	0.13	0.27	4	0.70
RMS, 7-limit	8.770	5.945	0.08	0.41	6	0.27

that Gene Ward Smith singled out by the peaks of the Riemann zeta function on the critical line (5, 7, 10, 12, 19, 22, 27, 31, 41, 53, 72, 99, 118, 130, 152, 171; 80), 5 are in the top 20 of the 7-limit RMS list and most are high in the relative lists. The winner on almost every measure is 171.

Do prime sizes stand out? No. Of the 196 sizes, 44 are prime, so a random top 20 would hold about 4.5 primes. Table 15.8 compares the mean score of prime and composite sizes with a permutation test and a Mann–Whitney test, and counts primes in each top 20 against the hypergeometric law. No test comes near significance for the scale-free (relative) measures. On the absolute measures the primes look slightly worse on average, but that is because primes are more common among small n, whose steps are large; the difference is not significant either. Figure 15.6 shows the relative error for every size.

There is a theorem behind this answer. The relative errors of n-EDO depend only on the fractional parts of $n\log_2 3$, $n\log_2 5$ and $n\log_2 7$. The numbers $1, \log_2 3, \log_2 5, \log_2 7$ are linearly independent over the rationals (a relation would give $2^a 3^b 5^c 7^d = 1$), and by Vinogradov's theorem on exponential sums over primes, the vectors $(p\log_2 3, p\log_2 5, p\log_2 7) \bmod 1$, p prime, are equidistributed in the cube [97], exactly as they are for all integers n by Weyl's theorem. So

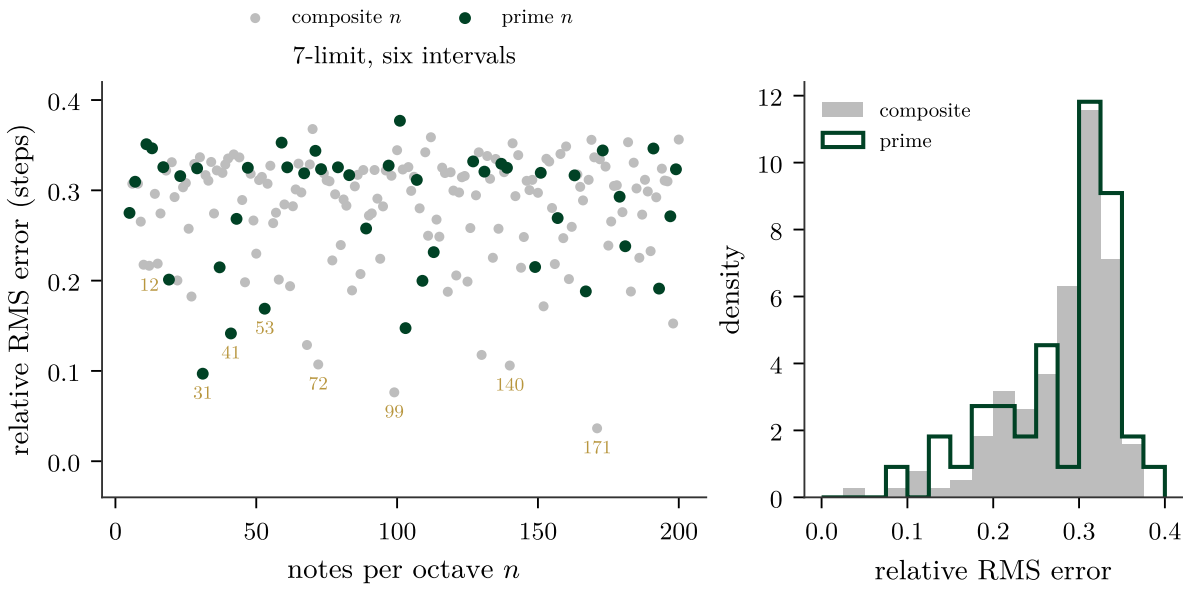


Figure 15.6. The relative 7-limit error (in steps) of every equal temperament from 5 to 200 notes, prime sizes in green (left), and the distributions for prime and composite sizes (right).

in the long run, the relative errors of prime-sized temperaments have the same distribution as those of all temperaments. Prime sizes do have one real musical property, noted in Chapter 4: every step size generates the whole scale. That is a fact about the scale's structure, not about its tuning accuracy.

Status. Answered: no. Prime sizes do not stand out in tuning accuracy, by any of the eight measures, and an equidistribution theorem says they cannot stand out in the long run.

15.8 Status after this work

Table 15.9 sums up where the six questions stand.

Table 15.9. The six questions of Section 14.3 after this chapter.

question	what we did	status
1. Prime convergents of π	12 million certified terms; q_n and p_n tested to $n = 20{,}000$; 15 prime numerators, 10 denominators; counts inside the heuristic's 95% range; e's divisibility analyzed	open; data typical
2. Fibonacci and Pell primes	known lists compared with the Grantham–Granville heuristic; divisibility facts proved	open
3. The heuristic in a model	exact expectation proved, so the constant $2\ln 2$ holds in expectation; for almost every x the count lies between $\ln N$ and $2\ln N$ (using deep cited theorems); finitely many n with p_n, q_n both prime (proved)	partly answered
4. Gauss–Kuzmin for π	calibrated tests on 12 million terms: values, tail, pairs, blocks, Khinchin, Lévy, records	open; data typical
5. Spectrum of the correlation matrix	Gram structure, determinant, mean and variance, geometric mean, lower bound for the top eigenvalue proved; GOE-like spacings observed	partly answered
6. Prime-sized temperaments	eight measures, top-20 lists, permutation tests	answered: no

For the video. Open with the six questions on screen and a stamp for each: two "still open" (Fibonacci primes, Gauss–Kuzmin for π), one "open, more data" (π's primes), two "partly answered" and one "answered: no." Spend most of the time on Question 3: the cylinder picture, the two expansions of $7/2$, and the line "in expectation, the constant is exactly $2\ln 2$." Then the matrix: mean 1, variance 2, and the eigenvalue 1 at the large primes. End with the temperaments: 171 wins, and primes are not special. About 12 minutes.

APPENDIX A

Algorithms

This appendix gives the main algorithms in pseudocode. The Python scripts of Appendix C implement each one.

Algorithm 1: Certified continued-fraction terms of a real constant

Input: a procedure that evaluates x to d decimal digits; the number of terms n
Output: the first n terms of x, each guaranteed correct

repeat
  evaluate x at precision d and at precision $d + 200$;
  expand both: $a \leftarrow \lfloor y \rfloor$, $y \leftarrow 1/(y - a)$, repeated;
  keep the common prefix of the two term lists, minus its last term;
  if *fewer than n terms are kept* **then** $d \leftarrow 1.6\,d$;
until *at least n terms are certified*;

Algorithm 2: Convergents (Proposition 2.1)

Input: terms $a_0, a_1, \ldots, a_N$

$p_{-1}, q_{-1}, p_0, q_0 \leftarrow 1, 0, a_0, 1$;
for $n \leftarrow 1$ **to** N **do**
  $p_n \leftarrow a_n p_{n-1} + p_{n-2}$; $q_n \leftarrow a_n q_{n-1} + q_{n-2}$;

Algorithm 3: The rhythm of a number (standard words, Theorem 3.5)

Input: terms $a_1, a_2, \ldots, a_k$ of $\alpha = [0; a_1, a_2, \ldots]$
Output: the level words $s_0, \ldots, s_k$; s_n has q_n beats and p_n hits

$s_{-1} \leftarrow$ `x`; $s_0 \leftarrow$ `.`;
$s_1 \leftarrow s_0^{a_1 - 1}\, s_{-1}$;
for $n \leftarrow 2$ **to** k **do**
  $s_n \leftarrow s_{n-1}^{a_n}\, s_{n-2}$; `// repeat, then close`

Algorithm 4: Bjorklund's algorithm for the Euclidean rhythm $E(k, n)$

Input: $1 \le k < n$

$A \leftarrow k$ groups `[x]`; $B \leftarrow n-k$ groups `[.]`;
while *B has more than one group* **do**
  $m \leftarrow \min(|A|, |B|)$;
  append the ith group of B to the ith group of A, for $i = 1, \dots, m$;
  the unused groups (from A if $|A| > m$, else from B) become the new B;
return *the concatenation of the groups of A and then of B*

Algorithm 5: Lehmer–Powers square search on the continued fraction of $\sqrt{kN}$

Input: odd composite N, multiplier k

$a_0 \leftarrow \lfloor \sqrt{kN} \rfloor$; $P \leftarrow 0$; $Q \leftarrow 1$; $a \leftarrow a_0$; $A_{-1} \leftarrow 1$; $A_0 \leftarrow a_0 \bmod N$;
for $n \leftarrow 1, 2, \dots$ **do**
  $P \leftarrow aQ - P$; $Q \leftarrow (kN - P^2)/Q$; $a \leftarrow \lfloor (a_0 + P)/Q \rfloor$;
  if *n is even and $Q = s^2$ for an integer s* **then**
    $g \leftarrow \gcd(A_{n-1} - s, N)$;
    if $1 < g < N$ **then return** g;
  $A_n \leftarrow (aA_{n-1} + A_{n-2}) \bmod N$;

Algorithm 6: Counting prime convergents of random reals (Chapter 8)

Input: number of reals M, number of convergents N

for $i \leftarrow 1$ **to** M **do**
  draw a uniform integer $r \in [1, 10^D)$ with $D = \lfloor 1.6N \rfloor + 20$;
  expand $r/10^D$ by exact Euclidean division; keep the first $N+1$ terms;
  compute p_n, q_n for $n \le N$ (Algorithm 2);
  foreach $n \le N$ **do**
    $E \leftarrow E + \zeta(2)/\ln q_n$; `// heuristic expectation`
    if *q_n has no prime factor below 3000 and passes Baillie–PSW* **then**
      count q_n as prime

Algorithm 7: Hausdorff dimension of a continued-fraction Cantor set

Input: allowed terms A; number of Chebyshev nodes m

nodes $x_1, \dots, x_m$ on $[0, 1]$; interpolation matrix V^{-1} for polynomials of degree $< m$;
`rho(`s`)`: matrix $M = \sum_{a \in A} \operatorname{diag}\big((a + x_i)^{-2s}\big)\, V_a V^{-1}$, where V_a evaluates the basis at $1/(a + x_i)$; **return** *spectral radius of M*;
bisect on $s \in [0, 1]$ for $\rho(s) = 1$;

Algorithm 8: Number variance of an unfolded point set

Input: sorted points $u_1 < \cdots < u_N$ with mean spacing 1; window lengths L; trials T

foreach L **do**
- draw T uniform window starts t in $[u_1, u_N - L]$;
- count the points in each window $[t, t + L)$ by binary search;
- $\Sigma^2(L) \leftarrow$ sample variance of the counts;

APPENDIX B

Computational methods

Software. All computations use Python 3 with mpmath 1.3 (arbitrary-precision reals and the zeros of the zeta function), sympy (factorizations, small primality tests), gmpy2 (fast big-integer arithmetic and the Baillie–PSW test), NumPy and SciPy (linear algebra, statistics), and matplotlib (figures). The typesetting uses XeLaTeX with BibTeX.

Continued fractions. The terms of π come from π evaluated to 11,600 decimal digits, expanded by the Gauss map at two working precisions; only terms on which the two expansions agree, less one, are kept (Algorithm 1), which certifies the first 10,000 terms. The constants e, $\sqrt{2}$, φ and $\log_2(3/2)$ are expanded the same way to 4,001 terms, with the precision raised automatically when needed. The pattern of e and the constant terms of $\sqrt{2}$ and φ are asserted, which checks the method.

Rhythms. Rhythms are built with Algorithm 3 and compared, letter for letter, with the floor rule (3.1) evaluated in multiple precision: 140,000 beats for $\pi - 3$ and complete words for 200 random term sequences.

Primality. A number is called prime if it is below 3,000 and in the list of small primes, or if it has no prime factor below 3,000 (one gcd with the primorial) and passes the Baillie–PSW probable-prime test of gmpy2. For the Fibonacci and Pell indices our lists agree with the OEIS [75, 79], and for π with [77, 78].

Long expansions. Chapter 15 uses 12,000,001 terms of π, from a 43-million-bit interval around π expanded by a divide-and-conquer form of Euclid's algorithm that keeps only the terms common to both ends of the interval (Section 15.1); the expansion is repeated at a second precision and checked against the published record terms. Primality tests there use trial division by the primes below 10^5 before Baillie–PSW. Random reals in that chapter are random fractions with 2,600 or more binary digits, expanded exactly.

Random reals. A random real is represented by $r/10^D$ with r a uniformly random integer of $D = 1{,}620$ digits (Algorithm 6). Its first 1,000 terms are shared by every real in $[r/10^D, (r+1)/10^D)$, as explained in Chapter 8. The generator is Python's `random.Random` with a fixed seed, so the sample is reproducible.

Zeta zeros. The first 5,000 zeros were computed with `mpmath.zetazero` at 20 digits; the script checks that `mpmath.nzeros` counts exactly 5,000 zeros up to the last one, so none is missing, and recomputes a few at higher precision. The explicit-formula figure uses the first 100 zeros.

Random matrices. GUE matrices are $H = (A + A^*)/2$ with A a 1000×1000 matrix of independent standard complex Gaussians, so that $E|H_{ij}|^2 = 1$ off the diagonal; eigenvalues are divided by $\sqrt{1000}$ (semicircle of radius 2) and unfolded with the semicircle distribution function, keeping the central half.

Statistics. Chi-square tests use SciPy's χ^2 survival function; the Kolmogorov–Smirnov distance is the two-sample statistic; Poisson tail probabilities use SciPy. The pair-correlation histogram counts all pairs within 3 mean spacings among the first 6,000 points of each set; the number variance uses 1,500 to 4,000 random windows per length (Algorithm 8).

Structure factors. $S(k)$ is computed directly from (6.1) on a grid of 40,000 wavenumbers, with positions measured from the start of the window; the figures use a Hann taper (weights $2\sin^2(\pi x/\ell)$ across the window of length ℓ) to suppress side lobes, while the peak values quoted in the text are untapered and evaluated exactly at the rational wavenumbers.

Circle map. Rotation numbers are averages over 2,520 iterations (a multiple of every $q \le 10$, so locked orbits give exact fractions) after a transient of 400; step widths are measured on a grid of 40,001 values of Ω.

Dimension. Algorithm 7 with 40 and 60 Chebyshev nodes; the two agree to 13 digits for $E_{\{1,2\}}$, and the 60-node value matches the published $0.53128050627720514\ldots$ [46] to the twelve digits we print.

APPENDIX C

Reproducibility

The full source of this work, including the LaTeX files, the bibliography and every script, is kept with the companion site's source under `docs/papers/rhythm-of-numbers/`. Running the scripts in order regenerates every figure (`paper/figures/`) and every number in the text: each script writes a file of LaTeX macros to `paper/gen/`, and the text uses only those macros for computed values. Scripts stop with an error if any asserted fact fails.

Building. With Python 3 and the packages of Appendix B installed, run `sh run_all.sh` in `compute/`, then `latexmk -xelatex main.tex` in `paper/`. Random choices use fixed seeds, so a rebuild reproduces the same numbers, except that the GUE samples (seeded too) and timings may differ across library versions in the last digit.

Provenance of non-computed facts. Dates, names and bibliographic details were checked against the sources listed in the bibliography. The musical facts about the Suite come from the companion site's data file `data/music/suite.yaml`, compiled from the engraved score.

Table C.1. The scripts, in the order `run_all.sh` runs them, with what each computes.

script	computes
`common.py`	shared helpers: certified continued fractions, convergents, standard words, the floor rule, macro output, figure style
`c1_cf.py`	terms of π, e, $\sqrt{2}$, φ, $\log_2(3/2)$; the starting guess; convergent table; Legendre and best-approximation checks; nested-block checks; beats 32,996 and 33,102; three-gap check; Gauss–Kuzmin, Khinchin and Lévy statistics
`c2_music.py`	equal temperaments, comma, thirds; Bjorklund vs. Christoffel; Euclidean table; circle map, tongues, staircase and Farey rule; Farey counts
`c3_coprime.py`	orchard, Euler product, correlation formula and matrix, Balmer, residue bias to 10^7, sunflower and wheels, structure factors, explicit formula
`c4_tools.py`	Lehmer–Powers factoring, necklaces, the prime hunt and its Poisson tails
`zeros.py`	the first 5,000 zeta zeros (about one hour)
`c5_spacing.py`	spacing distributions, pair correlation, number variance, the Fourier picture of the zeros
`c6_fractals.py`, `c6b_gcf.py`	complete quotients, Ford circles, Stern–Brocot, $?(x)$, dimensions, Mandelbrot bulbs, Fibonacci word, six pictures; patterned fractions for π
`c7_primeconv.py`, `c7b_primeconv_figs.py`	prime convergents of five constants and 400 random reals (several minutes); divisibility shares; figures and tables
`c8_suite.py`	the Suite's forms checked against the continued fractions; form figure
`c9_misc_figs.py`, `c10_connections.py`	remaining figures; drum-mode ratios
`c11_openq_picf.py`	12 million certified terms of π by divide-and-conquer, two precisions, checked against the OEIS records (about two minutes)
`c11_openq_pisearch.py`	probable-prime q_n and p_n of π for $n \le 20{,}000$ (resumable; about 45 minutes)
`c11_openq_stats.py`	Gauss–Kuzmin, pair, tail and block tests, Khinchin, Lévy and Galambos for 12 million terms, calibrated on 1,000 random reals (about 3 minutes)
`c11_openq_model.py`	the exact expectation of Theorem 15.3 and its check by brute force and by 20,000 random reals; the densities for e (about 2 minutes)
`c11_openq_matrix.py`	spectra of the correlation matrix to $m = 6{,}000$, the top eigenvalue to $m = 10^6$ (about 2 minutes)
`c11_openq_edo.py`, `c11_openq_primes.py`	equal-temperament rankings and tests; tables and figure for π's prime convergents and for Fibonacci and Pell primes

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